



ŘEDITELSTVÍ SILNIC A DÁLNIC ČR

NDIC - DATEX II Situation Publication - Common

Release 1.1rev1

Národní Dopravní Informační Centrum

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CONTENTS

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Publisher
NDIC

Author
Jan Vlčinský, Petr Bureš

The specification defines format for publishing common traffic information (accidents etc.).

INTRODUCTION

This document shall be used by publisher and / or subscriber. Subscriber shall use it for format suitability evaluation and for possible implementation of received NDIC data import into own information system. Publisher shall use it as a specification of data to be published format and for possible implementation of data export from own information system into NDIC.

Here you can find:

- conceptual description of published information
- information about location referencing methods used
- W3C XML schema of the format
- data sample(s)

Following topics are out of scope of this document and if present, shall be considered as informative only:

- transfer protocol incl. access point, life cycle etc.
- related data formats
- typical data origin
- typical data use

1.1 General terms

Following terms are used in this document:

format

description of data structure for transferring some kind of information

protocol

description of data transfer process and rules

NDIC

national traffic information centre

DATEX II model

class model for describing information related to the road traffic. The model in form of UML is platform independent.

One of specific platforms for expressing the model is XML.

The model is extensive (it consists of few thousands of structural items representing term incl. definition) For reference data (UML class model in form of Enterprise Architect EAP file, XML schema etc.) see [?]. For additional information see [?].

(DATEX II) profile

simplified class model, still sufficient for specific needs. Profile is created from model by selecting only specific parts.

(DATEX II) extension

extended DATEX II model, adding data structures missing in standard model but needed for specific needs.

B level extension

Extension of DATEX II model, which is backward compatible with the original DATEX II model, even though it contains additional structures. [?]

C level extension

Extension of DATEX II model, which is not backward compatible with the original DATEX II model.

publication

an element of DATEX II model, serving as an envelope for transmitted content of certain kind. Every sent DATEX II message contains exactly one element of publication type. There are various types of publications, for example *SituationPublication* [?], *ElaboratedDataPublication* [?], *PredefinedLocationsPublication* [?] etc.

(W3C) XML schema

language, which allows definition of rules for a XML document structure.

XSD

XML document containing XML schema. Sometimes it is synonym for XML schema itself.

uri

uniform resource identifier, text, used as unique identifier. In this document, every format has it's own uri.

CONCEPTS

Described format allows publishing general information about traffic situation. Common traffic information include, in particular, accidents, restrictions, closures, traffic conditions, road activities, etc.

2.1 Publication type SituationPublication

Format is based on DATEX II publication of type *SituationPublication* [?]. The profile uses most of data structures, provided by this type of publication.

```
1 <?xml version="1.0" encoding="utf-8"?>
2 <d2LogicalModel modelBaseVersion="2" xmlns="http://datex2.eu/schema/2/2_0"
3 xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
4 xsi:schemaLocation="http://datex2.eu/schema/2/2_0 ../schema/schema.xsd">
5   <exchange>
6     <supplierIdentification>
7       <country>cz</country>
8       <nationalIdentifier>NDIC</nationalIdentifier>
9     </supplierIdentification>
10  </exchange>
11  <payloadPublication lang="cs" xsi:type="SituationPublication">
12    <feedType>x-format:cz-ndic_d2-common-v1.1</feedType>
13    <publicationTime>2017-07-21T10:47:04+02:00</publicationTime>
14    <publicationCreator>
15      <country>cz</country>
16      <nationalIdentifier>PČR</nationalIdentifier>
17    </publicationCreator>
```

2.2 Event description (Situation a SituationRecords)

SituationPublication requires description by means of class *Situation*, which aggregates instances of class *SituationRecord* (or its descendants). *Situation* also contains elements with general service availability information.

Element *situation* describes one particular event, more particular events means more *situation* elements. Since the situation described may evolve over time, the element is versioned and has unique identifier. Each *situation* element contains one or more *situationRecord* elements.

The *situationRecord* element may, in the element *situation*, occur several times (each element describing different aspect of the situation, i.e. accident in one record leading to a queue described in another record). The *situationRecord* element may evolve over time, so the element is versioned and has unique identifier.

The traffic situation publication in particular includes:

- Increased traffic levels, i.e. queuing traffic, stationary traffic
- Accidents

- Activities at the roadside,
- Operating conditions
 - road conditions related to weather (snow, water, etc.)
 - other road condition (sand, oil, spilled load, etc.)
 - conditions given by the environment (precipitation, visibility, wind, etc.)
- Operator Actions - Roadwork - Communication management
 - Detours
 - Speed limitation
 - Changing of traffic organization

For situation description, most of the structures, provided by *SituationPublication* class can be used. For more details see W3C XML schema or [?].

```

18 <situation id="ff808181-53b7-4de8-0154-d27e38fb58ea" version="1">
19   <situationVersionTime>2017-07-21T10:46:18+02:00</situationVersionTime>
20   <headerInformation>
21     <confidentiality>noRestriction</confidentiality>
22     <informationStatus>real</informationStatus>
23     <urgency>urgent</urgency>
24   </headerInformation>
25   <situationRecord id="82998aae-7bde-419a-89a0-0aba6d79015d"
26     version="1" xsi:type="Accident">
27     <situationRecordCreationTime>2017-07-21T10:47:04+02:00</
↪situationRecordCreationTime>
28     <situationRecordVersionTime>2017-07-21T10:46:18+02:00</
↪situationRecordVersionTime>
29     <probabilityOfOccurrence>certain</probabilityOfOccurrence>
30     <validity>
31       <validityStatus>definedByValidityTimeSpec</validityStatus>
32       <validityTimeSpecification>
33         <overallStartTime>2017-07-21T10:45:00+02:00</overallStartTime>
34         <overallEndTime>2017-07-21T12:45:00+02:00</overallEndTime>
35       </validityTimeSpecification>
36     </validity>
37     <generalPublicComment>
38       <comment>
39         <values>
40           <value lang="cs">Od 21.7.2017 10:45 do 12:45; na trase
41             Štěrboholy - Troja v ulici Jižní spojka v obci Praha; nehoda
42             nákladního vozidla; 2 havarovaná vozidla; probíhá vyšetřování
43             nehody, nebezpečí; OA x NA, bez zranění osob.</value>
44         </values>
45       </comment>
46     </generalPublicComment>
47     <groupOfLocations xsi:type="Linear">
48       <alertCLinear xsi:type="AlertCMethod2Linear">
49         <alertCLocationCountryCode>CZ</alertCLocationCountryCode>
50         <alertCLocationTableNumber>25</alertCLocationTableNumber>
51         <alertCLocationTableVersion>6.0</alertCLocationTableVersion>
52         <alertCDirection>
53           <alertCDirectionCoded>negative</alertCDirectionCoded>
54         </alertCDirection>
55         <alertCMethod2PrimaryPointLocation>
56           <alertCLocation>

```

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```

57         <specificLocation>1377</specificLocation>
58     </alertCLocation>
59 </alertCMethod2PrimaryPointLocation>
60 <alertCMethod2SecondaryPointLocation>
61     <alertCLocation>
62         <specificLocation>4110</specificLocation>
63     </alertCLocation>
64 </alertCMethod2SecondaryPointLocation>
65 </alertCLinear>
66 <globalNetworkLinear>
67     <networkVersion>
68         <values>
69             <value lang="cs">15.06</value>
70         </values>
71     </networkVersion>
72 <linearGeometryType>continuous</linearGeometryType>
73 <startPoint>
74     <sjtskPointCoordinates>
75         <sjtskX>-741263</sjtskX>
76         <sjtskY>-1048786</sjtskY>
77     </sjtskPointCoordinates>
78 </startPoint>
79 <endPoint>
80     <sjtskPointCoordinates>
81         <sjtskX>-741263</sjtskX>
82         <sjtskY>-1048786</sjtskY>
83     </sjtskPointCoordinates>
84 </endPoint>
85 <linearWithinLinearGNElement>
86     <sectionId>2376306</sectionId>
87     <directionRelativeOnLinearSection>aligned</
88     <directionRelativeOnLinearSection>
89         <orderOfSection>1</orderOfSection>
90         <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
91             <percentageDistanceAlong>0</percentageDistanceAlong>
92         </fromPoint>
93         <toPoint xsi:type="PercentageDistanceAlongLinearElement">
94             <percentageDistanceAlong>1</percentageDistanceAlong>
95         </toPoint>
96     </linearWithinLinearGNElement>
97 </globalNetworkLinear>
98 </groupOfLocations>
99 <accidentType>accidentInvolvingHeavyLorries</accidentType>
100 <groupOfVehiclesInvolved>
101     <numberOfVehicles>1</numberOfVehicles>
102     <vehicleCharacteristics>
103         <vehicleType>lorry</vehicleType>
104     </vehicleCharacteristics>
105 </groupOfVehiclesInvolved>
106 </situationRecord>
107 <situationRecord id="a16f6186-57f6-45a5-a294-472e8b45c8d5"
108     version="1" xsi:type="VehicleObstruction">
109     <situationRecordCreationTime>2017-07-21T10:47:04+02:00</
110     <situationRecordCreationTime>
111     <situationRecordVersionTime>2017-07-21T10:46:18+02:00</
112     <situationRecordVersionTime>

```

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```

110 <probabilityOfOccurrence>certain</probabilityOfOccurrence>
111 <source>
112   <sourceIdentification>PCR</sourceIdentification>
113   <sourceName>
114     <values>
115       <value lang="cs">Policie České republiky</value>
116     </values>
117   </sourceName>
118 </source>
119 <validity>
120   <validityStatus>definedByValidityTimeSpec</validityStatus>
121   <validityTimeSpecification>
122     <overallStartTime>2017-07-21T10:45:00+02:00</overallStartTime>
123     <overallEndTime>2017-07-21T12:45:00+02:00</overallEndTime>
124   </validityTimeSpecification>
125 </validity>
126 <groupOfLocations xsi:type="Linear">
127   <alertCLinear xsi:type="AlertCMethod2Linear">
128     <alertCLocationCountryCode>CZ</alertCLocationCountryCode>
129     <alertCLocationTableNumber>25</alertCLocationTableNumber>
130     <alertCLocationTableVersion>6.0</alertCLocationTableVersion>
131     <alertCDirection>
132       <alertCDirectionCoded>negative</alertCDirectionCoded>
133     </alertCDirection>
134     <alertCMethod2PrimaryPointLocation>
135       <alertCLocation>
136         <specificLocation>1377</specificLocation>
137       </alertCLocation>
138     </alertCMethod2PrimaryPointLocation>
139     <alertCMethod2SecondaryPointLocation>
140       <alertCLocation>
141         <specificLocation>4110</specificLocation>
142       </alertCLocation>
143     </alertCMethod2SecondaryPointLocation>
144   </alertCLinear>
145   <globalNetworkLinear>
146     <networkVersion>
147       <values>
148         <value lang="cs">15.06</value>
149       </values>
150     </networkVersion>
151     <linearGeometryType>continuous</linearGeometryType>
152     <startPoint>
153       <sjtskPointCoordinates>
154         <sjtskX>-741263</sjtskX>
155         <sjtskY>-1048786</sjtskY>
156       </sjtskPointCoordinates>
157     </startPoint>
158     <endPoint>
159       <sjtskPointCoordinates>
160         <sjtskX>-741263</sjtskX>
161         <sjtskY>-1048786</sjtskY>
162       </sjtskPointCoordinates>
163     </endPoint>
164     <linearWithinLinearGNElement>
165       <sectionId>2376306</sectionId>

```

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```

166         <directionRelativeOnLinearSection>aligned</
↪directionRelativeOnLinearSection>
167         <orderOfSection>1</orderOfSection>
168         <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
169             <percentageDistanceAlong>0</percentageDistanceAlong>
170         </fromPoint>
171         <toPoint xsi:type="PercentageDistanceAlongLinearElement">
172             <percentageDistanceAlong>1</percentageDistanceAlong>
173         </toPoint>
174         </linearWithinLinearGNElement>
175         </globalNetworkLinear>
176         </groupOfLocations>
177         <vehicleObstructionType>brokenDownVehicle</vehicleObstructionType>
178     </situationRecord>
179     <situationRecord id="d491c9bb-b156-457d-907d-b47b8fd06ceb"
180     version="1" xsi:type="AuthorityOperation">
181         <situationRecordCreationTime>2017-07-21T10:47:04+02:00</
↪situationRecordCreationTime>
182         <situationRecordVersionTime>2017-07-21T10:46:18+02:00</
↪situationRecordVersionTime>
183         <probabilityOfOccurrence>certain</probabilityOfOccurrence>
184         <source>
185             <sourceIdentification>PCR</sourceIdentification>
186             <sourceName>
187                 <values>
188                     <value lang="cs">Policie České republiky</value>
189                 </values>
190             </sourceName>
191         </source>
192         <validity>
193             <validityStatus>definedByValidityTimeSpec</validityStatus>
194             <validityTimeSpecification>
195                 <overallStartTime>2017-07-21T10:45:00+02:00</overallStartTime>
196                 <overallEndTime>2017-07-21T12:45:00+02:00</overallEndTime>
197             </validityTimeSpecification>
198         </validity>
199         <groupOfLocations xsi:type="Linear">
200             <alertCLinear xsi:type="AlertCMethod2Linear">
201                 <alertCLocationCountryCode>CZ</alertCLocationCountryCode>
202                 <alertCLocationTableNumber>25</alertCLocationTableNumber>
203                 <alertCLocationTableVersion>6.0</alertCLocationTableVersion>
204                 <alertCDirection>
205                     <alertCDirectionCoded>negative</alertCDirectionCoded>
206                 </alertCDirection>
207                 <alertCMethod2PrimaryPointLocation>
208                     <alertCLocation>
209                         <specificLocation>1377</specificLocation>
210                     </alertCLocation>
211                 </alertCMethod2PrimaryPointLocation>
212                 <alertCMethod2SecondaryPointLocation>
213                     <alertCLocation>
214                         <specificLocation>4110</specificLocation>
215                     </alertCLocation>
216                 </alertCMethod2SecondaryPointLocation>
217             </alertCLinear>
218         </groupOfLocations>

```

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```

219     <networkVersion>
220       <values>
221         <value lang="cs">15.06</value>
222       </values>
223     </networkVersion>
224     <linearGeometryType>continuous</linearGeometryType>
225     <startPoint>
226       <sjtskPointCoordinates>
227         <sjtskX>-741263</sjtskX>
228         <sjtskY>-1048786</sjtskY>
229       </sjtskPointCoordinates>
230     </startPoint>
231     <endPoint>
232       <sjtskPointCoordinates>
233         <sjtskX>-741263</sjtskX>
234         <sjtskY>-1048786</sjtskY>
235       </sjtskPointCoordinates>
236     </endPoint>
237     <linearWithinLinearGNElement>
238       <sectionId>2376306</sectionId>
239       <directionRelativeOnLinearSection>aligned</
↪ directionRelativeOnLinearSection>
240       <orderOfSection>1</orderOfSection>
241       <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
242         <percentageDistanceAlong>0</percentageDistanceAlong>
243       </fromPoint>
244       <toPoint xsi:type="PercentageDistanceAlongLinearElement">
245         <percentageDistanceAlong>1</percentageDistanceAlong>
246       </toPoint>
247     </linearWithinLinearGNElement>
248     </globalNetworkLinear>
249   </groupOfLocations>
250   <authorityOperationType>accidentInvestigationWork</authorityOperationType>
251 </situationRecord>
252 </situation>

```

2.3 Location description (Point or linear road network segment)

Location is a point or linear segment, both referenced to a road network segment. Location of type area and single point locations without a road segment reference are not used. The location may be described by means of several methods (at least by one). Each *SituationRecord* has to have its own location description. Within one *Situation*, the *SituationRecord* location descriptions, may differ in i.e. one *SituationRecord* describing point source that caused a linear problem (second *SituationRecord*).

```

47   <groupOfLocations xsi:type="Linear">
48     <alertCLinear xsi:type="AlertCMethod2Linear">
49       <alertCLocationCountryCode>CZ</alertCLocationCountryCode>
50       <alertCLocationTableNumber>25</alertCLocationTableNumber>
51       <alertCLocationTableVersion>6.0</alertCLocationTableVersion>
52       <alertCDirection>
53         <alertCDirectionCoded>negative</alertCDirectionCoded>
54       </alertCDirection>
55       <alertCMethod2PrimaryPointLocation>
56         <alertCLocation>

```

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```

57         <specificLocation>1377</specificLocation>
58     </alertCLocation>
59 </alertCMethod2PrimaryPointLocation>
60 <alertCMethod2SecondaryPointLocation>
61     <alertCLocation>
62         <specificLocation>4110</specificLocation>
63     </alertCLocation>
64 </alertCMethod2SecondaryPointLocation>
65 </alertCLinear>
66 <globalNetworkLinear>
67     <networkVersion>
68         <values>
69             <value lang="cs">15.06</value>
70         </values>
71     </networkVersion>
72     <linearGeometryType>continuous</linearGeometryType>
73     <startPoint>
74         <sjtskPointCoordinates>
75             <sjtskX>-741263</sjtskX>
76             <sjtskY>-1048786</sjtskY>
77         </sjtskPointCoordinates>
78     </startPoint>
79     <endPoint>
80         <sjtskPointCoordinates>
81             <sjtskX>-741263</sjtskX>
82             <sjtskY>-1048786</sjtskY>
83         </sjtskPointCoordinates>
84     </endPoint>
85     <linearWithinLinearGNElement>
86         <sectionId>2376306</sectionId>
87         <directionRelativeOnLinearSection>aligned</
88     <directionRelativeOnLinearSection>
89         <orderOfSection>1</orderOfSection>
90         <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
91             <percentageDistanceAlong>0</percentageDistanceAlong>
92         </fromPoint>
93         <toPoint xsi:type="PercentageDistanceAlongLinearElement">
94             <percentageDistanceAlong>1</percentageDistanceAlong>
95         </toPoint>
96     </linearWithinLinearGNElement>
97 </globalNetworkLinear>
</groupOfLocations>

```

The position description methods are further described in the chapter “Location description”.

TIME VALIDITY

Time validity is stated independently for every single *SituationRecord* described in a situation and is defined by a time interval between two time instants. The class *Validity* is used with an instance of class *OverallPeriod* in the role of *validityTimeSpecification* and elements *overallStartTime* and *overallStopTime* to define start and end of the interval.

Setting a week-day and recurring time validity within the specified period, e.g. for recurring intermittent nightly tunnel closure, is not allowed. For such information, splitting to more messages valid in respective times is used.

Used time validity intervals may fall into past, present or future.

```
30      <validity>
31        <validityStatus>definedByValidityTimeSpec</validityStatus>
32        <validityTimeSpecification>
33          <overallStartTime>2017-07-21T10:45:00+02:00</overallStartTime>
34          <overallEndTime>2017-07-21T12:45:00+02:00</overallEndTime>
35        </validityTimeSpecification>
36      </validity>
```

LOCATION DESCRIPTION

The locations of the events varies thus its description can not be prepared in advance. For this reason, description by reference to *PrefefinedLocationPublication* can not be used. Element *groupOfLocations* with type *Point* or *Linear* (descendants of *NetworkLocation*) contains description of one physical location in one or more location referencing methods.

```
<groupOfLocations xsi:type="Linear">
```

Three location description methods for a linear location and three location description methods for point location can be used to describe the location. The method use depends on method availability at the location. At least one location description should be used. Point description without a linear reference is not allowed.

Following is a selection of most used location referencing methods.

4.1 Location description by means of ALERT-C

The method describes position by ALERT-C codes, by class *alertCLinear* of the *AlertCMethod2Linear* type for linear locations and by class *alertCPoint* of the *AlertCMethod2Point* type for point locations. Other methods of ALERT-C description are not used. The use of this method depends on the presence of the ALERT-C location at a given location.

The *alertCLinear* method uses the primary location (element *alertCMethod2PrimaryPointLocation*), secondary location (element *alertCMethod2SecondaryPointLocation*) and direction (*alertCDirection*) element to describe the event. Of course, the method also includes elements that allows to identify a particular location table.

Element *alertCLinear* can appear more than once in a situation location description. This could happen for long events that must be described by more than one ALERT-C route. ALERT-C allows to chain (reference between) locations only within one route.

The primary location is, by the ALERT-C principle, the place where the accident occurred or where the driver leaves the problem area. The secondary location is the end of the traffic queue or the place where the driver first enters the problem area.

This method is not optimal because the RDS-TMC uses to describe the event location the primary location, direction and extent, looking up the secondary and intermittent locations in the location table. Therefore, if the recipient systems are adapted to RDS-TMC event reconstruction, it will be necessary to modify them.

The *alertCPoint* method is used to describe the point location (the inside of a junction or a tunnel, a short section). This method uses the primary location (element *alertCMethod2PrimaryPointLocation*) and the direction (*alertCDirection*) to describe the event. Of course, the method also includes elements that allow to identify a particular location table.

```
<alertCLinear xsi:type="AlertCMethod2Linear">
  <alertCLocationCountryCode>CZ</alertCLocationCountryCode>
  <alertCLocationTableNumber>25</alertCLocationTableNumber>
  <alertCLocationTableVersion>6.0</alertCLocationTableVersion>
  <alertCDirection>
```

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```

67     <alertCDirectionCoded>negative</alertCDirectionCoded>
68 </alertCDirection>
69 <alertCMethod2PrimaryPointLocation>
70   <alertCLocation>
71     <specificLocation>1347</specificLocation>
72   </alertCLocation>
73 </alertCMethod2PrimaryPointLocation>
74 <alertCMethod2SecondaryPointLocation>
75   <alertCLocation>
76     <specificLocation>1347</specificLocation>
77   </alertCLocation>
78 </alertCMethod2SecondaryPointLocation>
79 </alertCLinear>

```

However, if it is necessary to express the short location covered by one ALERT-C point also by other methods, then the *alertCLinear* method has to be used (where the secondary location is equal to the primary), since the methods describing one location must be of same type e.g. point or linear.

4.2 Location description using a textual description of the road network

The method uses the “address” of the street or road (name, area, number, etc.) and the distance of the event location from the beginning and the end of the road. For description of linear locations, the *linearWithinLinearElement* class is used and for point locations the *pointAlongLinearElement* class is used. The use of this method depends on the presence of a proper description of the road (ideal for motorways).

The *roadNumber* element describes the number of the affected road, element *DirectionRelativeOnLinearSection* describes the direction of affected traffic, and the *fromPoint* and *toPoint* elements describes distance of the start and the end of the event from the beginning of the road.

```

80   <linearWithinLinearElement>
81     <directionRelativeOnLinearSection>opposite</
82   directionRelativeOnLinearSection>
83     <linearElement>
84       <roadNumber>D11</roadNumber>
85     </linearElement>
86     <fromPoint xsi:type="DistanceFromLinearElementStart">
87       <distanceAlong>8016</distanceAlong>
88     </fromPoint>
89     <toPoint xsi:type="DistanceFromLinearElementStart">
90       <distanceAlong>7942</distanceAlong>
91     </toPoint>
92   </linearWithinLinearElement>

```

The method for description of a point event *PointAlongLinearElement* differs by simply describing one distance from the beginning of the identified road instead of two.

4.3 Location description using the Global Network extension

This is a DATEX II model extension type B.

The method uses for location description identifiers of affected sections of the road network, specified by Global Network (GN) database, and event start and end coordinates (in the S-JTSK reference system). The last and initial segments are truncated by the relative distance of the start and end points of the event relative to the GN segment.

The *globalNetworkLinear* class is used to describe the linear location and *pointWithinLinearGNElement* for point location description.

The *globalNetworkLinear* element describes the location of one event on the network. It contains the network version identifier *networkVersion*, the points bounding the beginning and end of the event (*startPoint* and *endPoint* in coordinates, here S-JTSK). And one or more *linearWithinLinearGNElement* elements that describe specific segments of the GN database by their direction, identifier, location order, and relative truncation (for the first and last sections).

This is native means of location referencing by NDIC.

```

92      <globalNetworkLinear>
93        <networkVersion>
94          <values>
95            <value lang="cs">15.06</value>
96          </values>
97        </networkVersion>
98        <linearGeometryType>continuous</linearGeometryType>
99        <startPoint>
100          <sjtskPointCoordinates>
101            <sjtskX>-722190</sjtskX>
102            <sjtskY>-1041872</sjtskY>
103          </sjtskPointCoordinates>
104        </startPoint>
105        <endPoint>
106          <sjtskPointCoordinates>
107            <sjtskX>-722190</sjtskX>
108            <sjtskY>-1041872</sjtskY>
109          </sjtskPointCoordinates>
110        </endPoint>
111        <linearWithinLinearGNElement>
112          <sectionId>1397666</sectionId>
113          <directionRelativeOnLinearSection>opposite</
114      ↪ directionRelativeOnLinearSection>
115          <orderOfSection>1</orderOfSection>
116          <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
117            <percentageDistanceAlong>0</percentageDistanceAlong>
118          </fromPoint>
119          <toPoint xsi:type="PercentageDistanceAlongLinearElement">
120            <percentageDistanceAlong>1</percentageDistanceAlong>
121          </toPoint>
122        </linearWithinLinearGNElement>
      </globalNetworkLinear>

```

The method for description of a point location *pointWithinLinearGNElement* only describes one distance from the beginning of the identified segment of GN database instead of two distances. Is very similar to the *pointAlongLinearElement* method.

4.4 Location description using Coordinates

Point location description without a reference to the linear is not allowed.

Both methods *pointByCoordinates* and *pointByMultiCoordinates* could be used only within the framework of the above defined methods.

The point location is described using geographic coordinates. Two methods may be used based on used coordinate reference system.

Method *pointByCoordinates* is used for description of position in WGS-84 coordinate system.

Method *pointByMultiCoordinates* is used to describe location by one or more coordinate systems, one of which being Czech system S-JTSK and second WGS-84. This is a DATEX II model extension type B.

```
<startPoint>
  <sjtskPointCoordinates>
    <sjtskX>-741263</sjtskX>
    <sjtskY>-1048786</sjtskY>
  </sjtskPointCoordinates>
</startPoint>
```

APPENDIX A: MESSAGE EXAMPLES

5.1 accident.xml - accident of two vehicles

Publication contains exactly one message about an accident of two vehicles, where at least one of them is a truck.

```
<?xml version="1.0" encoding="utf-8"?>
<d2LogicalModel modelBaseVersion="2" xmlns="http://datex2.eu/schema/2/2_0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://datex2.eu/schema/2/2_0 ../schema/schema.xsd">
  <exchange>
    <supplierIdentification>
      <country>cz</country>
      <nationalIdentifier>NDIC</nationalIdentifier>
    </supplierIdentification>
  </exchange>
  <payloadPublication lang="cs" xsi:type="SituationPublication">
    <feedType>x-format:cz-ndic_d2-common-v1.1</feedType>
    <publicationTime>2017-07-21T10:47:04+02:00</publicationTime>
    <publicationCreator>
      <country>cz</country>
      <nationalIdentifier>PČR</nationalIdentifier>
    </publicationCreator>
    <situation id="ff808181-53b7-4de8-0154-d27e38fb58ea" version="1">
      <situationVersionTime>2017-07-21T10:46:18+02:00</situationVersionTime>
      <headerInformation>
        <confidentiality>noRestriction</confidentiality>
        <informationStatus>real</informationStatus>
        <urgency>urgent</urgency>
      </headerInformation>
      <situationRecord id="82998aae-7bde-419a-89a0-0aba6d79015d"
version="1" xsi:type="Accident">
        <situationRecordCreationTime>2017-07-21T10:47:04+02:00</
↪situationRecordCreationTime>
        <situationRecordVersionTime>2017-07-21T10:46:18+02:00</
↪situationRecordVersionTime>
        <probabilityOfOccurrence>certain</probabilityOfOccurrence>
        <validity>
          <validityStatus>definedByValidityTimeSpec</validityStatus>
          <validityTimeSpecification>
            <overallStartTime>2017-07-21T10:45:00+02:00</overallStartTime>
            <overallEndTime>2017-07-21T12:45:00+02:00</overallEndTime>
          </validityTimeSpecification>
        </validity>
        <generalPublicComment>
          <comment>
```

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```

<values>
  <value lang="cs">Od 21.7.2017 10:45 do 12:45; na trase
    Štěrboholy - Troja v ulici Jižní spojka v obci Praha; nehoda
    nákladního vozidla; 2 havarovaná vozidla; probíhá vyšetřování
    nehody, nebezpečí; OA x NA, bez zranění osob.</value>
</values>
</comment>
</generalPublicComment>
<groupOfLocations xsi:type="Linear">
  <alertCLinear xsi:type="AlertCMethod2Linear">
    <alertCLocationCountryCode>CZ</alertCLocationCountryCode>
    <alertCLocationTableNumber>25</alertCLocationTableNumber>
    <alertCLocationTableVersion>6.0</alertCLocationTableVersion>
    <alertCDirection>
      <alertCDirectionCoded>negative</alertCDirectionCoded>
    </alertCDirection>
    <alertCMethod2PrimaryPointLocation>
      <alertCLocation>
        <specificLocation>1377</specificLocation>
      </alertCLocation>
    </alertCMethod2PrimaryPointLocation>
    <alertCMethod2SecondaryPointLocation>
      <alertCLocation>
        <specificLocation>4110</specificLocation>
      </alertCLocation>
    </alertCMethod2SecondaryPointLocation>
  </alertCLinear>
  <globalNetworkLinear>
    <networkVersion>
      <values>
        <value lang="cs">15.06</value>
      </values>
    </networkVersion>
    <linearGeometryType>continuous</linearGeometryType>
    <startPoint>
      <sjtskPointCoordinates>
        <sjtskX>-741263</sjtskX>
        <sjtskY>-1048786</sjtskY>
      </sjtskPointCoordinates>
    </startPoint>
    <endPoint>
      <sjtskPointCoordinates>
        <sjtskX>-741263</sjtskX>
        <sjtskY>-1048786</sjtskY>
      </sjtskPointCoordinates>
    </endPoint>
    <linearWithinLinearGNElement>
      <sectionId>2376306</sectionId>
      <directionRelativeOnLinearSection>aligned</
->directionRelativeOnLinearSection>
      <orderOfSection>1</orderOfSection>
      <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
        <percentageDistanceAlong>0</percentageDistanceAlong>
      </fromPoint>
      <toPoint xsi:type="PercentageDistanceAlongLinearElement">
        <percentageDistanceAlong>1</percentageDistanceAlong>

```

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```

        </toPoint>
      </linearWithinLinearGNElement>
    </globalNetworkLinear>
  </groupOfLocations>
  <accidentType>accidentInvolvingHeavyLorries</accidentType>
  <groupOfVehiclesInvolved>
    <numberOfVehicles>1</numberOfVehicles>
    <vehicleCharacteristics>
      <vehicleType>lorry</vehicleType>
    </vehicleCharacteristics>
  </groupOfVehiclesInvolved>
</situationRecord>
<situationRecord id="a16f6186-57f6-45a5-a294-472e8b45c8d5"
version="1" xsi:type="VehicleObstruction">
  <situationRecordCreationTime>2017-07-21T10:47:04+02:00</
↪situationRecordCreationTime>
  <situationRecordVersionTime>2017-07-21T10:46:18+02:00</
↪situationRecordVersionTime>
  <probabilityOfOccurrence>certain</probabilityOfOccurrence>
  <source>
    <sourceIdentification>PCR</sourceIdentification>
    <sourceName>
      <values>
        <value lang="cs">Policie České republiky</value>
      </values>
    </sourceName>
  </source>
  <validity>
    <validityStatus>definedByValidityTimeSpec</validityStatus>
    <validityTimeSpecification>
      <overallStartTime>2017-07-21T10:45:00+02:00</overallStartTime>
      <overallEndTime>2017-07-21T12:45:00+02:00</overallEndTime>
    </validityTimeSpecification>
  </validity>
  <groupOfLocations xsi:type="Linear">
    <alertCLinear xsi:type="AlertCMethod2Linear">
      <alertCLocationCountryCode>CZ</alertCLocationCountryCode>
      <alertCLocationTableNumber>25</alertCLocationTableNumber>
      <alertCLocationTableVersion>6.0</alertCLocationTableVersion>
      <alertCDirection>
        <alertCDirectionCoded>negative</alertCDirectionCoded>
      </alertCDirection>
      <alertCMethod2PrimaryPointLocation>
        <alertCLocation>
          <specificLocation>1377</specificLocation>
        </alertCLocation>
      </alertCMethod2PrimaryPointLocation>
      <alertCMethod2SecondaryPointLocation>
        <alertCLocation>
          <specificLocation>4110</specificLocation>
        </alertCLocation>
      </alertCMethod2SecondaryPointLocation>
    </alertCLinear>
  </groupOfLocations>
  <networkVersion>
    <values>

```

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```

        <value lang="cs">15.06</value>
      </values>
    </networkVersion>
    <linearGeometryType>continuous</linearGeometryType>
    <startPoint>
      <sjtskPointCoordinates>
        <sjtskX>-741263</sjtskX>
        <sjtskY>-1048786</sjtskY>
      </sjtskPointCoordinates>
    </startPoint>
    <endPoint>
      <sjtskPointCoordinates>
        <sjtskX>-741263</sjtskX>
        <sjtskY>-1048786</sjtskY>
      </sjtskPointCoordinates>
    </endPoint>
    <linearWithinLinearGNElement>
      <sectionId>2376306</sectionId>
      <directionRelativeOnLinearSection>aligned</
    <directionRelativeOnLinearSection>
      <orderOfSection>1</orderOfSection>
      <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
        <percentageDistanceAlong>0</percentageDistanceAlong>
      </fromPoint>
      <toPoint xsi:type="PercentageDistanceAlongLinearElement">
        <percentageDistanceAlong>1</percentageDistanceAlong>
      </toPoint>
    </linearWithinLinearGNElement>
  </globalNetworkLinear>
</groupOfLocations>
<vehicleObstructionType>brokenDownVehicle</vehicleObstructionType>
</situationRecord>
<situationRecord id="d491c9bb-b156-457d-907d-b47b8fd06ceb"
version="1" xsi:type="AuthorityOperation">
  <situationRecordCreationTime>2017-07-21T10:47:04+02:00</
    <situationRecordCreationTime>
  <situationRecordVersionTime>2017-07-21T10:46:18+02:00</
    <situationRecordVersionTime>
  <probabilityOfOccurrence>certain</probabilityOfOccurrence>
  <source>
    <sourceIdentification>PCR</sourceIdentification>
    <sourceName>
      <values>
        <value lang="cs">Policie České republiky</value>
      </values>
    </sourceName>
  </source>
  <validity>
    <validityStatus>definedByValidityTimeSpec</validityStatus>
    <validityTimeSpecification>
      <overallStartTime>2017-07-21T10:45:00+02:00</overallStartTime>
      <overallEndTime>2017-07-21T12:45:00+02:00</overallEndTime>
    </validityTimeSpecification>
  </validity>
</groupOfLocations xsi:type="Linear">
  <alertCLinear xsi:type="AlertCMethod2Linear">

```

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```

<alertCLocationCountryCode>CZ</alertCLocationCountryCode>
<alertCLocationTableNumber>25</alertCLocationTableNumber>
<alertCLocationTableVersion>6.0</alertCLocationTableVersion>
<alertCDirection>
  <alertCDirectionCoded>negative</alertCDirectionCoded>
</alertCDirection>
<alertCMethod2PrimaryPointLocation>
  <alertCLocation>
    <specificLocation>1377</specificLocation>
  </alertCLocation>
</alertCMethod2PrimaryPointLocation>
<alertCMethod2SecondaryPointLocation>
  <alertCLocation>
    <specificLocation>4110</specificLocation>
  </alertCLocation>
</alertCMethod2SecondaryPointLocation>
</alertCLinear>
<globalNetworkLinear>
  <networkVersion>
    <values>
      <value lang="cs">15.06</value>
    </values>
  </networkVersion>
  <linearGeometryType>continuous</linearGeometryType>
  <startPoint>
    <sjtskPointCoordinates>
      <sjtskX>-741263</sjtskX>
      <sjtskY>-1048786</sjtskY>
    </sjtskPointCoordinates>
  </startPoint>
  <endPoint>
    <sjtskPointCoordinates>
      <sjtskX>-741263</sjtskX>
      <sjtskY>-1048786</sjtskY>
    </sjtskPointCoordinates>
  </endPoint>
  <linearWithinLinearGNElement>
    <sectionId>2376306</sectionId>
    <directionRelativeOnLinearSection>aligned</
->directionRelativeOnLinearSection>
    <orderOfSection>1</orderOfSection>
    <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
      <percentageDistanceAlong>0</percentageDistanceAlong>
    </fromPoint>
    <toPoint xsi:type="PercentageDistanceAlongLinearElement">
      <percentageDistanceAlong>1</percentageDistanceAlong>
    </toPoint>
  </linearWithinLinearGNElement>
</globalNetworkLinear>
</groupOfLocations>
<authorityOperationType>accidentInvestigationWork</authorityOperationType>
</situationRecord>
</situation>
</payloadPublication>
</d2LogicalModel>

```

5.1.1 Root element and publication envelope

Root element *d2LogicalModel* declares, that this message is using DATEX II.

Element *exchange* contains provider identifier.

Element *payloadPublication* is an envelope for actual information payload containing general information and one or more elements *situation*, (in our case there is one).

Attribute *xsi:type="SituationPublication"* in element *payloadPublication* determines, that this envelope is of type Situation publication.

Element *feedType* states publikacation schema by its uri.

Element *publicationTime* states publication time of this publication as a whole.

Element *publicationCreator* identifies creator of the publication.

5.1.2 Description of one specific situation

Element *situation* describes a particular event, more particular events means more *situation* elements. Each *situation* element may contain one or more individual sub-messages, elements *situationRecord*. In this case occurs in the element *situation* just once.

Element *situationVersionTime* sets the time of the publication of this version of the situation.

Element *headerInformation* describes how to handle the information described.

Event description (what happened)

Attribute *xsi:type="Accident"* of element *situationRecord* defines event type. In DATEX II model it can be verified, that given attribute is of type of descendant of class *SituationRecord* in following descendant sequence *Accident* -> *TrafficElement* -> *SituationRecord*.

Element *probabilityOfOccurrence* states by value *certain*, that the event has certainly happened.

Element *accidentType* defines the type of accident. In this case value *accidentInvolvingHeavyLorries* states, that a heavy truck is involved in this accident.

Element *vehicleInvolved* partially repeats previous information by giving characteristics of the participating vehicle being *lorry*.

Time validity

Estimated time validity is given in a structure *validity* by values of validity interval start *overallStartTime* and validity interval end *overallEndTime*.

Event location

Event location is described in element *groupOfLocations* directly by embedded description. Reference to predefined locations is not used. Attribute *xsi:type="Linear"* states, that given element is of type linear location, which can be expressed by different location referencing methods.

The position of linear type is represented by three descriptions. First, using element *alertCLinear* of type *AlertCMethod2Linear* and then described using Global Network links in element *globalNetworkLinear*.

All location referencing methods used are describing the same location. Because the position of event is located within a ALERT-C segment, the *Linear* location type has to be used, with the result that *startPoint* and *endPoint* in the *globalNetworkLinear* element will be the same.

Additional content

The document states time of its first version in *situationRecordCreationTime* and time of current version in *situationRecordVersionTime* for each *situationRecord*.

Element *source* lists the the creator of this information, who is responsible for its accuracy.

5.2 brokendownvehicles.xml - two broken down lorries

Publication contains information about two broken down lorries, which fully block traffic on given road.

```
<?xml version="1.0" encoding="utf-8"?>
<d2LogicalModel xmlns="http://datex2.eu/schema/2/2_0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" modelBaseVersion="2"
xsi:schemaLocation="http://datex2.eu/schema/2/2_0 ../schema/schema.xsd">
  <exchange>
    <supplierIdentification>
      <country>cz</country>
      <nationalIdentifier>NDIC</nationalIdentifier>
    </supplierIdentification>
  </exchange>
  <payloadPublication xsi:type="SituationPublication" lang="cz">
    <feedType>x-format:cz-ndic_d2-common-v1.1</feedType>
    <publicationTime>2017-07-24T13:42:12+02:00</publicationTime>
    <publicationCreator>
      <country>cz</country>
      <nationalIdentifier>NDIC</nationalIdentifier>
    </publicationCreator>
    <situation id="ff808181-49aa-aeaa-0149-e1d266c74728" version="2">
      <headerInformation>
        <confidentiality>noRestriction</confidentiality>
        <informationStatus>real</informationStatus>
        <urgency>urgent</urgency>
      </headerInformation>
      <situationRecord xsi:type="VehicleObstruction"
id="2143f305-3090-4e03-8078-50e721605614"
version="1">
        <situationRecordCreationTime>2017-07-24T13:40:00+02:00</
↪situationRecordCreationTime>
        <situationRecordVersionTime>2017-07-24T13:42:12+02:00</
↪situationRecordVersionTime>
        <probabilityOfOccurrence>certain</probabilityOfOccurrence>
        <source>
          <sourceIdentification>PCR</sourceIdentification>
          <sourceName>
            <values>
              <value lang="cs">Policie České republiky</value>
            </values>
          </sourceName>
        </source>
        <validity>
          <validityStatus>definedByValidityTimeSpec</validityStatus>
          <validityTimeSpecification>
            <overallStartTime>2017-07-24T13:40:00+02:00</overallStartTime>
            <overallEndTime>2017-07-24T15:45:00+02:00</overallEndTime>
          </validityTimeSpecification>
        </validity>
      </situationRecord>
    </situation>
  </payloadPublication>
</d2LogicalModel>
```

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```

</validity>
<impact>
  <trafficConstrictionType>roadBlocked</trafficConstrictionType>
</impact>
<generalPublicComment>
  <comment>
    <values>
      <value lang="cs">2 havarovaná nákladní vozidla</value>
    </values>
  </comment>
</generalPublicComment>
<groupOfLocations xsi:type="Linear">
  <supplementaryPositionalDescription>
    <affectedCarriagewayAndLanes>
      <carriageway>mainCarriageway</carriageway>
      <lane>hardShoulder</lane>
    </affectedCarriagewayAndLanes>
  </supplementaryPositionalDescription>
  <alertCLinear xsi:type="AlertCMethod2Linear">
    <alertCLocationCountryCode>CZ</alertCLocationCountryCode>
    <alertCLocationTableNumber>25</alertCLocationTableNumber>
    <alertCLocationTableVersion>6.0</alertCLocationTableVersion>
    <alertCDirection>
      <alertCDirectionCoded>negative</alertCDirectionCoded>
    </alertCDirection>
    <alertCMethod2PrimaryPointLocation>
      <alertCLocation>
        <specificLocation>1347</specificLocation>
      </alertCLocation>
    </alertCMethod2PrimaryPointLocation>
    <alertCMethod2SecondaryPointLocation>
      <alertCLocation>
        <specificLocation>1347</specificLocation>
      </alertCLocation>
    </alertCMethod2SecondaryPointLocation>
  </alertCLinear>
  <linearWithinLinearElement>
    <directionRelativeOnLinearSection>opposite</
→directionRelativeOnLinearSection>
    <linearElement>
      <roadNumber>D11</roadNumber>
    </linearElement>
    <fromPoint xsi:type="DistanceFromLinearElementStart">
      <distanceAlong>8016</distanceAlong>
    </fromPoint>
    <toPoint xsi:type="DistanceFromLinearElementStart">
      <distanceAlong>7942</distanceAlong>
    </toPoint>
  </linearWithinLinearElement>
</globalNetworkLinear>
  <networkVersion>
    <values>
      <value lang="cs">15.06</value>
    </values>
  </networkVersion>
  <linearGeometryType>continuous</linearGeometryType>

```

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```

    <startPoint>
      <sjtskPointCoordinates>
        <sjtskX>-722190</sjtskX>
        <sjtskY>-1041872</sjtskY>
      </sjtskPointCoordinates>
    </startPoint>
    <endPoint>
      <sjtskPointCoordinates>
        <sjtskX>-722190</sjtskX>
        <sjtskY>-1041872</sjtskY>
      </sjtskPointCoordinates>
    </endPoint>
    <linearWithinLinearGNElement>
      <sectionId>1397666</sectionId>
      <directionRelativeOnLinearSection>opposite</
    directionRelativeOnLinearSection>
      <orderOfSection>1</orderOfSection>
      <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
        <percentageDistanceAlong>0</percentageDistanceAlong>
      </fromPoint>
      <toPoint xsi:type="PercentageDistanceAlongLinearElement">
        <percentageDistanceAlong>1</percentageDistanceAlong>
      </toPoint>
    </linearWithinLinearGNElement>
  </globalNetworkLinear>
</groupOfLocations>
<numberOfObstructions>2</numberOfObstructions>
<vehicleObstructionType>brokenDownVehicle</vehicleObstructionType>
<obstructingVehicle>
  <vehicleCharacteristics>
    <vehicleType>lorry</vehicleType>
  </vehicleCharacteristics>
</obstructingVehicle>
</situationRecord>
</situation>
</payloadPublication>
</d2LogicalModel>

```

5.2.1 Publication envelope, time validity and location

Publication envelope, time validity and location is using identical structures as previous example.

The only difference is in description of one concrete situation in element *situationRecord*.

Event description (what happened)

Attribute *xsi:type="VehicleObstruction"* of element *situationRecord* defines the type of event. In DATEX II model it can be verified, that given attribute is of type of descendant of class *SituationRecord* in following descendant sequence *VehicleObstruction* -> *Obstruction* -> *TrafficElement* -> *SituationRecord*.

Element *probabilityOfOccurrence* has value *certain* which means, that the event certainly happened.

Element *impact* states in *trafficConstrictionType*, that the accident resulted in the road being totally impassable.

Element *numberOfObstructions* by value of 2 states that there are 2 obstacles.

Element *vehicleObstructionType* by value *brokenDownVehicle* specifies the type of obstacle being “broken down vehicles”.

Element *obstructingVehicle* extends the previous information by value *lorry*, which means, that lorry or lorries are involved.

Time validity

As in previous example.

Event location

It differs from previous example in using one more method, instead of two, three are used. First, using element *alertCLinear* of type *AlertCMethod2Linear*, then using a general description of the position on the road in element *linearWithinLinearElement* and then described using Global Network links in element *globalNetworkLinear*.

The same case occurs as in the previous example, with respect to the ALERT-C location that is linear (can not be point), methods are also selected as linear.

All location referencing methods used are describing the same location.

Additional content

As in previous example.

APPENDIX B: W3C XML SCHEMA

XML structure of a transmitted message is defined by following W3C XML schema:

6.1 *schema.xsd*

```

1 <?xml version="1.0" encoding="utf-8" standalone="no"?>
2 <xs:schema elementFormDefault="qualified" attributeFormDefault="unqualified"
  ↳ xmlns:D2LogicalModel="http://datex2.eu/schema/2/2_0" version="2.3" targetNamespace=
  ↳ "http://datex2.eu/schema/2/2_0" xmlns:xs="http://www.w3.org/2001/XMLSchema">
3   <xs:complexType name="_ExtensionType">
4     <xs:sequence>
5       <xs:any namespace="##any" processContents="lax" minOccurs="0" maxOccurs=
  ↳ "unbounded" />
6     </xs:sequence>
7   </xs:complexType>
8   <xs:complexType name="_IntermediatePointOnLinearElement">
9     <xs:sequence>
10      <xs:element name="referent" type="D2LogicalModel:Referent" minOccurs="1"
  ↳ maxOccurs="1" />
11    </xs:sequence>
12    <xs:attribute name="index" type="xs:int" use="required" />
13  </xs:complexType>
14  <xs:complexType name="_LinearExtensionType">
15    <xs:sequence>
16      <xs:element name="openlrExtendedLinear" type=
  ↳ "D2LogicalModel:OpenlrExtendedLinear" minOccurs="0" />
17      <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs=
  ↳ "unbounded" />
18    </xs:sequence>
19  </xs:complexType>
20  <xs:complexType name="_PointExtensionType">
21    <xs:sequence>
22      <xs:element name="openlrExtendedPoint" type="D2LogicalModel:OpenlrExtendedPoint
  ↳ " minOccurs="0" />
23      <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs=
  ↳ "unbounded" />
24    </xs:sequence>
25  </xs:complexType>
26  <xs:complexType name="AbnormalTraffic">
27    <xs:complexContent>
28      <xs:extension base="D2LogicalModel:TrafficElement">
29        <xs:sequence>
30          <xs:element name="abnormalTrafficType" type=
  ↳ "D2LogicalModel:AbnormalTrafficTypeEnum" minOccurs="0" maxOccurs="1" />

```

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```

31      <xs:element name="relativeTrafficFlow" type=
↪ "D2LogicalModel:RelativeTrafficFlowEnum" minOccurs="0" maxOccurs="1" />
32      <xs:element name="trafficTrendType" type=
↪ "D2LogicalModel:TrafficTrendTypeEnum" minOccurs="0" maxOccurs="1" />
33      <xs:element name="abnormalTrafficExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
34    </xs:sequence>
35  </xs:extension>
36 </xs:complexContent>
37 </xs:complexType>
38 <xs:simpleType name="AbnormalTrafficTypeEnum">
39   <xs:restriction base="xs:string">
40     <xs:enumeration value="stationaryTraffic" />
41     <xs:enumeration value="queuingTraffic" />
42     <xs:enumeration value="slowTraffic" />
43     <xs:enumeration value="heavyTraffic" />
44     <xs:enumeration value="unspecifiedAbnormalTraffic" />
45     <xs:enumeration value="other" />
46   </xs:restriction>
47 </xs:simpleType>
48 <xs:complexType name="Accident">
49   <xs:complexContent>
50     <xs:extension base="D2LogicalModel:TrafficElement">
51       <xs:sequence>
52         <xs:element name="accidentType" type="D2LogicalModel:AccidentTypeEnum" ↪
↪ minOccurs="1" maxOccurs="unbounded" />
53         <xs:element name="totalNumberOfVehiclesInvolved" type=
↪ "D2LogicalModel:NonNegativeInteger" minOccurs="0" maxOccurs="1" />
54         <xs:element name="vehicleInvolved" type="D2LogicalModel:Vehicle" minOccurs=
↪ "0" maxOccurs="unbounded" />
55         <xs:element name="groupOfVehiclesInvolved" type=
↪ "D2LogicalModel:GroupOfVehiclesInvolved" minOccurs="0" maxOccurs="unbounded" />
56         <xs:element name="groupOfPeopleInvolved" type=
↪ "D2LogicalModel:GroupOfPeopleInvolved" minOccurs="0" maxOccurs="unbounded" />
57         <xs:element name="accidentExtension" type="D2LogicalModel:_ExtensionType" ↪
↪ minOccurs="0" />
58       </xs:sequence>
59     </xs:extension>
60   </xs:complexContent>
61 </xs:complexType>
62 <xs:simpleType name="AccidentTypeEnum">
63   <xs:restriction base="xs:string">
64     <xs:enumeration value="accident" />
65     <xs:enumeration value="accidentInvolvingBicycles" />
66     <xs:enumeration value="accidentInvolvingBuses" />
67     <xs:enumeration value="accidentInvolvingHazardousMaterials" />
68     <xs:enumeration value="accidentInvolvingHeavyLorries" />
69     <xs:enumeration value="accidentInvolvingMassTransitVehicle" />
70     <xs:enumeration value="accidentInvolvingMopeds" />
71     <xs:enumeration value="accidentInvolvingMotorcycles" />
72     <xs:enumeration value="accidentInvolvingRadioactiveMaterial" />
73     <xs:enumeration value="accidentInvolvingTrain" />
74     <xs:enumeration value="chemicalSpillageAccident" />
75     <xs:enumeration value="collision" />
76     <xs:enumeration value="collisionWithAnimal" />
77     <xs:enumeration value="collisionWithObstruction" />

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78     <xs:enumeration value="collisionWithPerson" />
79     <xs:enumeration value="earlierAccident" />
80     <xs:enumeration value="fuelSpillageAccident" />
81     <xs:enumeration value="headOnCollision" />
82     <xs:enumeration value="headOnOrSideCollision" />
83     <xs:enumeration value="jackknifedArticulatedLorry" />
84     <xs:enumeration value="jackknifedCaravan" />
85     <xs:enumeration value="jackknifedTrailer" />
86     <xs:enumeration value="multipleVehicleCollision" />
87     <xs:enumeration value="multivehicleAccident" />
88     <xs:enumeration value="oilSpillageAccident" />
89     <xs:enumeration value="overturnedHeavyLorry" />
90     <xs:enumeration value="overturnedTrailer" />
91     <xs:enumeration value="overturnedVehicle" />
92     <xs:enumeration value="rearCollision" />
93     <xs:enumeration value="secondaryAccident" />
94     <xs:enumeration value="seriousAccident" />
95     <xs:enumeration value="sideCollision" />
96     <xs:enumeration value="vehicleOffRoad" />
97     <xs:enumeration value="vehicleSpunAround" />
98     <xs:enumeration value="other" />
99   </xs:restriction>
100 </xs:simpleType>
101 <xs:complexType name="Activity" abstract="true">
102   <xs:complexContent>
103     <xs:extension base="D2LogicalModel:TrafficElement">
104       <xs:sequence>
105         <xs:element name="activityExtension" type="D2LogicalModel:_ExtensionType"
106         ↪ minOccurs="0" />
107       </xs:sequence>
108     </xs:extension>
109   </xs:complexContent>
110 </xs:complexType>
111 <xs:complexType name="AffectedCarriagewayAndLanes">
112   <xs:sequence>
113     <xs:element name="carriageway" type="D2LogicalModel:CarriagewayEnum" minOccurs=
114     ↪ "1" maxOccurs="1" />
115     <xs:element name="lane" type="D2LogicalModel:LaneEnum" minOccurs="0" maxOccurs=
116     ↪ "unbounded" />
117     <xs:element name="footpath" type="D2LogicalModel:Boolean" minOccurs="0"
118     ↪ maxOccurs="1" />
119     <xs:element name="lengthAffected" type="D2LogicalModel:MetresAsFloat" minOccurs=
120     ↪ "0" maxOccurs="1" />
121     <xs:element name="affectedCarriagewayAndLanesExtension" type="D2LogicalModel:_
122     ↪ ExtensionType" minOccurs="0" />
123   </xs:sequence>
124 </xs:complexType>
125 <xs:complexType name="AlertCDirection">
126   <xs:sequence>
127     <xs:element name="alertCDirectionCoded" type="D2LogicalModel:AlertCDirectionEnum
128     ↪ " minOccurs="1" maxOccurs="1" />
129     <xs:element name="alertCDirectionNamed" type="D2LogicalModel:MultilingualString
130     ↪ " minOccurs="0" maxOccurs="1" />
131     <xs:element name="alertCDirectionSense" type="D2LogicalModel:Boolean" minOccurs=
132     ↪ "0" maxOccurs="1" />
133     <xs:element name="alertCDirectionExtension" type="D2LogicalModel:_ExtensionType

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125   <minOccurs="0" />
126 </xs:sequence>
127 </xs:complexType>
128 <xs:simpleType name="AlertCDirectionEnum">
129   <xs:restriction base="xs:string">
130     <xs:enumeration value="both" />
131     <xs:enumeration value="negative" />
132     <xs:enumeration value="positive" />
133     <xs:enumeration value="unknown" />
134   </xs:restriction>
135 </xs:simpleType>
136 <xs:complexType name="AlertCLinear" abstract="true">
137   <xs:sequence>
138     <xs:element name="alertCLocationCountryCode" type="D2LogicalModel:String"
139       minOccurs="1" maxOccurs="1" />
140     <xs:element name="alertCLocationTableNumber" type="D2LogicalModel:String"
141       minOccurs="1" maxOccurs="1" />
142     <xs:element name="alertCLocationTableVersion" type="D2LogicalModel:String"
143       minOccurs="1" maxOccurs="1" />
144     <xs:element name="alertCLinearExtension" type="D2LogicalModel:_ExtensionType"
145       minOccurs="0" />
146   </xs:sequence>
147 </xs:complexType>
148 <xs:complexType name="AlertCLinearByCode">
149   <xs:complexContent>
150     <xs:extension base="D2LogicalModel:AlertCLinear">
151       <xs:sequence>
152         <xs:element name="alertCDirection" type="D2LogicalModel:AlertCDirection" />
153         <xs:element name="locationCodeForLinearLocation" type=
154           "D2LogicalModel:AlertCLocation" />
155         <xs:element name="alertCLinearByCodeExtension" type="D2LogicalModel:_
156           ExtensionType" minOccurs="0" />
157       </xs:sequence>
158     </xs:extension>
159   </xs:complexContent>
160 </xs:complexType>
161 <xs:complexType name="AlertCLocation">
162   <xs:sequence>
163     <xs:element name="alertCLocationName" type="D2LogicalModel:MultilingualString"
164       minOccurs="0" maxOccurs="1" />
165     <xs:element name="specificLocation" type="D2LogicalModel:AlertCLocationCode"
166       minOccurs="1" maxOccurs="1" />
167     <xs:element name="alertCLocationExtension" type="D2LogicalModel:_ExtensionType"
168       minOccurs="0" />
169   </xs:sequence>
170 </xs:complexType>
171 <xs:simpleType name="AlertCLocationCode">
172   <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
173 </xs:simpleType>
174 <xs:complexType name="AlertCMethod2Linear">
175   <xs:complexContent>
176     <xs:extension base="D2LogicalModel:AlertCLinear">
177       <xs:sequence>
178         <xs:element name="alertCDirection" type="D2LogicalModel:AlertCDirection" />
179         <xs:element name="alertCMethod2PrimaryPointLocation" type=
180           "D2LogicalModel:AlertCMethod2PrimaryPointLocation" />

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170     <xs:element name="alertCMethod2SecondaryPointLocation" type=
↪ "D2LogicalModel:AlertCMethod2SecondaryPointLocation" />
171     <xs:element name="alertCMethod2LinearExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
172     </xs:sequence>
173     </xs:extension>
174     </xs:complexContent>
175 </xs:complexType>
176 <xs:complexType name="AlertCMethod2Point">
177     <xs:complexContent>
178         <xs:extension base="D2LogicalModel:AlertCPoint">
179             <xs:sequence>
180                 <xs:element name="alertCDirection" type="D2LogicalModel:AlertCDirection" />
181                 <xs:element name="alertCMethod2PrimaryPointLocation" type=
↪ "D2LogicalModel:AlertCMethod2PrimaryPointLocation" />
182                 <xs:element name="alertCMethod2PointExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
183             </xs:sequence>
184         </xs:extension>
185     </xs:complexContent>
186 </xs:complexType>
187 <xs:complexType name="AlertCMethod2PrimaryPointLocation">
188     <xs:sequence>
189         <xs:element name="alertCLocation" type="D2LogicalModel:AlertCLocation" />
190         <xs:element name="alertCMethod2PrimaryPointLocationExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
191     </xs:sequence>
192 </xs:complexType>
193 <xs:complexType name="AlertCMethod2SecondaryPointLocation">
194     <xs:sequence>
195         <xs:element name="alertCLocation" type="D2LogicalModel:AlertCLocation" />
196         <xs:element name="alertCMethod2SecondaryPointLocationExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
197     </xs:sequence>
198 </xs:complexType>
199 <xs:complexType name="AlertCPoint" abstract="true">
200     <xs:sequence>
201         <xs:element name="alertCLocationCountryCode" type="D2LogicalModel:String"
↪ minOccurs="1" maxOccurs="1" />
202         <xs:element name="alertCLocationTableNumber" type="D2LogicalModel:String"
↪ minOccurs="1" maxOccurs="1" />
203         <xs:element name="alertCLocationTableVersion" type="D2LogicalModel:String"
↪ minOccurs="1" maxOccurs="1" />
204         <xs:element name="alertCPointExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
205     </xs:sequence>
206 </xs:complexType>
207 <xs:simpleType name="AngleInDegrees">
208     <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
209 </xs:simpleType>
210 <xs:complexType name="AnimalPresenceObstruction">
211     <xs:complexContent>
212         <xs:extension base="D2LogicalModel:Obstruction">
213             <xs:sequence>
214                 <xs:element name="alive" type="D2LogicalModel:Boolean" minOccurs="0"
↪ maxOccurs="1" />

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215     <xs:element name="animalPresenceType" type=
↪ "D2LogicalModel:AnimalPresenceTypeEnum" minOccurs="1" maxOccurs="1" />
216     <xs:element name="animalPresenceObstructionExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
217     </xs:sequence>
218   </xs:extension>
219 </xs:complexContent>
220 </xs:complexType>
221 <xs:simpleType name="AnimalPresenceTypeEnum">
222   <xs:restriction base="xs:string">
223     <xs:enumeration value="animalsOnTheRoad" />
224     <xs:enumeration value="herdOfAnimalsOnTheRoad" />
225     <xs:enumeration value="largeAnimalsOnTheRoad" />
226   </xs:restriction>
227 </xs:simpleType>
228 <xs:complexType name="ApplicationRateValue">
229   <xs:complexContent>
230     <xs:extension base="D2LogicalModel:DataValue">
231       <xs:sequence>
232         <xs:element name="applicationRate" type=
↪ "D2LogicalModel:IntensityKilogramsPerSquareMetre" minOccurs="1" maxOccurs="1" />
233         <xs:element name="applicationRateValueExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
234       </xs:sequence>
235     </xs:extension>
236   </xs:complexContent>
237 </xs:complexType>
238 <xs:complexType name="Area">
239   <xs:complexContent>
240     <xs:extension base="D2LogicalModel:Location">
241       <xs:sequence>
242         <xs:element name="areaExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
243       </xs:sequence>
244     </xs:extension>
245   </xs:complexContent>
246 </xs:complexType>
247 <xs:complexType name="AreaDestination">
248   <xs:complexContent>
249     <xs:extension base="D2LogicalModel:Destination">
250       <xs:sequence>
251         <xs:element name="area" type="D2LogicalModel:Area" />
252         <xs:element name="areaDestinationExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
253       </xs:sequence>
254     </xs:extension>
255   </xs:complexContent>
256 </xs:complexType>
257 <xs:complexType name="AuthorityOperation">
258   <xs:complexContent>
259     <xs:extension base="D2LogicalModel:Activity">
260       <xs:sequence>
261         <xs:element name="authorityOperationType" type=
↪ "D2LogicalModel:AuthorityOperationTypeEnum" minOccurs="1" maxOccurs="1" />
262         <xs:element name="authorityOperationExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />

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263     </xs:sequence>
264   </xs:extension>
265 </xs:complexContent>
266 </xs:complexType>
267 <xs:simpleType name="AuthorityOperationTypeEnum">
268   <xs:restriction base="xs:string">
269     <xs:enumeration value="accidentInvestigationWork" />
270     <xs:enumeration value="bombSquadInAction" />
271     <xs:enumeration value="civilEmergency" />
272     <xs:enumeration value="customsOperation" />
273     <xs:enumeration value="juridicalReconstruction" />
274     <xs:enumeration value="policeCheckPoint" />
275     <xs:enumeration value="policeInvestigation" />
276     <xs:enumeration value="roadOperatorCheckPoint" />
277     <xs:enumeration value="survey" />
278     <xs:enumeration value="transportOfVip" />
279     <xs:enumeration value="undefinedAuthorityActivity" />
280     <xs:enumeration value="vehicleInspectionCheckPoint" />
281     <xs:enumeration value="vehicleWeighing" />
282     <xs:enumeration value="weighInMotion" />
283     <xs:enumeration value="other" />
284   </xs:restriction>
285 </xs:simpleType>
286 <xs:complexType name="AxleFlowValue">
287   <xs:complexContent>
288     <xs:extension base="D2LogicalModel:DataValue">
289       <xs:sequence>
290         <xs:element name="axleFlowRate" type="D2LogicalModel:AxlesPerHour"
291 ↪ minOccurs="1" maxOccurs="1" />
292         <xs:element name="axleFlowValueExtension" type="D2LogicalModel:_
293 ↪ ExtensionType" minOccurs="0" />
294       </xs:sequence>
295     </xs:extension>
296   </xs:complexContent>
297 </xs:complexType>
298 <xs:complexType name="AxleSpacing">
299   <xs:sequence>
300     <xs:element name="axleSpacing" type="D2LogicalModel:MetresAsFloat" minOccurs="1
301 ↪ " maxOccurs="1" />
302     <xs:element name="axleSpacingSequenceIdentifier" type=
303 ↪ "D2LogicalModel:NonNegativeInteger" minOccurs="1" maxOccurs="1" />
304     <xs:element name="axleSpacingExtension" type="D2LogicalModel:_ExtensionType"
305 ↪ minOccurs="0" />
306   </xs:sequence>
307 </xs:complexType>
308 <xs:simpleType name="AxlesPerHour">
309   <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
310 </xs:simpleType>
311 <xs:complexType name="AxleWeight">
312   <xs:sequence>
313     <xs:element name="axlePositionIdentifier" type=
314 ↪ "D2LogicalModel:NonNegativeInteger" minOccurs="1" maxOccurs="1" />
315     <xs:element name="axleWeight" type="D2LogicalModel:Tonnes" minOccurs="0"
316 ↪ maxOccurs="1" />
317     <xs:element name="maximumPermittedAxleWeight" type="D2LogicalModel:Tonnes"
318 ↪ minOccurs="0" maxOccurs="1" />

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311     <xs:element name="axleWeightExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
312   </xs:sequence>
313 </xs:complexType>
314 <xs:simpleType name="Boolean">
315   <xs:restriction base="xs:boolean" />
316 </xs:simpleType>
317 <xs:simpleType name="CarParkConfigurationEnum">
318   <xs:restriction base="xs:string">
319     <xs:enumeration value="multiStorey" />
320     <xs:enumeration value="parkAndRide" />
321     <xs:enumeration value="singleLevel" />
322     <xs:enumeration value="underground" />
323   </xs:restriction>
324 </xs:simpleType>
325 <xs:complexType name="CarParks">
326   <xs:complexContent>
327     <xs:extension base="D2LogicalModel:NonRoadEventInformation">
328       <xs:sequence>
329         <xs:element name="carParkConfiguration" type=
↪ "D2LogicalModel:CarParkConfigurationEnum" minOccurs="0" maxOccurs="1" />
330         <xs:element name="carParkIdentity" type="D2LogicalModel:String" minOccurs="1
↪ " maxOccurs="1" />
331         <xs:element name="carParkOccupancy" type="D2LogicalModel:Percentage"
↪ minOccurs="0" maxOccurs="1" />
332         <xs:element name="carParkStatus" type="D2LogicalModel:CarParkStatusEnum"
↪ minOccurs="0" maxOccurs="1" />
333         <xs:element name="numberOfVacantParkingSpaces" type=
↪ "D2LogicalModel:NonNegativeInteger" minOccurs="0" maxOccurs="1" />
334         <xs:element name="occupiedSpaces" type="D2LogicalModel:NonNegativeInteger"
↪ minOccurs="0" maxOccurs="1" />
335         <xs:element name="totalCapacity" type="D2LogicalModel:NonNegativeInteger"
↪ minOccurs="0" maxOccurs="1" />
336         <xs:element name="carParksExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
337       </xs:sequence>
338     </xs:extension>
339   </xs:complexContent>
340 </xs:complexType>
341 <xs:simpleType name="CarParkStatusEnum">
342   <xs:restriction base="xs:string">
343     <xs:enumeration value="carParkClosed" />
344     <xs:enumeration value="allCarParksFull" />
345     <xs:enumeration value="carParkFacilityFaulty" />
346     <xs:enumeration value="carParkFull" />
347     <xs:enumeration value="carParkStatusUnknown" />
348     <xs:enumeration value="enoughSpacesAvailable" />
349     <xs:enumeration value="multiStoryCarParksFull" />
350     <xs:enumeration value="noMoreParkingSpacesAvailable" />
351     <xs:enumeration value="noParkAndRideInformation" />
352     <xs:enumeration value="noParkingAllowed" />
353     <xs:enumeration value="noParkingInformationAvailable" />
354     <xs:enumeration value="normalParkingRestrictionsLifted" />
355     <xs:enumeration value="onlyAFewSpacesAvailable" />
356     <xs:enumeration value="parkAndRideServiceNotOperating" />
357     <xs:enumeration value="parkAndRideServiceOperating" />

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358     <xs:enumeration value="specialParkingRestrictionsInForce" />
359   </xs:restriction>
360 </xs:simpleType>
361 <xs:simpleType name="CarriagewayEnum">
362   <xs:restriction base="xs:string">
363     <xs:enumeration value="connectingCarriageway" />
364     <xs:enumeration value="entrySlipRoad" />
365     <xs:enumeration value="exitSlipRoad" />
366     <xs:enumeration value="flyover" />
367     <xs:enumeration value="leftHandFeederRoad" />
368     <xs:enumeration value="leftHandParallelCarriageway" />
369     <xs:enumeration value="mainCarriageway" />
370     <xs:enumeration value="oppositeCarriageway" />
371     <xs:enumeration value="parallelCarriageway" />
372     <xs:enumeration value="rightHandFeederRoad" />
373     <xs:enumeration value="rightHandParallelCarriageway" />
374     <xs:enumeration value="roundabout" />
375     <xs:enumeration value="serviceRoad" />
376     <xs:enumeration value="slipRoads" />
377     <xs:enumeration value="underpass" />
378   </xs:restriction>
379 </xs:simpleType>
380 <xs:complexType name="Cause" abstract="true">
381   <xs:sequence>
382     <xs:element name="causeExtension" type="D2LogicalModel:_ExtensionType"
383     ↪ minOccurs="0" />
384   </xs:sequence>
385 </xs:complexType>
386 <xs:simpleType name="CauseTypeEnum">
387   <xs:restriction base="xs:string">
388     <xs:enumeration value="accident" />
389     <xs:enumeration value="congestion" />
390     <xs:enumeration value="earlierAccident" />
391     <xs:enumeration value="earlierEvent" />
392     <xs:enumeration value="earlierIncident" />
393     <xs:enumeration value="equipmentFailure" />
394     <xs:enumeration value="excessiveHeat" />
395     <xs:enumeration value="frost" />
396     <xs:enumeration value="holidayTraffic" />
397     <xs:enumeration value="infrastructureFailure" />
398     <xs:enumeration value="largeNumbersOfVisitors" />
399     <xs:enumeration value="obstruction" />
400     <xs:enumeration value="pollutionAlert" />
401     <xs:enumeration value="poorWeather" />
402     <xs:enumeration value="problemsAtBorderPost" />
403     <xs:enumeration value="problemsAtCustomPost" />
404     <xs:enumeration value="problemsOnLocalRoads" />
405     <xs:enumeration value="radioactiveLeakAlert" />
406     <xs:enumeration value="roadsideEvent" />
407     <xs:enumeration value="rubberNecking" />
408     <xs:enumeration value="securityIncident" />
409     <xs:enumeration value="shearWeightOfTraffic" />
410     <xs:enumeration value="technicalProblems" />
411     <xs:enumeration value="terrorism" />
412     <xs:enumeration value="toxicCloudAlert" />
413     <xs:enumeration value="vandalism" />

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413     <xs:enumeration value="other" />
414   </xs:restriction>
415 </xs:simpleType>
416 <xs:complexType name="Comment">
417   <xs:sequence>
418     <xs:element name="comment" type="D2LogicalModel:MultilingualString" minOccurs="1
419 ↪ " maxOccurs="1" />
420     <xs:element name="commentDateTime" type="D2LogicalModel:DateTime" minOccurs="0"
421 ↪ maxOccurs="1" />
422     <xs:element name="commentType" type="D2LogicalModel:CommentTypeEnum" minOccurs=
423 ↪ "0" maxOccurs="1" />
424     <xs:element name="commentExtension" type="D2LogicalModel:_ExtensionType"
425 ↪ minOccurs="0" />
426   </xs:sequence>
427 </xs:complexType>
428 <xs:simpleType name="CommentTypeEnum">
429   <xs:restriction base="xs:string">
430     <xs:enumeration value="abnormalLoadMovementNote" />
431     <xs:enumeration value="dataProcessingNote" />
432     <xs:enumeration value="description" />
433     <xs:enumeration value="internalNote" />
434     <xs:enumeration value="locationDescriptor" />
435     <xs:enumeration value="warning" />
436     <xs:enumeration value="other" />
437   </xs:restriction>
438 </xs:simpleType>
439 <xs:simpleType name="ComparisonOperatorEnum">
440   <xs:restriction base="xs:string">
441     <xs:enumeration value="equalTo" />
442     <xs:enumeration value="greaterThan" />
443     <xs:enumeration value="greaterThanOrEqualTo" />
444     <xs:enumeration value="lessThan" />
445     <xs:enumeration value="lessThanOrEqualTo" />
446   </xs:restriction>
447 </xs:simpleType>
448 <xs:simpleType name="ComplianceOptionEnum">
449   <xs:restriction base="xs:string">
450     <xs:enumeration value="advisory" />
451     <xs:enumeration value="mandatory" />
452   </xs:restriction>
453 </xs:simpleType>
454 <xs:simpleType name="ComputationMethodEnum">
455   <xs:restriction base="xs:string">
456     <xs:enumeration value="arithmeticAverageOfSamplesBasedOnAFixedNumberOfSamples" /
457 ↪ >
458     <xs:enumeration value="arithmeticAverageOfSamplesInATimePeriod" />
459     <xs:enumeration value="harmonicAverageOfSamplesInATimePeriod" />
460     <xs:enumeration value="medianOfSamplesInATimePeriod" />
461     <xs:enumeration value="movingAverageOfSamples" />
462   </xs:restriction>
463 </xs:simpleType>
464 <xs:simpleType name="ConcentrationKilogramsPerCubicMetre">
465   <xs:restriction base="D2LogicalModel:Float" />
466 </xs:simpleType>
467 <xs:complexType name="ConcentrationOfVehiclesValue">
468   <xs:complexContent>

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464     <xs:extension base="D2LogicalModel:DataValue">
465         <xs:sequence>
466             <xs:element name="concentrationOfVehicles" type=
467 ↪ "D2LogicalModel:ConcentrationVehiclesPerKilometre" minOccurs="1" maxOccurs="1" />
468             <xs:element name="concentrationOfVehiclesValueExtension" type=
469 ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
470         </xs:sequence>
471     </xs:extension>
472 </xs:complexContent>
473 </xs:complexType>
474 <xs:simpleType name="ConcentrationVehiclesPerKilometre">
475     <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
476 </xs:simpleType>
477 <xs:complexType name="Conditions">
478     <xs:complexContent>
479         <xs:extension base="D2LogicalModel:TrafficElement">
480             <xs:sequence>
481                 <xs:element name="drivingConditionType" type=
482 ↪ "D2LogicalModel:DrivingConditionTypeEnum" minOccurs="0" maxOccurs="1" />
483                 <xs:element name="conditionsExtension" type="D2LogicalModel:_ExtensionType"
484 ↪ minOccurs="0" />
485             </xs:sequence>
486         </xs:extension>
487     </xs:complexContent>
488 </xs:complexType>
489 <xs:simpleType name="ConfidentialityValueEnum">
490     <xs:restriction base="xs:string">
491         <xs:enumeration value="internalUse" />
492         <xs:enumeration value="noRestriction" />
493         <xs:enumeration value="restrictedToAuthorities" />
494         <xs:enumeration value="restrictedToAuthoritiesAndTrafficOperators" />
495         <xs:enumeration value="restrictedToAuthoritiesTrafficOperatorsAndPublishers" />
496         <xs:enumeration value="restrictedToAuthoritiesTrafficOperatorsAndVms" />
497     </xs:restriction>
498 </xs:simpleType>
499 <xs:complexType name="ConstructionWorks">
500     <xs:complexContent>
501         <xs:extension base="D2LogicalModel:Roadworks">
502             <xs:sequence>
503                 <xs:element name="constructionWorkType" type=
504 ↪ "D2LogicalModel:ConstructionWorkTypeEnum" minOccurs="0" maxOccurs="1" />
505                 <xs:element name="constructionWorksExtension" type="D2LogicalModel:_
506 ↪ ExtensionType" minOccurs="0" />
507             </xs:sequence>
508         </xs:extension>
509     </xs:complexContent>
510 </xs:complexType>
511 <xs:simpleType name="ConstructionWorkTypeEnum">
512     <xs:restriction base="xs:string">
513         <xs:enumeration value="blastingWork" />
514         <xs:enumeration value="constructionWork" />
515         <xs:enumeration value="demolitionWork" />
516         <xs:enumeration value="roadImprovementOrUpgrading" />
517         <xs:enumeration value="roadWideningWork" />
518     </xs:restriction>
519 </xs:simpleType>

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514 <xs:simpleType name="CountryEnum">
515   <xs:restriction base="xs:string">
516     <xs:enumeration value="at" />
517     <xs:enumeration value="be" />
518     <xs:enumeration value="bg" />
519     <xs:enumeration value="ch" />
520     <xs:enumeration value="cs" />
521     <xs:enumeration value="cy" />
522     <xs:enumeration value="cz" />
523     <xs:enumeration value="de" />
524     <xs:enumeration value="dk" />
525     <xs:enumeration value="ee" />
526     <xs:enumeration value="es" />
527     <xs:enumeration value="fi" />
528     <xs:enumeration value="fo" />
529     <xs:enumeration value="fr" />
530     <xs:enumeration value="gb" />
531     <xs:enumeration value="gg" />
532     <xs:enumeration value="gi" />
533     <xs:enumeration value="gr" />
534     <xs:enumeration value="hr" />
535     <xs:enumeration value="hu" />
536     <xs:enumeration value="ie" />
537     <xs:enumeration value="im" />
538     <xs:enumeration value="is" />
539     <xs:enumeration value="it" />
540     <xs:enumeration value="je" />
541     <xs:enumeration value="li" />
542     <xs:enumeration value="lt" />
543     <xs:enumeration value="lu" />
544     <xs:enumeration value="lv" />
545     <xs:enumeration value="ma" />
546     <xs:enumeration value="mc" />
547     <xs:enumeration value="mk" />
548     <xs:enumeration value="mt" />
549     <xs:enumeration value="nl" />
550     <xs:enumeration value="no" />
551     <xs:enumeration value="pl" />
552     <xs:enumeration value="pt" />
553     <xs:enumeration value="ro" />
554     <xs:enumeration value="se" />
555     <xs:enumeration value="si" />
556     <xs:enumeration value="sk" />
557     <xs:enumeration value="sm" />
558     <xs:enumeration value="tr" />
559     <xs:enumeration value="va" />
560     <xs:enumeration value="other" />
561   </xs:restriction>
562 </xs:simpleType>
563 <xs:simpleType name="CubicMetres">
564   <xs:restriction base="D2LogicalModel:Float" />
565 </xs:simpleType>
566 <xs:element name="d2LogicalModel" type="D2LogicalModel:D2LogicalModel">
567   <xs:unique name="_d2LogicalModelSituationRecordConstraint">
568     <xs:selector xpath="."/><xs:field xpath="@id" />
569   </xs:unique>

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570     <xs:field xpath="@version" />
571 </xs:unique>
572 <xs:unique name="_d2LogicalModelSituationConstraint">
573   <xs:selector xpath="//D2LogicalModel:situation" />
574   <xs:field xpath="@id" />
575   <xs:field xpath="@version" />
576 </xs:unique>
577 </xs:element>
578 <xs:complexType name="D2LogicalModel">
579   <xs:sequence>
580     <xs:element name="exchange" type="D2LogicalModel:Exchange" />
581     <xs:element name="payloadPublication" type="D2LogicalModel:PayloadPublication"
582     ↪ minOccurs="0" />
583     <xs:element name="d2LogicalModelExtension" type="D2LogicalModel:_ExtensionType"
584     ↪ minOccurs="0" />
585   </xs:sequence>
586   <xs:attribute name="modelBaseVersion" use="required" fixed="2" />
587 </xs:complexType>
588 <xs:simpleType name="DangerousGoodsRegulationsEnum">
589   <xs:restriction base="xs:string">
590     <xs:enumeration value="adr" />
591     <xs:enumeration value="iataIcao" />
592     <xs:enumeration value="imoImdg" />
593     <xs:enumeration value="railroadDangerousGoodsBook" />
594   </xs:restriction>
595 </xs:simpleType>
596 <xs:complexType name="DataValue" abstract="true">
597   <xs:sequence>
598     <xs:element name="dataError" type="D2LogicalModel:Boolean" minOccurs="0"
599     ↪ maxOccurs="1" />
600     <xs:element name="reasonForDataError" type="D2LogicalModel:MultilingualString"
601     ↪ minOccurs="0" maxOccurs="1" />
602     <xs:element name="dataValueExtension" type="D2LogicalModel:_ExtensionType"
603     ↪ minOccurs="0" />
604   </xs:sequence>
605   <xs:attribute name="accuracy" type="D2LogicalModel:Percentage" use="optional" />
606   <xs:attribute name="computationalMethod" type=
607   ↪ "D2LogicalModel:ComputationMethodEnum" use="optional" />
608   <xs:attribute name="numberOfIncompleteInputs" type=
609   ↪ "D2LogicalModel:NonNegativeInteger" use="optional" />
610   <xs:attribute name="numberOfInputValuesUsed" type=
611   ↪ "D2LogicalModel:NonNegativeInteger" use="optional" />
612   <xs:attribute name="smoothingFactor" type="D2LogicalModel:Float" use="optional" />
613   <xs:attribute name="standardDeviation" type="D2LogicalModel:Float" use="optional"
614   ↪ />
615   <xs:attribute name="supplierCalculatedDataQuality" type="D2LogicalModel:Percentage
616   ↪ " use="optional" />
617 </xs:complexType>
618 <xs:simpleType name="DateTime">
619   <xs:restriction base="xs:dateTime" />
620 </xs:simpleType>
621 <xs:complexType name="DateTimeValue">
622   <xs:complexContent>
623     <xs:extension base="D2LogicalModel:DataValue">
624       <xs:sequence>
625         <xs:element name="dateTime" type="D2LogicalModel:DateTime" minOccurs="1"

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616     <xs:element name="dateTimeValueExtension" type="D2LogicalModel:_
617     </xs:sequence>
618     </xs:extension>
619     </xs:complexContent>
620   </xs:complexType>
621   <xs:complexType name="Delays">
622     <xs:sequence>
623       <xs:element name="delaysType" type="D2LogicalModel:DelaysTypeEnum" minOccurs="0"
624       <xs:element name="delayTimeValue" type="D2LogicalModel:Seconds" minOccurs="0"
625       <xs:element name="delaysExtension" type="D2LogicalModel:_ExtensionType"
626     </xs:sequence>
627   </xs:complexType>
628   <xs:simpleType name="DelaysTypeEnum">
629     <xs:restriction base="xs:string">
630       <xs:enumeration value="delays" />
631       <xs:enumeration value="delaysOfUncertainDuration" />
632       <xs:enumeration value="longDelays" />
633       <xs:enumeration value="veryLongDelays" />
634     </xs:restriction>
635   </xs:simpleType>
636   <xs:complexType name="Destination" abstract="true">
637     <xs:sequence>
638       <xs:element name="destinationExtension" type="D2LogicalModel:_ExtensionType"
639     </xs:sequence>
640   </xs:complexType>
641   <xs:complexType name="DirectionBearingValue">
642     <xs:complexContent>
643       <xs:extension base="D2LogicalModel:DataValue">
644         <xs:sequence>
645           <xs:element name="directionBearing" type="D2LogicalModel:AngleInDegrees"
646           <xs:element name="directionBearingValueExtension" type="D2LogicalModel:_
647         </xs:sequence>
648       </xs:extension>
649     </xs:complexContent>
650   </xs:complexType>
651   <xs:simpleType name="DirectionCompassEnum">
652     <xs:restriction base="xs:string">
653       <xs:enumeration value="east" />
654       <xs:enumeration value="eastNorthEast" />
655       <xs:enumeration value="eastSouthEast" />
656       <xs:enumeration value="north" />
657       <xs:enumeration value="northEast" />
658       <xs:enumeration value="northNorthEast" />
659       <xs:enumeration value="northNorthWest" />
660       <xs:enumeration value="northWest" />
661       <xs:enumeration value="south" />
662       <xs:enumeration value="southEast" />
663       <xs:enumeration value="southSouthEast" />

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664     <xs:enumeration value="southSouthWest" />
665     <xs:enumeration value="southWest" />
666     <xs:enumeration value="west" />
667     <xs:enumeration value="westNorthWest" />
668     <xs:enumeration value="westSouthWest" />
669   </xs:restriction>
670 </xs:simpleType>
671 <xs:complexType name="DirectionCompassValue">
672   <xs:complexContent>
673     <xs:extension base="D2LogicalModel:DataValue">
674       <xs:sequence>
675         <xs:element name="directionCompass" type=
676 ↪ "D2LogicalModel:DirectionCompassEnum" minOccurs="1" maxOccurs="1" />
677         <xs:element name="directionCompassValueExtension" type="D2LogicalModel:_
678 ↪ ExtensionType" minOccurs="0" />
679       </xs:sequence>
680     </xs:extension>
681   </xs:complexContent>
682 </xs:complexType>
683 <xs:simpleType name="DirectionEnum">
684   <xs:restriction base="xs:string">
685     <xs:enumeration value="allDirections" />
686     <xs:enumeration value="bothWays" />
687     <xs:enumeration value="clockwise" />
688     <xs:enumeration value="anticlockwise" />
689     <xs:enumeration value="innerRing" />
690     <xs:enumeration value="outerRing" />
691     <xs:enumeration value="northBound" />
692     <xs:enumeration value="northEastBound" />
693     <xs:enumeration value="eastBound" />
694     <xs:enumeration value="southEastBound" />
695     <xs:enumeration value="southBound" />
696     <xs:enumeration value="southWestBound" />
697     <xs:enumeration value="westBound" />
698     <xs:enumeration value="northWestBound" />
699     <xs:enumeration value="inboundTowardsTown" />
700     <xs:enumeration value="outboundFromTown" />
701     <xs:enumeration value="unknown" />
702     <xs:enumeration value="opposite" />
703     <xs:enumeration value="other" />
704   </xs:restriction>
705 </xs:simpleType>
706 <xs:complexType name="DistanceAlongLinearElement" abstract="true">
707   <xs:sequence>
708     <xs:element name="distanceAlongLinearElementExtension" type="D2LogicalModel:_
709 ↪ ExtensionType" minOccurs="0" />
710   </xs:sequence>
711 </xs:complexType>
712 <xs:complexType name="DistanceFromLinearElementReferent">
713   <xs:complexContent>
714     <xs:extension base="D2LogicalModel:DistanceAlongLinearElement">
715       <xs:sequence>
716         <xs:element name="distanceAlong" type="D2LogicalModel:MetresAsFloat" ↪
717 ↪ minOccurs="1" maxOccurs="1" />
718         <xs:element name="fromReferent" type="D2LogicalModel:Referent" />
719         <xs:element name="towardsReferent" type="D2LogicalModel:Referent" minOccurs=

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716   <xs:element name="distanceFromLinearElementReferentExtension" type=
717   <xs:sequence>
718   </xs:extension>
719   </xs:complexContent>
720 </xs:complexType>
721 <xs:complexType name="DistanceFromLinearElementStart">
722   <xs:complexContent>
723     <xs:extension base="D2LogicalModel:DistanceAlongLinearElement">
724       <xs:sequence>
725         <xs:element name="distanceAlong" type="D2LogicalModel:MetresAsFloat"
726         <xs:element name="distanceFromLinearElementStartExtension" type=
727         </xs:sequence>
728         </xs:extension>
729         </xs:complexContent>
730 </xs:complexType>
731 <xs:complexType name="DisturbanceActivity">
732   <xs:complexContent>
733     <xs:extension base="D2LogicalModel:Activity">
734       <xs:sequence>
735         <xs:element name="disturbanceActivityType" type=
736         <xs:element name="disturbanceActivityExtension" type="D2LogicalModel:_
737         </xs:sequence>
738         </xs:extension>
739         </xs:complexContent>
740 </xs:complexType>
741 <xs:simpleType name="DisturbanceActivityTypeEnum">
742   <xs:restriction base="xs:string">
743     <xs:enumeration value="airRaid" />
744     <xs:enumeration value="altercationOfVehicleOccupants" />
745     <xs:enumeration value="assault" />
746     <xs:enumeration value="assetDestruction" />
747     <xs:enumeration value="attack" />
748     <xs:enumeration value="attackOnVehicle" />
749     <xs:enumeration value="blockadeOrBarrier" />
750     <xs:enumeration value="bombAlert" />
751     <xs:enumeration value="crowd" />
752     <xs:enumeration value="demonstration" />
753     <xs:enumeration value="evacuation" />
754     <xs:enumeration value="filterBlockade" />
755     <xs:enumeration value="goSlowOperation" />
756     <xs:enumeration value="gunfireOnRoadway" />
757     <xs:enumeration value="illVehicleOccupants" />
758     <xs:enumeration value="march" />
759     <xs:enumeration value="publicDisturbance" />
760     <xs:enumeration value="radioactiveLeakAlert" />
761     <xs:enumeration value="riot" />
762     <xs:enumeration value="sabotage" />
763     <xs:enumeration value="securityAlert" />
764     <xs:enumeration value="securityIncident" />
765     <xs:enumeration value="sightseersObstructingAccess" />

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766     <xs:enumeration value="strike" />
767     <xs:enumeration value="terroristIncident" />
768     <xs:enumeration value="theft" />
769     <xs:enumeration value="toxicCloudAlert" />
770     <xs:enumeration value="unspecifiedAlert" />
771     <xs:enumeration value="other" />
772 </xs:restriction>
773 </xs:simpleType>
774 <xs:simpleType name="DrivingConditionTypeEnum">
775     <xs:restriction base="xs:string">
776         <xs:enumeration value="impossible" />
777         <xs:enumeration value="hazardous" />
778         <xs:enumeration value="normal" />
779         <xs:enumeration value="passableWithCare" />
780         <xs:enumeration value="unknown" />
781         <xs:enumeration value="veryHazardous" />
782         <xs:enumeration value="winterConditions" />
783         <xs:enumeration value="other" />
784     </xs:restriction>
785 </xs:simpleType>
786 <xs:complexType name="ElaboratedDataFault">
787     <xs:complexContent>
788         <xs:extension base="D2LogicalModel:Fault">
789             <xs:sequence>
790                 <xs:element name="elaboratedDataFault" type=
791 ↪ "D2LogicalModel:ElaboratedDataFaultEnum" minOccurs="1" maxOccurs="1" />
792                 <xs:element name="elaboratedDataFaultExtension" type="D2LogicalModel:_
793 ↪ ExtensionType" minOccurs="0" />
794             </xs:sequence>
795         </xs:extension>
796     </xs:complexContent>
797 </xs:complexType>
798 <xs:simpleType name="ElaboratedDataFaultEnum">
799     <xs:restriction base="xs:string">
800         <xs:enumeration value="intermittentDataValues" />
801         <xs:enumeration value="noDataValuesAvailable" />
802         <xs:enumeration value="spuriousUnreliableDataValues" />
803         <xs:enumeration value="unspecifiedOrUnknownFault" />
804         <xs:enumeration value="other" />
805     </xs:restriction>
806 </xs:simpleType>
807 <xs:complexType name="EnvironmentalObstruction">
808     <xs:complexContent>
809         <xs:extension base="D2LogicalModel:Obstruction">
810             <xs:sequence>
811                 <xs:element name="depth" type="D2LogicalModel:MetresAsFloat" minOccurs="0"
812 ↪ maxOccurs="1" />
813                 <xs:element name="environmentalObstructionType" type=
814 ↪ "D2LogicalModel:EnvironmentalObstructionTypeEnum" minOccurs="1" maxOccurs="1" />
815                 <xs:element name="environmentalObstructionExtension" type="D2LogicalModel:_
816 ↪ ExtensionType" minOccurs="0" />
817             </xs:sequence>
818         </xs:extension>
819     </xs:complexContent>
820 </xs:complexType>
821 <xs:simpleType name="EnvironmentalObstructionTypeEnum">

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817 <xs:restriction base="xs:string">
818   <xs:enumeration value="avalanches" />
819   <xs:enumeration value="earthquakeDamage" />
820   <xs:enumeration value="fallenTrees" />
821   <xs:enumeration value="fallingIce" />
822   <xs:enumeration value="fallingLightIceOrSnow" />
823   <xs:enumeration value="flashFloods" />
824   <xs:enumeration value="flooding" />
825   <xs:enumeration value="forestFire" />
826   <xs:enumeration value="grassFire" />
827   <xs:enumeration value="landslips" />
828   <xs:enumeration value="mudSlide" />
829   <xs:enumeration value="sewerOverflow" />
830   <xs:enumeration value="rockfalls" />
831   <xs:enumeration value="seriousFire" />
832   <xs:enumeration value="smokeOrFumes" />
833   <xs:enumeration value="stormDamage" />
834   <xs:enumeration value="subsidence" />
835   <xs:enumeration value="other" />
836 </xs:restriction>
837 </xs:simpleType>
838 <xs:complexType name="EquipmentOrSystemFault">
839   <xs:complexContent>
840     <xs:extension base="D2LogicalModel:TrafficElement">
841       <xs:sequence>
842         <xs:element name="equipmentOrSystemFaultType" type=
843 ↪ "D2LogicalModel:EquipmentOrSystemFaultTypeEnum" minOccurs="1" maxOccurs="1" />
844         <xs:element name="faultyEquipmentOrSystemType" type=
845 ↪ "D2LogicalModel:EquipmentOrSystemTypeEnum" minOccurs="1" maxOccurs="1" />
846         <xs:element name="equipmentOrSystemFaultExtension" type="D2LogicalModel:_
847 ↪ ExtensionType" minOccurs="0" />
848       </xs:sequence>
849     </xs:extension>
850   </xs:complexContent>
851 </xs:complexType>
852 <xs:simpleType name="EquipmentOrSystemFaultTypeEnum">
853   <xs:restriction base="xs:string">
854     <xs:enumeration value="notWorking" />
855     <xs:enumeration value="outOfService" />
856     <xs:enumeration value="workingIncorrectly" />
857     <xs:enumeration value="workingIntermittently" />
858   </xs:restriction>
859 </xs:simpleType>
860 <xs:simpleType name="EquipmentOrSystemTypeEnum">
861   <xs:restriction base="xs:string">
862     <xs:enumeration value="anprCameras" />
863     <xs:enumeration value="automatedTollSystem" />
864     <xs:enumeration value="cctvCameras" />
865     <xs:enumeration value="emergencyRoadsideTelephones" />
866     <xs:enumeration value="galleryLights" />
867     <xs:enumeration value="laneControlSigns" />
868     <xs:enumeration value="levelCrossing" />
869     <xs:enumeration value="matrixSigns" />
870     <xs:enumeration value="rampControls" />
871     <xs:enumeration value="roadsideCommunicationsSystem" />
872     <xs:enumeration value="roadsidePowerSystem" />

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870     <xs:enumeration value="speedControlSigns" />
871     <xs:enumeration value="streetLighting" />
872     <xs:enumeration value="temporaryTrafficLights" />
873     <xs:enumeration value="tollGates" />
874     <xs:enumeration value="trafficLightSets" />
875     <xs:enumeration value="trafficSignals" />
876     <xs:enumeration value="tunnelLights" />
877     <xs:enumeration value="tunnelVentilation" />
878     <xs:enumeration value="variableMessageSigns" />
879     <xs:enumeration value="other" />
880 </xs:restriction>
881 </xs:simpleType>
882 <xs:complexType name="Exchange">
883   <xs:sequence>
884     <xs:element name="supplierIdentification" type=
885 ↪ "D2LogicalModel:InternationalIdentifier" />
886     <xs:element name="exchangeExtension" type="D2LogicalModel:_ExtensionType"
887 ↪ minOccurs="0" />
888   </xs:sequence>
889 </xs:complexType>
890 <xs:complexType name="Fault">
891   <xs:sequence>
892     <xs:element name="faultIdentifier" type="D2LogicalModel:String" minOccurs="0"
893 ↪ maxOccurs="1" />
894     <xs:element name="faultDescription" type="D2LogicalModel:String" minOccurs="0"
895 ↪ maxOccurs="1" />
896     <xs:element name="faultCreationTime" type="D2LogicalModel:DateTime" minOccurs="0"
897 ↪ maxOccurs="1" />
898     <xs:element name="faultLastUpdateTime" type="D2LogicalModel:DateTime" minOccurs=
899 ↪ "1" maxOccurs="1" />
900     <xs:element name="faultSeverity" type="D2LogicalModel:FaultSeverityEnum"
901 ↪ minOccurs="0" maxOccurs="1" />
902     <xs:element name="faultExtension" type="D2LogicalModel:_ExtensionType"
903 ↪ minOccurs="0" />
904   </xs:sequence>
905 </xs:complexType>
906 <xs:simpleType name="FaultSeverityEnum">
907   <xs:restriction base="xs:string">
908     <xs:enumeration value="low" />
909     <xs:enumeration value="medium" />
910     <xs:enumeration value="high" />
911     <xs:enumeration value="unknown" />
912   </xs:restriction>
913 </xs:simpleType>
914 <xs:simpleType name="Float">
915   <xs:restriction base="xs:float" />
916 </xs:simpleType>
917 <xs:complexType name="FloatingPointMetreDistanceValue">
918   <xs:complexContent>
919     <xs:extension base="D2LogicalModel:DataValue">
920       <xs:sequence>
921         <xs:element name="floatingPointMetreDistance" type=
922 ↪ "D2LogicalModel:MetresAsFloat" minOccurs="1" maxOccurs="1" />
923         <xs:element name="floatingPointMetreDistanceValueExtension" type=
924 ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
925       </xs:sequence>

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916     </xs:extension>
917   </xs:complexContent>
918 </xs:complexType>
919 <xs:complexType name="GeneralInstructionOrMessageToRoadUsers">
920   <xs:complexContent>
921     <xs:extension base="D2LogicalModel:NetworkManagement">
922       <xs:sequence>
923         <xs:element name="generalInstructionToRoadUsersType" type=
924 ↪ "D2LogicalModel:GeneralInstructionToRoadUsersTypeEnum" minOccurs="0" maxOccurs="1" /
925 ↪ >
926         <xs:element name="generalMessageToRoadUsers" type=
927 ↪ "D2LogicalModel:MultilingualString" minOccurs="0" maxOccurs="1" />
928         <xs:element name="generalInstructionOrMessageToRoadUsersExtension" type=
929 ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
930       </xs:sequence>
931     </xs:extension>
932   </xs:complexContent>
933 </xs:complexType>
934 <xs:simpleType name="GeneralInstructionToRoadUsersTypeEnum">
935   <xs:restriction base="xs:string">
936     <xs:enumeration value="allowEmergencyVehiclesToPass" />
937     <xs:enumeration value="approachWithCare" />
938     <xs:enumeration value="avoidTheArea" />
939     <xs:enumeration value="closeAllWindowsTurnOffHeaterAndVents" />
940     <xs:enumeration value="crossJunctionWithCare" />
941     <xs:enumeration value="doNotAllowUnnecessaryGaps" />
942     <xs:enumeration value="doNotLeaveYourVehicle" />
943     <xs:enumeration value="doNotThrowOutAnyBurningObjects" />
944     <xs:enumeration value="doNotUseNavigationSystems" />
945     <xs:enumeration value="driveCarefully" />
946     <xs:enumeration value="driveWithExtremeCaution" />
947     <xs:enumeration value="flashYourLights" />
948     <xs:enumeration value="followTheVehicleInFrontSmoothly" />
949     <xs:enumeration value="increaseNormalFollowingDistance" />
950     <xs:enumeration value="inEmergencyWaitForPatrolService" />
951     <xs:enumeration value="keepYourDistance" />
952     <xs:enumeration value="leaveYourVehicleProceedToNextSafePlace" />
953     <xs:enumeration value="noNakedFlames" />
954     <xs:enumeration value="noOvertaking" />
955     <xs:enumeration value="noSmoking" />
956     <xs:enumeration value="noStopping" />
957     <xs:enumeration value="noUturns" />
958     <xs:enumeration value="observeAmberAlert" />
959     <xs:enumeration value="observeSignals" />
960     <xs:enumeration value="observeSigns" />
961     <xs:enumeration value="onlyTravelIfAbsolutelyNecessary" />
962     <xs:enumeration value="overtakeWithCare" />
963     <xs:enumeration value="pullOverToTheEdgeOfTheRoadway" />
964     <xs:enumeration value="stopAtNextSafePlace" />
965     <xs:enumeration value="stopAtNextServiceArea" />
966     <xs:enumeration value="switchOffEngine" />
967     <xs:enumeration value="switchOffMobilePhonesAndTwoWayRadios" />
968     <xs:enumeration value="testYourBrakes" />
969     <xs:enumeration value="useBusService" />
970     <xs:enumeration value="useFogLights" />
971     <xs:enumeration value="useHazardWarningLights" />

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968     <xs:enumeration value="useHeadlights" />
969     <xs:enumeration value="useRailService" />
970     <xs:enumeration value="useTramService" />
971     <xs:enumeration value="useUndergroundService" />
972     <xs:enumeration value="waitForEscortVehicle" />
973     <xs:enumeration value="other" />
974   </xs:restriction>
975 </xs:simpleType>
976 <xs:complexType name="GeneralNetworkManagement">
977   <xs:complexContent>
978     <xs:extension base="D2LogicalModel:NetworkManagement">
979       <xs:sequence>
980         <xs:element name="generalNetworkManagementType" type=
981 ↪ "D2LogicalModel:GeneralNetworkManagementTypeEnum" minOccurs="1" maxOccurs="1" />
982         <xs:element name="trafficManuallyDirectedBy" type=
983 ↪ "D2LogicalModel:PersonCategoryEnum" minOccurs="0" maxOccurs="1" />
984         <xs:element name="generalNetworkManagementExtension" type="D2LogicalModel:_
985 ↪ ExtensionType" minOccurs="0" />
986       </xs:sequence>
987     </xs:extension>
988   </xs:complexContent>
989 </xs:complexType>
990 <xs:simpleType name="GeneralNetworkManagementTypeEnum">
991   <xs:restriction base="xs:string">
992     <xs:enumeration value="bridgeSwingInOperation" />
993     <xs:enumeration value="convoyService" />
994     <xs:enumeration value="obstacleSignalling" />
995     <xs:enumeration value="rampMeteringInOperation" />
996     <xs:enumeration value="temporaryTrafficLights" />
997     <xs:enumeration value="tollGatesOpen" />
998     <xs:enumeration value="trafficBeingManuallyDirected" />
999     <xs:enumeration value="trafficHeld" />
1000     <xs:enumeration value="other" />
1001   </xs:restriction>
1002 </xs:simpleType>
1003 <xs:complexType name="GeneralObstruction">

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1004   <xs:complexContent>
1005     <xs:extension base="D2LogicalModel:Obstruction">
1006       <xs:sequence>
1007         <xs:element name="obstructionType" type="D2LogicalModel:ObstructionTypeEnum
1008 ↪ " minOccurs="1" maxOccurs="unbounded" />
1009         <xs:element name="groupOfPeopleInvolved" type=
1010 ↪ "D2LogicalModel:GroupOfPeopleInvolved" minOccurs="0" maxOccurs="unbounded" />
1011         <xs:element name="generalObstructionExtension" type="D2LogicalModel:_
1012 ↪ ExtensionType" minOccurs="0" />
1013       </xs:sequence>
1014     </xs:extension>
1015   </xs:complexContent>
1016 </xs:complexType>
1017 <xs:simpleType name="GeometryTypeEnum">
1018   <xs:restriction base="xs:string" />
1019 </xs:simpleType>
1020 <xs:complexType name="GlobalNetworkLinear">
1021   <xs:sequence>
1022     <xs:element name="networkVersion" type="D2LogicalModel:MultilingualString"

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1017   <minOccurs="1" maxOccurs="1" />
      <xs:element name="linearGeometryType" type="D2LogicalModel:GeometryTypeEnum"
1018   <minOccurs="1" maxOccurs="1" />
      <xs:element name="startPoint" type="D2LogicalModel:PointByMultiCoordinates"
1019   <minOccurs="0" />
      <xs:element name="endPoint" type="D2LogicalModel:PointByMultiCoordinates"
1020   <minOccurs="0" />
      <xs:element name="linearWithinLinearGNElement" type=
1021   "D2LogicalModel:LinearWithinLinearGNElement" maxOccurs="unbounded" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="GrossWeightCharacteristic">
1023    <xs:sequence>
1024      <xs:element name="comparisonOperator" type=
1025      "D2LogicalModel:ComparisonOperatorEnum" minOccurs="1" maxOccurs="1" />
1026      <xs:element name="grossVehicleWeight" type="D2LogicalModel:Tonnes" minOccurs="1"
1027      " maxOccurs="1" />
1028      <xs:element name="grossWeightCharacteristicExtension" type="D2LogicalModel:_
1029      ExtensionType" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="GroupOfLocations" abstract="true">
1031    <xs:sequence>
1032      <xs:element name="groupOfLocationsExtension" type="D2LogicalModel:_ExtensionType
1033      " minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="GroupOfPeopleInvolved">
1036    <xs:sequence>
1037      <xs:element name="numberOfPeople" type="D2LogicalModel:NonNegativeInteger"
1038      <minOccurs="0" maxOccurs="1" />
1039      <xs:element name="injuryStatus" type="D2LogicalModel:InjuryStatusTypeEnum"
1040      <minOccurs="0" maxOccurs="1" />
1041      <xs:element name="involvementRole" type="D2LogicalModel:InvolvementRolesEnum"
1042      <minOccurs="0" maxOccurs="1" />
1043      <xs:element name="categoryOfPeopleInvolved" type=
1044      "D2LogicalModel:PersonCategoryEnum" minOccurs="0" maxOccurs="1" />
1045      <xs:element name="groupOfPeopleInvolvedExtension" type="D2LogicalModel:_
1046      ExtensionType" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="GroupOfVehiclesInvolved">
1048    <xs:sequence>
1049      <xs:element name="numberOfVehicles" type="D2LogicalModel:NonNegativeInteger"
1050      <minOccurs="0" maxOccurs="1" />
1051      <xs:element name="vehicleStatus" type="D2LogicalModel:VehicleStatusEnum"
1052      <minOccurs="0" maxOccurs="1" />
1053      <xs:element name="vehicleCharacteristics" type=
1054      "D2LogicalModel:VehicleCharacteristics" minOccurs="0" />
1055      <xs:element name="groupOfVehiclesInvolvedExtension" type="D2LogicalModel:_
1056      ExtensionType" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="HazardousMaterials">
    <xs:sequence>
      <xs:element name="chemicalName" type="D2LogicalModel:MultilingualString"

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1055   <minOccurs="1" maxOccurs="1" />
1056   <xs:element name="dangerousGoodsFlashPoint" type=
1057   <xs:element name="dangerousGoodsRegulations" type=
1058   <xs:element name="hazardCodeIdentification" type="D2LogicalModel:String"
1059   <minOccurs="0" maxOccurs="1" />
1060   <xs:element name="hazardCodeVersionNumber" type=
1061   <xs:element name="hazardSubstanceItemPageNumber" type="D2LogicalModel:String"
1062   <minOccurs="0" maxOccurs="1" />
1063   <xs:element name="tremCardNumber" type="D2LogicalModel:String" minOccurs="0"
1064   <maxOccurs="1" />
1065   <xs:element name="undgNumber" type="D2LogicalModel:String" minOccurs="0"
1066   <maxOccurs="1" />
1067   <xs:element name="volumeOfDangerousGoods" type="D2LogicalModel:CubicMetres"
1068   <minOccurs="0" maxOccurs="1" />
1069   <xs:element name="weightOfDangerousGoods" type="D2LogicalModel:Tonnes"
1070   <minOccurs="0" maxOccurs="1" />
1071   <xs:element name="hazardousMaterialsExtension" type="D2LogicalModel:_
1072   <ExtensionType" minOccurs="0" />
1073   </xs:sequence>
1074   </xs:complexType>
1075   <xs:complexType name="HeaderInformation">
1076   <xs:sequence>
1077   <xs:element name="confidentiality" type="D2LogicalModel:ConfidentialityValueEnum
1078   <minOccurs="1" maxOccurs="1" />
1079   <xs:element name="informationStatus" type="D2LogicalModel:InformationStatusEnum
1080   <minOccurs="1" maxOccurs="1" />
1081   <xs:element name="urgency" type="D2LogicalModel:UrgencyEnum" minOccurs="0"
1082   <maxOccurs="1" />
1083   <xs:element name="headerInformationExtension" type="D2LogicalModel:_
1084   <ExtensionType" minOccurs="0" />
1085   </xs:sequence>
1086   </xs:complexType>
1087   <xs:complexType name="HeaviestAxleWeightCharacteristic">
1088   <xs:sequence>
1089   <xs:element name="comparisonOperator" type=
1090   <xs:element name="heaviestAxleWeight" type="D2LogicalModel:Tonnes" minOccurs="1
1091   <maxOccurs="1" />
1092   <xs:element name="heaviestAxleWeightCharacteristicExtension" type=
1093   <xs:element name="heightCharacteristicExtension" type="D2LogicalModel:_
1094   <ExtensionType" minOccurs="0" />
1095   </xs:sequence>
1096   </xs:complexType>
1097   <xs:simpleType name="HeightGradeEnum">

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1090     <xs:restriction base="xs:string">
1091       <xs:enumeration value="aboveGrade" />
1092       <xs:enumeration value="atGrade" />
1093       <xs:enumeration value="belowGrade" />
1094     </xs:restriction>
1095   </xs:simpleType>
1096   <xs:complexType name="Impact">
1097     <xs:sequence>
1098       <xs:element name="capacityRemaining" type="D2LogicalModel:Percentage" minOccurs=
1099 ↪ "0" maxOccurs="1" />
1100       <xs:element name="numberOfLanesRestricted" type=
1101 ↪ "D2LogicalModel:NonNegativeInteger" minOccurs="0" maxOccurs="1" />
1102       <xs:element name="numberOfOperationalLanes" type=
1103 ↪ "D2LogicalModel:NonNegativeInteger" minOccurs="0" maxOccurs="1" />
1104       <xs:element name="originalNumberOfLanes" type="D2LogicalModel:NonNegativeInteger
1105 ↪ " minOccurs="0" maxOccurs="1" />
1106       <xs:element name="trafficConstrictionType" type=
1107 ↪ "D2LogicalModel:TrafficConstrictionTypeEnum" minOccurs="0" maxOccurs="1" />
1108       <xs:element name="delays" type="D2LogicalModel:Delays" minOccurs="0" />
1109       <xs:element name="impactExtension" type="D2LogicalModel:_ExtensionType"
1110 ↪ minOccurs="0" />
1111     </xs:sequence>
1112   </xs:complexType>
1113   <xs:simpleType name="InformationStatusEnum">
1114     <xs:restriction base="xs:string">
1115       <xs:enumeration value="real" />
1116       <xs:enumeration value="securityExercise" />
1117       <xs:enumeration value="technicalExercise" />
1118       <xs:enumeration value="test" />
1119     </xs:restriction>
1120   </xs:simpleType>
1121   <xs:complexType name="InfrastructureDamageObstruction">
1122     <xs:complexContent>
1123       <xs:extension base="D2LogicalModel:Obstruction">
1124         <xs:sequence>
1125           <xs:element name="infrastructureDamageType" type=
1126 ↪ "D2LogicalModel:InfrastructureDamageTypeEnum" minOccurs="1" maxOccurs="1" />
1127           <xs:element name="infrastructureDamageObstructionExtension" type=
1128 ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
1129         </xs:sequence>
1130       </xs:extension>
1131     </xs:complexContent>
1132   </xs:complexType>
1133   <xs:simpleType name="InfrastructureDamageTypeEnum">
1134     <xs:restriction base="xs:string">
1135       <xs:enumeration value="burstPipe" />
1136       <xs:enumeration value="burstWaterMain" />
1137       <xs:enumeration value="collapsedSewer" />
1138       <xs:enumeration value="damagedBridge" />
1139       <xs:enumeration value="damagedCrashBarrier" />
1140       <xs:enumeration value="damagedFlyover" />
1141       <xs:enumeration value="damagedGallery" />
1142       <xs:enumeration value="damagedGantry" />
1143       <xs:enumeration value="damagedRoadSurface" />
1144       <xs:enumeration value="damagedTunnel" />
1145       <xs:enumeration value="damagedViaduct" />

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1138     <xs:enumeration value="fallenPowerCables" />
1139     <xs:enumeration value="gasLeak" />
1140     <xs:enumeration value="weakBridge" />
1141     <xs:enumeration value="other" />
1142   </xs:restriction>
1143 </xs:simpleType>
1144 <xs:simpleType name="InjuryStatusTypeEnum">
1145   <xs:restriction base="xs:string">
1146     <xs:enumeration value="dead" />
1147     <xs:enumeration value="injured" />
1148     <xs:enumeration value="seriouslyInjured" />
1149     <xs:enumeration value="slightlyInjured" />
1150     <xs:enumeration value="uninjured" />
1151     <xs:enumeration value="unknown" />
1152   </xs:restriction>
1153 </xs:simpleType>
1154 <xs:simpleType name="Integer">
1155   <xs:restriction base="xs:integer" />
1156 </xs:simpleType>
1157 <xs:complexType name="IntegerMetreDistanceValue">
1158   <xs:complexContent>
1159     <xs:extension base="D2LogicalModel:DataValue">
1160       <xs:sequence>
1161         <xs:element name="integerMetreDistance" type=
1162 ↪ "D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="1" maxOccurs="1" />
1163         <xs:element name="integerMetreDistanceValueExtension" type="D2LogicalModel:_
1164 ↪ ExtensionType" minOccurs="0" />
1165       </xs:sequence>
1166     </xs:extension>
1167   </xs:complexContent>
1168 </xs:complexType>
1169 <xs:simpleType name="IntensityKilogramsPerSquareMetre">
1170   <xs:restriction base="D2LogicalModel:Float" />
1171 </xs:simpleType>
1172 <xs:simpleType name="IntensityMillimetresPerHour">
1173   <xs:restriction base="D2LogicalModel:Float" />
1174 </xs:simpleType>
1175 <xs:complexType name="InternationalIdentifier">
1176   <xs:sequence>
1177     <xs:element name="country" type="D2LogicalModel:CountryEnum" minOccurs="1"
1178 ↪ maxOccurs="1" />
1179     <xs:element name="nationalIdentifier" type="D2LogicalModel:String" minOccurs="1"
1180 ↪ maxOccurs="1" />
1181     <xs:element name="internationalIdentifierExtension" type="D2LogicalModel:_
1182 ↪ ExtensionType" minOccurs="0" />
1183   </xs:sequence>
1184 </xs:complexType>
1185 <xs:simpleType name="InvolvementRolesEnum">
1186   <xs:restriction base="xs:string">
1187     <xs:enumeration value="cyclist" />
1188     <xs:enumeration value="pedestrian" />
1189     <xs:enumeration value="unknown" />
1190     <xs:enumeration value="vehicleDriver" />
1191     <xs:enumeration value="vehicleOccupant" />
1192     <xs:enumeration value="vehiclePassenger" />
1193     <xs:enumeration value="witness" />

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1189     </xs:restriction>
1190 </xs:simpleType>
1191 <xs:complexType name="KilogramsConcentrationValue">
1192   <xs:complexContent>
1193     <xs:extension base="D2LogicalModel:DataValue">
1194       <xs:sequence>
1195         <xs:element name="kilogramsConcentration" type=
1196 ↪ "D2LogicalModel:ConcentrationKilogramsPerCubicMetre" minOccurs="1" maxOccurs="1" />
1197         <xs:element name="kilogramsConcentrationValueExtension" type=
1198 ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
1199       </xs:sequence>
1200     </xs:extension>
1201   </xs:complexContent>
1202 </xs:complexType>
1203 <xs:simpleType name="KilometresPerHour">
1204   <xs:restriction base="D2LogicalModel:Float" />
1205 </xs:simpleType>
1206 <xs:simpleType name="LaneEnum">
1207   <xs:restriction base="xs:string">
1208     <xs:enumeration value="allLanesCompleteCarriageway" />
1209     <xs:enumeration value="busLane" />
1210     <xs:enumeration value="busStop" />
1211     <xs:enumeration value="carPoolLane" />
1212     <xs:enumeration value="centralReservation" />
1213     <xs:enumeration value="crawlerLane" />
1214     <xs:enumeration value="emergencyLane" />
1215     <xs:enumeration value="escapeLane" />
1216     <xs:enumeration value="expressLane" />
1217     <xs:enumeration value="hardShoulder" />
1218     <xs:enumeration value="heavyVehicleLane" />
1219     <xs:enumeration value="lane1" />
1220     <xs:enumeration value="lane2" />
1221     <xs:enumeration value="lane3" />
1222     <xs:enumeration value="lane4" />
1223     <xs:enumeration value="lane5" />
1224     <xs:enumeration value="lane6" />
1225     <xs:enumeration value="lane7" />
1226     <xs:enumeration value="lane8" />
1227     <xs:enumeration value="lane9" />
1228     <xs:enumeration value="layBy" />
1229     <xs:enumeration value="leftHandTurningLane" />
1230     <xs:enumeration value="leftLane" />
1231     <xs:enumeration value="localTrafficLane" />
1232     <xs:enumeration value="middleLane" />
1233     <xs:enumeration value="opposingLanes" />
1234     <xs:enumeration value="overtakingLane" />
1235     <xs:enumeration value="rightHandTurningLane" />
1236     <xs:enumeration value="rightLane" />
1237     <xs:enumeration value="rushHourLane" />
1238     <xs:enumeration value="setDownArea" />
1239     <xs:enumeration value="slowVehicleLane" />
1240     <xs:enumeration value="throughTrafficLane" />
1241     <xs:enumeration value="tidalFlowLane" />
1242     <xs:enumeration value="turningLane" />
1243     <xs:enumeration value="verge" />
1244   </xs:restriction>

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1243 </xs:simpleType>
1244 <xs:simpleType name="Language">
1245   <xs:restriction base="xs:language" />
1246 </xs:simpleType>
1247 <xs:complexType name="LengthCharacteristic">
1248   <xs:sequence>
1249     <xs:element name="comparisonOperator" type=
1250 ↪ "D2LogicalModel:ComparisonOperatorEnum" minOccurs="1" maxOccurs="1" />
1251     <xs:element name="vehicleLength" type="D2LogicalModel:MetresAsFloat" minOccurs=
1252 ↪ "1" maxOccurs="1" />
1253     <xs:element name="lengthCharacteristicExtension" type="D2LogicalModel:_
1254 ↪ ExtensionType" minOccurs="0" />
1255   </xs:sequence>
1256 </xs:complexType>
1257 <xs:complexType name="Linear">
1258   <xs:complexContent>
1259     <xs:extension base="D2LogicalModel:NetworkLocation">
1260       <xs:sequence>
1261         <xs:element name="alertCLinear" type="D2LogicalModel:AlertCLinear"
1262 ↪ minOccurs="0" />
1263         <xs:element name="linearWithinLinearElement" type=
1264 ↪ "D2LogicalModel:LinearWithinLinearElement" minOccurs="0" />
1265         <xs:element name="globalNetworkLinear" type=
1266 ↪ "D2LogicalModel:GlobalNetworkLinear" minOccurs="0" />
1267         <xs:element name="linearExtension" type="D2LogicalModel:_LinearExtensionType
1268 ↪ " minOccurs="0" />
1269       </xs:sequence>
1270     </xs:extension>
1271   </xs:complexContent>
1272 </xs:complexType>
1273 <xs:complexType name="LinearElement">
1274   <xs:sequence>
1275     <xs:element name="roadName" type="D2LogicalModel:MultilingualString" minOccurs=
1276 ↪ "0" maxOccurs="1" />
1277     <xs:element name="roadNumber" type="D2LogicalModel:String" minOccurs="0"
1278 ↪ maxOccurs="1" />
1279     <xs:element name="linearElementReferenceModel" type="D2LogicalModel:String"
1280 ↪ minOccurs="0" maxOccurs="1" />
1281     <xs:element name="linearElementReferenceModelVersion" type=
1282 ↪ "D2LogicalModel:String" minOccurs="0" maxOccurs="1" />
1283     <xs:element name="linearElementNature" type=
1284 ↪ "D2LogicalModel:LinearElementNatureEnum" minOccurs="0" maxOccurs="1" />
1285     <xs:element name="linearElementExtension" type="D2LogicalModel:_ExtensionType"
1286 ↪ minOccurs="0" />
1287   </xs:sequence>
1288 </xs:complexType>
1289 <xs:complexType name="LinearElementByCode">
1290   <xs:complexContent>
1291     <xs:extension base="D2LogicalModel:LinearElement">
1292       <xs:sequence>
1293         <xs:element name="linearElementIdentifier" type="D2LogicalModel:String"
1294 ↪ minOccurs="1" maxOccurs="1" />
1295         <xs:element name="linearElementByCodeExtension" type="D2LogicalModel:_
1296 ↪ ExtensionType" minOccurs="0" />
1297       </xs:sequence>
1298     </xs:extension>

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1284     </xs:complexContent>
1285 </xs:complexType>
1286 <xs:complexType name="LinearElementByPoints">
1287     <xs:complexContent>
1288         <xs:extension base="D2LogicalModel:LinearElement">
1289             <xs:sequence>
1290                 <xs:element name="startPointOfLinearElement" type="D2LogicalModel:Referent" />
1291                 <xs:element name="intermediatePointOnLinearElement" type="D2LogicalModel:_
1292                 <xs:element name="endPointOfLinearElement" type="D2LogicalModel:Referent" />
1293                 <xs:element name="linearElementByPointsExtension" type="D2LogicalModel:_
1294             </xs:sequence>
1295         </xs:extension>
1296     </xs:complexContent>
1297 </xs:complexType>
1298 <xs:simpleType name="LinearElementNatureEnum">
1299     <xs:restriction base="xs:string">
1300         <xs:enumeration value="road" />
1301         <xs:enumeration value="roadSection" />
1302         <xs:enumeration value="slipRoad" />
1303         <xs:enumeration value="other" />
1304     </xs:restriction>
1305 </xs:simpleType>
1306 <xs:simpleType name="LinearReferencingDirectionEnum">
1307     <xs:restriction base="xs:string">
1308         <xs:enumeration value="both" />
1309         <xs:enumeration value="opposite" />
1310         <xs:enumeration value="aligned" />
1311         <xs:enumeration value="unknown" />
1312     </xs:restriction>
1313 </xs:simpleType>
1314 <xs:complexType name="LinearWithinLinearElement">
1315     <xs:sequence>
1316         <xs:element name="administrativeAreaOfLinearSection" type=
1317         <xs:element name="directionBoundOnLinearSection" type=
1318         <xs:element name="directionRelativeOnLinearSection" type=
1319         <xs:element name="heightGradeOfLinearSection" type=
1320         <xs:element name="linearElement" type="D2LogicalModel:LinearElement" />
1321         <xs:element name="fromPoint" type="D2LogicalModel:DistanceAlongLinearElement" />
1322         <xs:element name="toPoint" type="D2LogicalModel:DistanceAlongLinearElement" />
1323         <xs:element name="linearWithinLinearElementExtension" type="D2LogicalModel:_
1324     </xs:sequence>
1325 </xs:complexType>
1326 <xs:complexType name="LinearWithinLinearGNElement">
1327     <xs:sequence>
1328         <xs:element name="sectionId" type="D2LogicalModel:NonNegativeInteger" minOccurs=
1329         <xs:element name="directionRelativeOnLinearSection" type=

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1330     <xs:element name="orderOfSection" type="D2LogicalModel:NonNegativeInteger"
↪ minOccurs="0" maxOccurs="1" />
1331     <xs:element name="fromPoint" type="D2LogicalModel:DistanceAlongLinearElement" />
1332     <xs:element name="toPoint" type="D2LogicalModel:DistanceAlongLinearElement" />
1333     <xs:element name="linearWithinLinearGNElementExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
1334   </xs:sequence>
1335 </xs:complexType>
1336 <xs:simpleType name="LoadTypeEnum">
1337   <xs:restriction base="xs:string">
1338     <xs:enumeration value="abnormalLoad" />
1339     <xs:enumeration value="ammunition" />
1340     <xs:enumeration value="chemicals" />
1341     <xs:enumeration value="combustibleMaterials" />
1342     <xs:enumeration value="corrosiveMaterials" />
1343     <xs:enumeration value="debris" />
1344     <xs:enumeration value="empty" />
1345     <xs:enumeration value="explosiveMaterials" />
1346     <xs:enumeration value="extraHighLoad" />
1347     <xs:enumeration value="extraLongLoad" />
1348     <xs:enumeration value="extraWideLoad" />
1349     <xs:enumeration value="fuel" />
1350     <xs:enumeration value="glass" />
1351     <xs:enumeration value="goods" />
1352     <xs:enumeration value="hazardousMaterials" />
1353     <xs:enumeration value="liquid" />
1354     <xs:enumeration value="livestock" />
1355     <xs:enumeration value="materials" />
1356     <xs:enumeration value="materialsDangerousForPeople" />
1357     <xs:enumeration value="materialsDangerousForTheEnvironment" />
1358     <xs:enumeration value="materialsDangerousForWater" />
1359     <xs:enumeration value="oil" />
1360     <xs:enumeration value="ordinary" />
1361     <xs:enumeration value="perishableProducts" />
1362     <xs:enumeration value="petrol" />
1363     <xs:enumeration value="pharmaceuticalMaterials" />
1364     <xs:enumeration value="radioactiveMaterials" />
1365     <xs:enumeration value="refuse" />
1366     <xs:enumeration value="toxicMaterials" />
1367     <xs:enumeration value="vehicles" />
1368     <xs:enumeration value="other" />
1369   </xs:restriction>
1370 </xs:simpleType>
1371 <xs:complexType name="Location" abstract="true">
1372   <xs:complexContent>
1373     <xs:extension base="D2LogicalModel:GroupOfLocations">
1374       <xs:sequence>
1375         <xs:element name="locationExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
1376       </xs:sequence>
1377     </xs:extension>
1378   </xs:complexContent>
1379 </xs:complexType>
1380 <xs:simpleType name="LocationDescriptorEnum">
1381   <xs:restriction base="xs:string">
1382     <xs:enumeration value="aroundABendInRoad" />

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1383     <xs:enumeration value="atMotorwayInterchange" />
1384     <xs:enumeration value="atRestArea" />
1385     <xs:enumeration value="atServiceArea" />
1386     <xs:enumeration value="atTollPlaza" />
1387     <xs:enumeration value="atTunnelEntryOrExit" />
1388     <xs:enumeration value="inbound" />
1389     <xs:enumeration value="inGallery" />
1390     <xs:enumeration value="inTheCentre" />
1391     <xs:enumeration value="inTheOppositeDirection" />
1392     <xs:enumeration value="inTunnel" />
1393     <xs:enumeration value="onBorder" />
1394     <xs:enumeration value="onBridge" />
1395     <xs:enumeration value="onConnector" />
1396     <xs:enumeration value="onElevatedSection" />
1397     <xs:enumeration value="onFlyover" />
1398     <xs:enumeration value="onIceRoad" />
1399     <xs:enumeration value="onLevelCrossing" />
1400     <xs:enumeration value="onLinkRoad" />
1401     <xs:enumeration value="onPass" />
1402     <xs:enumeration value="onRoundabout" />
1403     <xs:enumeration value="onTheLeft" />
1404     <xs:enumeration value="onTheRight" />
1405     <xs:enumeration value="onTheRoadway" />
1406     <xs:enumeration value="onUndergroundSection" />
1407     <xs:enumeration value="onUnderpass" />
1408     <xs:enumeration value="outbound" />
1409     <xs:enumeration value="overCrestOfHill" />
1410     <xs:enumeration value="withinJunction" />
1411 </xs:restriction>
1412 </xs:simpleType>
1413 <xs:simpleType name="MaintenanceVehicleActionsEnum">
1414     <xs:restriction base="xs:string">
1415         <xs:enumeration value="maintenanceVehiclesMergingIntoTrafficFlow" />
1416         <xs:enumeration value="saltAndGritSpreading" />
1417         <xs:enumeration value="slowMoving" />
1418         <xs:enumeration value="snowClearing" />
1419         <xs:enumeration value="stoppingToServiceEquipments" />
1420     </xs:restriction>
1421 </xs:simpleType>
1422 <xs:complexType name="MaintenanceVehicles">
1423     <xs:sequence>
1424         <xs:element name="numberOfMaintenanceVehicles" type=
1425         ↪ "D2LogicalModel:NonNegativeInteger" minOccurs="0" maxOccurs="1" />
1426         <xs:element name="maintenanceVehicleActions" type=
1427         ↪ "D2LogicalModel:MaintenanceVehicleActionsEnum" minOccurs="0" maxOccurs="unbounded" /
1428         ↪ >
1429         <xs:element name="maintenanceVehiclesExtension" type="D2LogicalModel:_
1430         ↪ ExtensionType" minOccurs="0" />
1431     </xs:sequence>
1432 </xs:complexType>
1433 <xs:complexType name="MaintenanceWorks">
1434     <xs:complexContent>
1435         <xs:extension base="D2LogicalModel:Roadworks">
1436             <xs:sequence>
1437                 <xs:element name="roadMaintenanceType" type=
1438                 ↪ "D2LogicalModel:RoadMaintenanceTypeEnum" minOccurs="1" maxOccurs="unbounded" />

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1434     <xs:element name="maintenanceWorksExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
1435     </xs:sequence>
1436   </xs:extension>
1437 </xs:complexContent>
1438 </xs:complexType>
1439 <xs:complexType name="MeasurementEquipmentFault">
1440   <xs:complexContent>
1441     <xs:extension base="D2LogicalModel:Fault">
1442       <xs:sequence>
1443         <xs:element name="measurementEquipmentFault" type=
↪"D2LogicalModel:MeasurementEquipmentFaultEnum" minOccurs="1" maxOccurs="1" />
1444         <xs:element name="measurementEquipmentFaultExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
1445       </xs:sequence>
1446     </xs:extension>
1447   </xs:complexContent>
1448 </xs:complexType>
1449 <xs:simpleType name="MeasurementEquipmentFaultEnum">
1450   <xs:restriction base="xs:string">
1451     <xs:enumeration value="intermittentDataValues" />
1452     <xs:enumeration value="noDataValuesAvailable" />
1453     <xs:enumeration value="spuriousUnreliableDataValues" />
1454     <xs:enumeration value="unspecifiedOrUnknownFault" />
1455     <xs:enumeration value="other" />
1456   </xs:restriction>
1457 </xs:simpleType>
1458 <xs:simpleType name="MetresAsFloat">
1459   <xs:restriction base="D2LogicalModel:Float" />
1460 </xs:simpleType>
1461 <xs:simpleType name="MetresAsNonNegativeInteger">
1462   <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
1463 </xs:simpleType>
1464 <xs:complexType name="Mobility">
1465   <xs:sequence>
1466     <xs:element name="mobilityType" type="D2LogicalModel:MobilityEnum" minOccurs="1
↪" maxOccurs="1" />
1467     <xs:element name="mobilityExtension" type="D2LogicalModel:_ExtensionType"
↪minOccurs="0" />
1468   </xs:sequence>
1469 </xs:complexType>
1470 <xs:simpleType name="MobilityEnum">
1471   <xs:restriction base="xs:string">
1472     <xs:enumeration value="mobile" />
1473     <xs:enumeration value="stationary" />
1474     <xs:enumeration value="unknown" />
1475   </xs:restriction>
1476 </xs:simpleType>
1477 <xs:complexType name="MultilingualString">
1478   <xs:sequence>
1479     <xs:element name="values">
1480       <xs:complexType>
1481         <xs:sequence>
1482           <xs:element name="value" type="D2LogicalModel:MultilingualStringValue"
↪maxOccurs="unbounded" />
1483         </xs:sequence>

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1484     </xs:complexType>
1485   </xs:element>
1486 </xs:sequence>
1487 </xs:complexType>
1488 <xs:complexType name="MultilingualStringValue">
1489   <xs:simpleContent>
1490     <xs:extension base="D2LogicalModel:MultilingualStringValue" type="D2LogicalModel:MultilingualStringValue" />
1491     <xs:attribute name="lang" type="xs:language" />
1492   </xs:extension>
1493 </xs:simpleContent>
1494 </xs:complexType>
1495 <xs:simpleType name="MultilingualStringValue" type="xs:string">
1496   <xs:restriction base="xs:string">
1497     <xs:maxLength value="1024" />
1498   </xs:restriction>
1499 </xs:simpleType>
1500 <xs:complexType name="NetworkLocation" abstract="true">
1501   <xs:complexContent>
1502     <xs:extension base="D2LogicalModel:Location">
1503       <xs:sequence>
1504         <xs:element name="supplementaryPositionalDescription" type="D2LogicalModel:SupplementaryPositionalDescription" minOccurs="0" />
1505         <xs:element name="destination" type="D2LogicalModel:Destination" minOccurs="0" />
1506         <xs:element name="networkLocationExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
1507       </xs:sequence>
1508     </xs:extension>
1509   </xs:complexContent>
1510 </xs:complexType>
1511 <xs:complexType name="NetworkManagement" abstract="true">
1512   <xs:complexContent>
1513     <xs:extension base="D2LogicalModel:OperatorAction">
1514       <xs:sequence>
1515         <xs:element name="complianceOption" type="D2LogicalModel:ComplianceOptionEnum" minOccurs="1" maxOccurs="1" />
1516         <xs:element name="applicableForTrafficDirection" type="D2LogicalModel:DirectionEnum" minOccurs="0" maxOccurs="unbounded" />
1517         <xs:element name="applicableForTrafficType" type="D2LogicalModel:TrafficTypeEnum" minOccurs="0" maxOccurs="unbounded" />
1518         <xs:element name="placesAtWhichApplicable" type="D2LogicalModel:PlacesEnum" minOccurs="0" maxOccurs="unbounded" />
1519         <xs:element name="forVehiclesWithCharacteristicsOf" type="D2LogicalModel:VehicleCharacteristics" minOccurs="0" maxOccurs="unbounded" />
1520         <xs:element name="networkManagementExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
1521       </xs:sequence>
1522     </xs:extension>
1523   </xs:complexContent>
1524 </xs:complexType>
1525 <xs:complexType name="NonManagedCause">
1526   <xs:complexContent>
1527     <xs:extension base="D2LogicalModel:Cause">
1528       <xs:sequence>
1529         <xs:element name="causeType" type="D2LogicalModel:CauseTypeEnum" minOccurs="0" maxOccurs="1" />

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1530     <xs:element name="nonManagedCauseExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
1531   </xs:sequence>
1532 </xs:extension>
1533 </xs:complexContent>
1534 </xs:complexType>
1535 <xs:simpleType name="NonNegativeInteger">
1536   <xs:restriction base="xs:nonNegativeInteger" />
1537 </xs:simpleType>
1538 <xs:complexType name="NonRoadEventInformation" abstract="true">
1539   <xs:complexContent>
1540     <xs:extension base="D2LogicalModel:SituationRecord">
1541       <xs:sequence>
1542         <xs:element name="nonRoadEventInformationExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
1543       </xs:sequence>
1544     </xs:extension>
1545   </xs:complexContent>
1546 </xs:complexType>
1547 <xs:complexType name="NonWeatherRelatedRoadConditions">
1548   <xs:complexContent>
1549     <xs:extension base="D2LogicalModel:RoadConditions">
1550       <xs:sequence>
1551         <xs:element name="nonWeatherRelatedRoadConditionType" type=
↪"D2LogicalModel:NonWeatherRelatedRoadConditionTypeEnum" minOccurs="1" maxOccurs=
↪"unbounded" />
1552         <xs:element name="nonWeatherRelatedRoadConditionsExtension" type=
↪"D2LogicalModel:_ExtensionType" minOccurs="0" />
1553       </xs:sequence>
1554     </xs:extension>
1555   </xs:complexContent>
1556 </xs:complexType>
1557 <xs:simpleType name="NonWeatherRelatedRoadConditionTypeEnum">
1558   <xs:restriction base="xs:string">
1559     <xs:enumeration value="dieselOnRoad" />
1560     <xs:enumeration value="leavesOnRoad" />
1561     <xs:enumeration value="looseChippings" />
1562     <xs:enumeration value="looseSandOnRoad" />
1563     <xs:enumeration value="mudOnRoad" />
1564     <xs:enumeration value="oilOnRoad" />
1565     <xs:enumeration value="petrolOnRoad" />
1566     <xs:enumeration value="roadSurfaceInPoorCondition" />
1567     <xs:enumeration value="slipperyRoad" />
1568     <xs:enumeration value="other" />
1569   </xs:restriction>
1570 </xs:simpleType>
1571 <xs:complexType name="NumberOfAxlesCharacteristic">
1572   <xs:sequence>
1573     <xs:element name="comparisonOperator" type=
↪"D2LogicalModel:ComparisonOperatorEnum" minOccurs="1" maxOccurs="1" />
1574     <xs:element name="numberOfAxles" type="D2LogicalModel:NonNegativeInteger"
↪minOccurs="1" maxOccurs="1" />
1575     <xs:element name="numberOfAxlesCharacteristicExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
1576   </xs:sequence>
1577 </xs:complexType>

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1578 <xs:complexType name="Obstruction" abstract="true">
1579   <xs:complexContent>
1580     <xs:extension base="D2LogicalModel:TrafficElement">
1581       <xs:sequence>
1582         <xs:element name="numberOfObstructions" type=
1579 ↪ "D2LogicalModel:NonNegativeInteger" minOccurs="0" maxOccurs="1" />
1583         <xs:element name="mobilityOfObstruction" type="D2LogicalModel:Mobility"
1584 ↪ minOccurs="0" />
1584         <xs:element name="obstructionExtension" type="D2LogicalModel:_ExtensionType
1585 ↪ " minOccurs="0" />
1585       </xs:sequence>
1586     </xs:extension>
1587   </xs:complexContent>
1588 </xs:complexType>
1589 <xs:simpleType name="ObstructionTypeEnum">
1590   <xs:restriction base="xs:string">
1591     <xs:enumeration value="airCrash" />
1592     <xs:enumeration value="childrenOnRoadway" />
1593     <xs:enumeration value="clearanceWork" />
1594     <xs:enumeration value="craneOperating" />
1595     <xs:enumeration value="cyclistsOnRoadway" />
1596     <xs:enumeration value="debris" />
1597     <xs:enumeration value="explosion" />
1598     <xs:enumeration value="explosionHazard" />
1599     <xs:enumeration value="hazardsOnTheRoad" />
1600     <xs:enumeration value="highSpeedChase" />
1601     <xs:enumeration value="houseFire" />
1602     <xs:enumeration value="incident" />
1603     <xs:enumeration value="industrialAccident" />
1604     <xs:enumeration value="objectOnTheRoad" />
1605     <xs:enumeration value="objectsFallingFromMovingVehicle" />
1606     <xs:enumeration value="obstructionOnTheRoad" />
1607     <xs:enumeration value="peopleOnRoadway" />
1608     <xs:enumeration value="railCrash" />
1609     <xs:enumeration value="recklessDriver" />
1610     <xs:enumeration value="rescueAndRecoveryWork" />
1611     <xs:enumeration value="severeFrostDamagedRoadway" />
1612     <xs:enumeration value="shedLoad" />
1613     <xs:enumeration value="snowAndIceDebris" />
1614     <xs:enumeration value="spillageOccurringFromMovingVehicle" />
1615     <xs:enumeration value="spillageOnTheRoad" />
1616     <xs:enumeration value="unprotectedAccidentArea" />
1617     <xs:enumeration value="other" />
1618   </xs:restriction>
1619 </xs:simpleType>
1620 <xs:complexType name="OccupancyChangeValue">
1621   <xs:complexContent>
1622     <xs:extension base="D2LogicalModel:DataValue">
1623       <xs:sequence>
1624         <xs:element name="occupancyChange" type="D2LogicalModel:Integer" minOccurs=
1625 ↪ "1" maxOccurs="1" />
1625         <xs:element name="occupancyChangeValueExtension" type="D2LogicalModel:_
1626 ↪ ExtensionType" minOccurs="0" />
1626       </xs:sequence>
1627     </xs:extension>
1628   </xs:complexContent>

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1629 </xs:complexType>
1630 <xs:complexType name="OpenlrBaseLocationReferencePoint" abstract="true">
1631   <xs:sequence>
1632     <xs:element name="openlrCoordinate" type="D2LogicalModel:PointCoordinates" />
1633     <xs:element name="openlrLineAttributes" type=
1634     ↪ "D2LogicalModel:OpenlrLineAttributes" />
1635     <xs:element name="openlrBaseLocationReferencePointExtension" type=
1636     ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
1637   </xs:sequence>
1638 </xs:complexType>
1639 <xs:complexType name="OpenlrBasePointLocation" abstract="true">
1640   <xs:sequence>
1641     <xs:element name="openlrSideOfRoad" type="D2LogicalModel:OpenlrSideOfRoadEnum"
1642     ↪ minOccurs="1" maxOccurs="1" />
1643     <xs:element name="openlrOrientation" type="D2LogicalModel:OpenlrOrientationEnum"
1644     ↪ minOccurs="1" maxOccurs="1" />
1645     <xs:element name="openlrPositiveOffset" type=
1646     ↪ "D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="0" maxOccurs="1" />
1647     <xs:element name="openlrLocationReferencePoint" type=
1648     ↪ "D2LogicalModel:OpenlrLocationReferencePoint" />
1649     <xs:element name="openlrLastLocationReferencePoint" type=
1650     ↪ "D2LogicalModel:OpenlrLastLocationReferencePoint" />
1651     <xs:element name="openlrBasePointLocationExtension" type="D2LogicalModel:_
1652     ↪ ExtensionType" minOccurs="0" />
1653   </xs:sequence>
1654 </xs:complexType>
1655 <xs:complexType name="OpenlrExtendedLinear">
1656   <xs:sequence>
1657     <xs:element name="firstDirection" type=
1658     ↪ "D2LogicalModel:OpenlrLineLocationReference" />
1659     <xs:element name="oppositeDirection" type=
1660     ↪ "D2LogicalModel:OpenlrLineLocationReference" minOccurs="0" />
1661   </xs:sequence>
1662 </xs:complexType>
1663 <xs:complexType name="OpenlrExtendedPoint">
1664   <xs:sequence>
1665     <xs:element name="openlrPointLocationReference" type=
1666     ↪ "D2LogicalModel:OpenlrPointLocationReference" />
1667   </xs:sequence>
1668 </xs:complexType>
1669 <xs:simpleType name="OpenlrFormOfWayEnum">
1670   <xs:restriction base="xs:string">
1671     <xs:enumeration value="undefined" />
1672     <xs:enumeration value="motorway" />
1673     <xs:enumeration value="multipleCarriageway" />
1674     <xs:enumeration value="singleCarriageway" />
1675     <xs:enumeration value="roundabout" />
1676     <xs:enumeration value="slipRoad" />
1677     <xs:enumeration value="trafficSquare" />
1678     <xs:enumeration value="other" />
1679   </xs:restriction>
1680 </xs:simpleType>
1681 <xs:simpleType name="OpenlrFunctionalRoadClassEnum">
1682   <xs:restriction base="xs:string">
1683     <xs:enumeration value="FRC0" />
1684     <xs:enumeration value="FRC1" />

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1674     <xs:enumeration value="FRC2" />
1675     <xs:enumeration value="FRC3" />
1676     <xs:enumeration value="FRC4" />
1677     <xs:enumeration value="FRC5" />
1678     <xs:enumeration value="FRC6" />
1679     <xs:enumeration value="FRC7" />
1680   </xs:restriction>
1681 </xs:simpleType>
1682 <xs:complexType name="OpenlrGeoCoordinate">
1683   <xs:sequence>
1684     <xs:element name="openlrCoordinate" type="D2LogicalModel:PointCoordinates" />
1685     <xs:element name="openlrGeoCoordinateExtension" type="D2LogicalModel:_
1686 ↪ ExtensionType" minOccurs="0" />
1687   </xs:sequence>
1688 </xs:complexType>
1689 <xs:complexType name="OpenlrLastLocationReferencePoint">
1690   <xs:complexContent>
1691     <xs:extension base="D2LogicalModel:OpenlrBaseLocationReferencePoint">
1692       <xs:sequence>
1693         <xs:element name="openlrLastLocationReferencePointExtension" type=
1694 ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
1695       </xs:sequence>
1696     </xs:extension>
1697   </xs:complexContent>
1698 </xs:complexType>
1699 <xs:complexType name="OpenlrLineAttributes">
1700   <xs:sequence>
1701     <xs:element name="openlrFunctionalRoadClass" type=
1702 ↪ "D2LogicalModel:OpenlrFunctionalRoadClassEnum" minOccurs="1" maxOccurs="1" />
1703     <xs:element name="openlrFormOfWay" type="D2LogicalModel:OpenlrFormOfWayEnum"
1704 ↪ minOccurs="1" maxOccurs="1" />
1705     <xs:element name="openlrBearing" type="D2LogicalModel:AngleInDegrees" minOccurs=
1706 ↪ "1" maxOccurs="1" />
1707     <xs:element name="openlrLineAttributesExtension" type="D2LogicalModel:_
1708 ↪ ExtensionType" minOccurs="0" />
1709   </xs:sequence>
1710 </xs:complexType>
1711 <xs:complexType name="OpenlrLineLocationReference">
1712   <xs:sequence>
1713     <xs:element name="openlrLocationReferencePoint" type=
1714 ↪ "D2LogicalModel:OpenlrLocationReferencePoint" maxOccurs="unbounded" />
1715     <xs:element name="openlrLastLocationReferencePoint" type=
1716 ↪ "D2LogicalModel:OpenlrLastLocationReferencePoint" />
1717     <xs:element name="openlrOffsets" type="D2LogicalModel:OpenlrOffsets" minOccurs=
1718 ↪ "0" />
1719     <xs:element name="openlrLineLocationReferenceExtension" type="D2LogicalModel:_
1720 ↪ ExtensionType" minOccurs="0" />
1721   </xs:sequence>
1722 </xs:complexType>
1723 <xs:complexType name="OpenlrLocationReferencePoint">
1724   <xs:complexContent>
1725     <xs:extension base="D2LogicalModel:OpenlrBaseLocationReferencePoint">
1726       <xs:sequence>
1727         <xs:element name="openlrPathAttributes" type=
1728 ↪ "D2LogicalModel:OpenlrPathAttributes" />
1729         <xs:element name="openlrLocationReferencePointExtension" type=

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1719     </xs:sequence>
1720   </xs:extension>
1721 </xs:complexContent>
1722 </xs:complexType>
1723 <xs:complexType name="OpenlrOffsets">
1724   <xs:sequence>
1725     <xs:element name="openlrPositiveOffset" type=
1726       <xs:element name="openlrNegativeOffset" type=
1727       <xs:element name="openlrOffsetsExtension" type="D2LogicalModel:_
1728     </xs:sequence>
1729   </xs:complexType>
1730   <xs:simpleType name="OpenlrOrientationEnum">
1731     <xs:restriction base="xs:string">
1732       <xs:enumeration value="noOrientationOrUnknown" />
1733       <xs:enumeration value="withLineDirection" />
1734       <xs:enumeration value="againstLineDirection" />
1735       <xs:enumeration value="both" />
1736     </xs:restriction>
1737   </xs:simpleType>
1738   <xs:complexType name="OpenlrPathAttributes">
1739     <xs:sequence>
1740       <xs:element name="openlrLowestFRCToNextLRPoint" type=
1741       <xs:element name="openlrDistanceToNextLRPoint" type=
1742       <xs:element name="openlrPathAttributesExtension" type="D2LogicalModel:_
1743     </xs:sequence>
1744   </xs:complexType>
1745   <xs:complexType name="OpenlrPointAlongLine">
1746     <xs:complexContent>
1747       <xs:extension base="D2LogicalModel:OpenlrBasePointLocation">
1748         <xs:sequence>
1749           <xs:element name="openlrPointAlongLineExtension" type="D2LogicalModel:_
1750         </xs:sequence>
1751       </xs:extension>
1752     </xs:complexContent>
1753   </xs:complexType>
1754   <xs:complexType name="OpenlrPointLocationReference">
1755     <xs:sequence>
1756       <xs:element name="openlrGeoCoordinate" type="D2LogicalModel:OpenlrGeoCoordinate
1757       <xs:element name="openlrPoiWithAccessPoint" type=
1758       <xs:element name="openlrPointAlongLine" type=
1759       <xs:element name="openlrPointLocationReferenceExtension" type="D2LogicalModel:_
1760     </xs:sequence>
1761   </xs:complexType>
1762   <xs:complexType name="OpenlrPoiWithAccessPoint">

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1763 <xs:complexContent>
1764 <xs:extension base="D2LogicalModel:OpenlrBasePointLocation">
1765 <xs:sequence>
1766 <xs:element name="openlrCoordinate" type="D2LogicalModel:PointCoordinates" /
1767 <xs:element name="openlrPoiWithAccessPointExtension" type="D2LogicalModel:_
1768 ExtensionType" minOccurs="0" />
1769 </xs:sequence>
1770 </xs:extension>
1771 </xs:complexContent>
1772 </xs:complexType>
1773 <xs:simpleType name="OpenlrSideOfRoadEnum">
1774 <xs:restriction base="xs:string">
1775 <xs:enumeration value="onRoadOrUnknown" />
1776 <xs:enumeration value="right" />
1777 <xs:enumeration value="left" />
1778 <xs:enumeration value="both" />
1779 </xs:restriction>
1780 </xs:simpleType>
1781 <xs:complexType name="OperatorAction">
1782 <xs:complexContent>
1783 <xs:extension base="D2LogicalModel:SituationRecord">
1784 <xs:sequence>
1785 <xs:element name="operatorActionExtension" type="D2LogicalModel:_
1786 ExtensionType" minOccurs="0" />
1787 </xs:sequence>
1788 </xs:extension>
1789 </xs:complexContent>
1790 </xs:complexType>
1791 <xs:complexType name="OverallPeriod">
1792 <xs:sequence>
1793 <xs:element name="overallStartTime" type="D2LogicalModel:DateTime" minOccurs="1"
1794 maxOccurs="1" />
1795 <xs:element name="overallEndTime" type="D2LogicalModel:DateTime" minOccurs="0"
1796 maxOccurs="1" />
1797 <xs:element name="overallPeriodExtension" type="D2LogicalModel:_ExtensionType"
1798 minOccurs="0" />
1799 </xs:sequence>
1800 </xs:complexType>
1801 <xs:simpleType name="PassengerCarUnitsPerHour">
1802 <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
1803 </xs:simpleType>
1804 <xs:complexType name="PayloadPublication" abstract="true">
1805 <xs:sequence>
1806 <xs:element name="feedType" type="D2LogicalModel:String" minOccurs="0"
1807 maxOccurs="1" />
1808 <xs:element name="publicationTime" type="D2LogicalModel:DateTime" minOccurs="1"
1809 maxOccurs="1" />
1810 <xs:element name="publicationCreator" type=
1811 "D2LogicalModel:InternationalIdentifier" />
1812 <xs:element name="payloadPublicationExtension" type="D2LogicalModel:_
1813 ExtensionType" minOccurs="0" />
1814 </xs:sequence>
1815 <xs:attribute name="lang" type="D2LogicalModel:Language" use="required" />
1816 </xs:complexType>
1817 <xs:complexType name="PcuFlowValue">

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1809     <xs:complexContent>
1810       <xs:extension base="D2LogicalModel:DataValue">
1811         <xs:sequence>
1812           <xs:element name="pcuFlowRate" type="D2LogicalModel:PassengerCarUnitsPerHour
1813             ↪" minOccurs="1" maxOccurs="1" />
1814           <xs:element name="pcuFlowValueExtension" type="D2LogicalModel:_ExtensionType
1815             ↪" minOccurs="0" />
1816         </xs:sequence>
1817       </xs:extension>
1818     </xs:complexContent>
1819   </xs:complexType>
1820   <xs:simpleType name="Percentage">
1821     <xs:restriction base="D2LogicalModel:Float" />
1822   </xs:simpleType>
1823   <xs:complexType name="PercentageDistanceAlongLinearElement">
1824     <xs:complexContent>
1825       <xs:extension base="D2LogicalModel:DistanceAlongLinearElement">
1826         <xs:sequence>
1827           <xs:element name="percentageDistanceAlong" type="D2LogicalModel:Percentage" ↪
1828             ↪minOccurs="1" maxOccurs="1" />
1829           <xs:element name="percentageDistanceAlongLinearElementExtension" type=
1830             ↪"D2LogicalModel:_ExtensionType" minOccurs="0" />
1831         </xs:sequence>
1832       </xs:extension>
1833     </xs:complexContent>
1834   </xs:complexType>
1835   <xs:simpleType name="PersonCategoryEnum">
1836     <xs:restriction base="xs:string">
1837       <xs:enumeration value="adult" />
1838       <xs:enumeration value="child" />
1839       <xs:enumeration value="emergencyServicesPerson" />
1840       <xs:enumeration value="fireman" />
1841       <xs:enumeration value="infant" />
1842       <xs:enumeration value="medicalStaff" />
1843       <xs:enumeration value="memberOfThePublic" />
1844       <xs:enumeration value="policeman" />
1845       <xs:enumeration value="politician" />
1846       <xs:enumeration value="publicTransportPassenger" />
1847       <xs:enumeration value="sickPerson" />
1848       <xs:enumeration value="trafficOfficer" />
1849       <xs:enumeration value="trafficWarden" />
1850       <xs:enumeration value="veryImportantPerson" />
1851     </xs:restriction>
1852   </xs:simpleType>
1853   <xs:simpleType name="PlacesEnum">
1854     <xs:restriction base="xs:string">
1855       <xs:enumeration value="aroundBendsInTheRoad" />
1856       <xs:enumeration value="atCustomsPosts" />
1857       <xs:enumeration value="atHighAltitudes" />
1858       <xs:enumeration value="atTollPlazas" />
1859       <xs:enumeration value="inBuiltUpAreas" />
1860       <xs:enumeration value="inContraflowSections" />
1861       <xs:enumeration value="inForestedAreas" />
1862       <xs:enumeration value="inGalleries" />
1863       <xs:enumeration value="inLowLyingAreas" />
1864       <xs:enumeration value="inRoadworksAreas" />

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1861     <xs:enumeration value="inRuralAreas" />
1862     <xs:enumeration value="inShadedAreas" />
1863     <xs:enumeration value="inTheCityCentre" />
1864     <xs:enumeration value="inTheInnerCityAreas" />
1865     <xs:enumeration value="inTunnels" />
1866     <xs:enumeration value="onBridges" />
1867     <xs:enumeration value="onDownHillSections" />
1868     <xs:enumeration value="onDualCarriagewaySections" />
1869     <xs:enumeration value="onElevatedSections" />
1870     <xs:enumeration value="onEnteringOrLeavingTunnels" />
1871     <xs:enumeration value="onEnteringTheCountry" />
1872     <xs:enumeration value="onFlyovers" />
1873     <xs:enumeration value="onLeavingTheCountry" />
1874     <xs:enumeration value="onMotorways" />
1875     <xs:enumeration value="onNonMotorways" />
1876     <xs:enumeration value="onPasses" />
1877     <xs:enumeration value="onRoundabouts" />
1878     <xs:enumeration value="onSingleCarriagewaySections" />
1879     <xs:enumeration value="onSlipRoads" />
1880     <xs:enumeration value="onUndergroundSections" />
1881     <xs:enumeration value="onUnderpasses" />
1882     <xs:enumeration value="onUpHillSections" />
1883     <xs:enumeration value="overTheCrestOfHills" />
1884     <xs:enumeration value="other" />
1885 </xs:restriction>
1886 </xs:simpleType>
1887 <xs:complexType name="Point">
1888   <xs:complexContent>
1889     <xs:extension base="D2LogicalModel:NetworkLocation">
1890       <xs:sequence>
1891         <xs:element name="alertCPoint" type="D2LogicalModel:AlertCPoint" minOccurs=
1892 ↪ "0" />
1893         <xs:element name="pointAlongLinearElement" type=
1894 ↪ "D2LogicalModel:PointAlongLinearElement" minOccurs="0" />
1895         <xs:element name="pointWithinLinearGNElement" type=
1896 ↪ "D2LogicalModel:PointWithinLinearGNElement" minOccurs="0" />
1897         <xs:element name="pointExtension" type="D2LogicalModel:_PointExtensionType"
1898 ↪ minOccurs="0" />
1899       </xs:sequence>
1900     </xs:extension>
1901   </xs:complexContent>
1902 </xs:complexType>
1903 <xs:complexType name="PointAlongLinearElement">
1904   <xs:sequence>
1905     <xs:element name="administrativeAreaOfPoint" type=
1906 ↪ "D2LogicalModel:MultilingualString" minOccurs="0" maxOccurs="1" />
1907     <xs:element name="directionBoundAtPoint" type="D2LogicalModel:DirectionEnum"
1908 ↪ minOccurs="0" maxOccurs="1" />
1909     <xs:element name="directionRelativeAtPoint" type=
1910 ↪ "D2LogicalModel:LinearReferencingDirectionEnum" minOccurs="0" maxOccurs="1" />
1911     <xs:element name="heightGradeOfPoint" type="D2LogicalModel:HeightGradeEnum"
1912 ↪ minOccurs="0" maxOccurs="1" />
1913     <xs:element name="linearElement" type="D2LogicalModel:LinearElement" />
1914     <xs:element name="distanceAlongLinearElement" type=
1915 ↪ "D2LogicalModel:DistanceAlongLinearElement" />
1916     <xs:element name="pointAlongLinearElementExtension" type="D2LogicalModel:_

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1908   </xs:sequence>
1909 </xs:complexType>
1910 <xs:complexType name="PointByMultiCoordinates">
1911   <xs:sequence>
1912     <xs:element name="sjtskPointCoordinates" type=
1913   < "D2LogicalModel:SJTSKPointCoordinates" minOccurs="0" />
1914     <xs:element name="pointCoordinates" type="D2LogicalModel:PointCoordinates"
1915   < minOccurs="0" />
1916     <xs:element name="pointByMultiCoordinatesExtension" type="D2LogicalModel:_
1917   < ExtensionType" minOccurs="0" />
1918   </xs:sequence>
1919 </xs:complexType>
1920 <xs:complexType name="PointCoordinates">
1921   <xs:sequence>
1922     <xs:element name="latitude" type="D2LogicalModel:Float" minOccurs="1" maxOccurs=
1923   < "1" />
1924     <xs:element name="longitude" type="D2LogicalModel:Float" minOccurs="1"
1925   < maxOccurs="1" />
1926     <xs:element name="pointCoordinatesExtension" type="D2LogicalModel:_ExtensionType
1927   < " minOccurs="0" />
1928   </xs:sequence>
1929 </xs:complexType>
1930 <xs:complexType name="PointDestination">
1931   <xs:complexContent>
1932     <xs:extension base="D2LogicalModel:Destination">
1933       <xs:sequence>
1934         <xs:element name="point" type="D2LogicalModel:Point" />
1935         <xs:element name="pointDestinationExtension" type="D2LogicalModel:_
1936   < ExtensionType" minOccurs="0" />
1937       </xs:sequence>
1938     </xs:extension>
1939 </xs:complexType>
1940 <xs:complexType name="PointWithinLinearGNElement">
1941   <xs:sequence>
1942     <xs:element name="sectionId" type="D2LogicalModel:NonNegativeInteger" minOccurs=
1943   < "0" maxOccurs="1" />
1944     <xs:element name="directionRelativeOnLinearSection" type=
1945   < "D2LogicalModel:LinearReferencingDirectionEnum" minOccurs="1" maxOccurs="1" />
1946     <xs:element name="networkVersion" type="D2LogicalModel:MultilingualString"
1947   < minOccurs="1" maxOccurs="1" />
1948     <xs:element name="distanceAlongLinearElement" type=
1949   < "D2LogicalModel:DistanceAlongLinearElement" />
1950     <xs:element name="pointWithinLinearGNElementExtension" type="D2LogicalModel:_
1951   < ExtensionType" minOccurs="0" />
1952   </xs:sequence>
1953 </xs:complexType>
1954 <xs:complexType name="PoorEnvironmentConditions">
1955   <xs:complexContent>
1956     <xs:extension base="D2LogicalModel:Conditions">
1957       <xs:sequence>
1958         <xs:element name="poorEnvironmentType" type=
1959   < "D2LogicalModel:PoorEnvironmentTypeEnum" minOccurs="1" maxOccurs="unbounded" />
1960         <xs:element name="precipitationDetail" type=
1961   < "D2LogicalModel:PrecipitationDetail" minOccurs="0" />

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1949     <xs:element name="visibility" type="D2LogicalModel:Visibility" minOccurs="0
1950     ↪ " />
1951     <xs:element name="temperature" type="D2LogicalModel:Temperature" minOccurs=
1952     ↪ "0" />
1953     <xs:element name="wind" type="D2LogicalModel:Wind" minOccurs="0" />
1954     <xs:element name="poorEnvironmentConditionsExtension" type="D2LogicalModel:_
1955     ↪ ExtensionType" minOccurs="0" />
1956     </xs:sequence>
1957     </xs:extension>
1958     </xs:complexType>
1959     <xs:simpleType name="PoorEnvironmentTypeEnum">
1960     <xs:restriction base="xs:string">
1961     <xs:enumeration value="badWeather" />
1962     <xs:enumeration value="blizzard" />
1963     <xs:enumeration value="blowingDust" />
1964     <xs:enumeration value="blowingSnow" />
1965     <xs:enumeration value="crosswinds" />
1966     <xs:enumeration value="damagingHail" />
1967     <xs:enumeration value="denseFog" />
1968     <xs:enumeration value="eclipse" />
1969     <xs:enumeration value="extremeCold" />
1970     <xs:enumeration value="extremeHeat" />
1971     <xs:enumeration value="fog" />
1972     <xs:enumeration value="freezingFog" />
1973     <xs:enumeration value="frost" />
1974     <xs:enumeration value="gales" />
1975     <xs:enumeration value="gustyWinds" />
1976     <xs:enumeration value="hail" />
1977     <xs:enumeration value="heavyFrost" />
1978     <xs:enumeration value="heavyRain" />
1979     <xs:enumeration value="heavySnowfall" />
1980     <xs:enumeration value="hurricaneForceWinds" />
1981     <xs:enumeration value="lowSunGlare" />
1982     <xs:enumeration value="moderateFog" />
1983     <xs:enumeration value="ozonePollution" />
1984     <xs:enumeration value="pollution" />
1985     <xs:enumeration value="patchyFog" />
1986     <xs:enumeration value="precipitationInTheArea" />
1987     <xs:enumeration value="rain" />
1988     <xs:enumeration value="rainChangingToSnow" />
1989     <xs:enumeration value="sandStorms" />
1990     <xs:enumeration value="severeExhaustPollution" />
1991     <xs:enumeration value="severeSmog" />
1992     <xs:enumeration value="showers" />
1993     <xs:enumeration value="sleet" />
1994     <xs:enumeration value="smogAlert" />
1995     <xs:enumeration value="smokeHazard" />
1996     <xs:enumeration value="snowChangingToRain" />
1997     <xs:enumeration value="snowfall" />
1998     <xs:enumeration value="sprayHazard" />
1999     <xs:enumeration value="stormForceWinds" />
2000     <xs:enumeration value="strongGustsOfWind" />
2001     <xs:enumeration value="strongWinds" />
2002     <xs:enumeration value="swarmsOfInsects" />
2003     <xs:enumeration value="temperatureFalling" />

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2002     <xs:enumeration value="thunderstorms" />
2003     <xs:enumeration value="tornadoes" />
2004     <xs:enumeration value="veryStrongGustsOfWind" />
2005     <xs:enumeration value="visibilityReduced" />
2006     <xs:enumeration value="whiteOut" />
2007     <xs:enumeration value="winterStorm" />
2008   </xs:restriction>
2009 </xs:simpleType>
2010 <xs:complexType name="PrecipitationDetail">
2011   <xs:sequence>
2012     <xs:element name="precipitationType" type="D2LogicalModel:PrecipitationTypeEnum
2013     ↪ " minOccurs="0" maxOccurs="1" />
2014     <xs:element name="precipitationIntensity" type=
2015     ↪ "D2LogicalModel:PrecipitationIntensityValue" minOccurs="0" />
2016     <xs:element name="depositionDepth" type=
2017     ↪ "D2LogicalModel:FloatingPointMetreDistanceValue" minOccurs="0" />
2018     <xs:element name="precipitationDetailExtension" type="D2LogicalModel:_
2019     ↪ ExtensionType" minOccurs="0" />
2020   </xs:sequence>
2021 </xs:complexType>
2022 <xs:complexType name="PrecipitationIntensityValue">
2023   <xs:complexContent>
2024     <xs:extension base="D2LogicalModel:DataValue">
2025       <xs:sequence>
2026         <xs:element name="millimetresPerHourIntensity" type=
2027         ↪ "D2LogicalModel:IntensityMillimetresPerHour" minOccurs="1" maxOccurs="1" />
2028         <xs:element name="precipitationIntensityValueExtension" type=
2029         ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
2030       </xs:sequence>
2031     </xs:extension>
2032   </xs:complexContent>
2033 </xs:complexType>
2034 <xs:simpleType name="PrecipitationTypeEnum">
2035   <xs:restriction base="xs:string">
2036     <xs:enumeration value="drizzle" />
2037     <xs:enumeration value="freezingRain" />
2038     <xs:enumeration value="hail" />
2039     <xs:enumeration value="rain" />
2040     <xs:enumeration value="sleet" />
2041     <xs:enumeration value="snow" />
2042   </xs:restriction>
2043 </xs:simpleType>
2044 <xs:simpleType name="ProbabilityOfOccurrenceEnum">
2045   <xs:restriction base="xs:string">
2046     <xs:enumeration value="certain" />
2047     <xs:enumeration value="probable" />
2048     <xs:enumeration value="riskOf" />
2049   </xs:restriction>
2050 </xs:simpleType>
2051 <xs:complexType name="PublicEvent">
2052   <xs:complexContent>
2053     <xs:extension base="D2LogicalModel:Activity">
2054       <xs:sequence>
2055         <xs:element name="publicEventType" type="D2LogicalModel:PublicEventTypeEnum
2056         ↪ " minOccurs="1" maxOccurs="1" />
2057         <xs:element name="publicEventExtension" type="D2LogicalModel:_ExtensionType

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2051   </xs:sequence>
2052   </xs:extension>
2053   </xs:complexContent>
2054 </xs:complexType>
2055 <xs:simpleType name="PublicEventTypeEnum">
2056   <xs:restriction base="xs:string">
2057     <xs:enumeration value="agriculturalShow" />
2058     <xs:enumeration value="airShow" />
2059     <xs:enumeration value="athleticsMeeting" />
2060     <xs:enumeration value="commercialEvent" />
2061     <xs:enumeration value="culturalEvent" />
2062     <xs:enumeration value="ballGame" />
2063     <xs:enumeration value="baseballGame" />
2064     <xs:enumeration value="basketballGame" />
2065     <xs:enumeration value="bicycleRace" />
2066     <xs:enumeration value="boatRace" />
2067     <xs:enumeration value="boatShow" />
2068     <xs:enumeration value="boxingTournament" />
2069     <xs:enumeration value="bullFight" />
2070     <xs:enumeration value="ceremonialEvent" />
2071     <xs:enumeration value="concert" />
2072     <xs:enumeration value="cricketMatch" />
2073     <xs:enumeration value="exhibition" />
2074     <xs:enumeration value="fair" />
2075     <xs:enumeration value="festival" />
2076     <xs:enumeration value="filmTVMaking" />
2077     <xs:enumeration value="footballMatch" />
2078     <xs:enumeration value="funfair" />
2079     <xs:enumeration value="gardeningOrFlowerShow" />
2080     <xs:enumeration value="golfTournament" />
2081     <xs:enumeration value="hockeyGame" />
2082     <xs:enumeration value="horseRaceMeeting" />
2083     <xs:enumeration value="internationalSportsMeeting" />
2084     <xs:enumeration value="majorEvent" />
2085     <xs:enumeration value="marathon" />
2086     <xs:enumeration value="market" />
2087     <xs:enumeration value="match" />
2088     <xs:enumeration value="motorShow" />
2089     <xs:enumeration value="motorSportRaceMeeting" />
2090     <xs:enumeration value="parade" />
2091     <xs:enumeration value="procession" />
2092     <xs:enumeration value="raceMeeting" />
2093     <xs:enumeration value="rugbyMatch" />
2094     <xs:enumeration value="severalMajorEvents" />
2095     <xs:enumeration value="show" />
2096     <xs:enumeration value="showJumping" />
2097     <xs:enumeration value="sportsMeeting" />
2098     <xs:enumeration value="stateOccasion" />
2099     <xs:enumeration value="tennisTournament" />
2100     <xs:enumeration value="tournament" />
2101     <xs:enumeration value="tradeFair" />
2102     <xs:enumeration value="waterSportsMeeting" />
2103     <xs:enumeration value="winterSportsMeeting" />
2104     <xs:enumeration value="other" />
2105   </xs:restriction>

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2106 </xs:simpleType>
2107 <xs:complexType name="Referent">
2108   <xs:sequence>
2109     <xs:element name="referentIdentifier" type="D2LogicalModel:String" minOccurs="1"
↪ " maxOccurs="1" />
2110     <xs:element name="referentName" type="D2LogicalModel:String" minOccurs="0"
↪ maxOccurs="1" />
2111     <xs:element name="referentType" type="D2LogicalModel:ReferentTypeEnum"
↪ minOccurs="1" maxOccurs="1" />
2112     <xs:element name="referentDescription" type="D2LogicalModel:MultilingualString"
↪ minOccurs="0" maxOccurs="1" />
2113     <xs:element name="pointCoordinates" type="D2LogicalModel:PointCoordinates"
↪ minOccurs="0" />
2114     <xs:element name="referentExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
2115   </xs:sequence>
2116 </xs:complexType>
2117 <xs:simpleType name="ReferentTypeEnum">
2118   <xs:restriction base="xs:string">
2119     <xs:enumeration value="boundary" />
2120     <xs:enumeration value="intersection" />
2121     <xs:enumeration value="referenceMarker" />
2122     <xs:enumeration value="landmark" />
2123     <xs:enumeration value="roadNode" />
2124   </xs:restriction>
2125 </xs:simpleType>
2126 <xs:simpleType name="RelativeTrafficFlowEnum">
2127   <xs:restriction base="xs:string">
2128     <xs:enumeration value="trafficVeryMuchHeavierThanNormal" />
2129     <xs:enumeration value="trafficHeavierThanNormal" />
2130     <xs:enumeration value="trafficFlowNormal" />
2131     <xs:enumeration value="trafficLighterThanNormal" />
2132     <xs:enumeration value="trafficVeryMuchLighterThanNormal" />
2133   </xs:restriction>
2134 </xs:simpleType>
2135 <xs:complexType name="ReroutingManagement">
2136   <xs:complexContent>
2137     <xs:extension base="D2LogicalModel:NetworkManagement">
2138       <xs:sequence>
2139         <xs:element name="reroutingManagementType" type=
↪ "D2LogicalModel:ReroutingManagementTypeEnum" minOccurs="1" maxOccurs="unbounded" />
2140         <xs:element name="reroutingItineraryDescription" type=
↪ "D2LogicalModel:MultilingualString" minOccurs="0" maxOccurs="1" />
2141         <xs:element name="signedRerouting" type="D2LogicalModel:Boolean" minOccurs=
↪ "0" maxOccurs="1" />
2142         <xs:element name="entry" type="D2LogicalModel:String" minOccurs="0"
↪ maxOccurs="1" />
2143         <xs:element name="exit" type="D2LogicalModel:String" minOccurs="0"
↪ maxOccurs="1" />
2144         <xs:element name="roadOrJunctionNumber" type="D2LogicalModel:String"
↪ minOccurs="0" maxOccurs="1" />
2145         <xs:element name="reroutingManagementExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
2146       </xs:sequence>
2147     </xs:extension>
2148   </xs:complexContent>

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2149 </xs:complexType>
2150 <xs:simpleType name="ReroutingManagementTypeEnum">
2151   <xs:restriction base="xs:string">
2152     <xs:enumeration value="doNotFollowDiversionSigns" />
2153     <xs:enumeration value="doNotUseEntry" />
2154     <xs:enumeration value="doNotUseExit" />
2155     <xs:enumeration value="doNotUseIntersectionOrJunction" />
2156     <xs:enumeration value="followDiversionSigns" />
2157     <xs:enumeration value="followLocalDiversion" />
2158     <xs:enumeration value="followSpecialMarkers" />
2159     <xs:enumeration value="useEntry" />
2160     <xs:enumeration value="useExit" />
2161     <xs:enumeration value="useIntersectionOrJunction" />
2162   </xs:restriction>
2163 </xs:simpleType>
2164 <xs:complexType name="RoadConditions" abstract="true">
2165   <xs:complexContent>
2166     <xs:extension base="D2LogicalModel:Conditions">
2167       <xs:sequence>
2168         <xs:element name="roadConditionsExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
2169       </xs:sequence>
2170     </xs:extension>
2171   </xs:complexContent>
2172 </xs:complexType>
2173 <xs:simpleType name="RoadMaintenanceTypeEnum">
2174   <xs:restriction base="xs:string">
2175     <xs:enumeration value="clearanceWork" />
2176     <xs:enumeration value="controlledAvalanche" />
2177     <xs:enumeration value="installationWork" />
2178     <xs:enumeration value="grassCuttingWork" />
2179     <xs:enumeration value="litterClearance" />
2180     <xs:enumeration value="maintenanceWork" />
2181     <xs:enumeration value="overheadWorks" />
2182     <xs:enumeration value="repairWork" />
2183     <xs:enumeration value="resurfacingWork" />
2184     <xs:enumeration value="roadMarkingWork" />
2185     <xs:enumeration value="roadsideWork" />
2186     <xs:enumeration value="roadworksClearance" />
2187     <xs:enumeration value="roadworks" />
2188     <xs:enumeration value="rockFallPreventativeMaintenance" />
2189     <xs:enumeration value="saltingInProgress" />
2190     <xs:enumeration value="snowploughsInUse" />
2191     <xs:enumeration value="sweepingOfRoad" />
2192     <xs:enumeration value="treeAndVegetationCuttingWork" />
2193     <xs:enumeration value="other" />
2194   </xs:restriction>
2195 </xs:simpleType>
2196 <xs:complexType name="RoadOperatorServiceDisruption">
2197   <xs:complexContent>
2198     <xs:extension base="D2LogicalModel:NonRoadEventInformation">
2199       <xs:sequence>
2200         <xs:element name="roadOperatorServiceDisruptionType" type=
↪ "D2LogicalModel:RoadOperatorServiceDisruptionTypeEnum" minOccurs="1" maxOccurs=
↪ "unbounded" />
2201         <xs:element name="roadOperatorServiceDisruptionExtension" type=

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2202     </xs:sequence>
2203   </xs:extension>
2204 </xs:complexContent>
2205 </xs:complexType>
2206 <xs:simpleType name="RoadOperatorServiceDisruptionTypeEnum">
2207   <xs:restriction base="xs:string">
2208     <xs:enumeration value="emergencyTelephoneNumberOutOfService" />
2209     <xs:enumeration value="informationServiceTelephoneNumberOutOfService" />
2210     <xs:enumeration value="noTrafficOfficerPatrolService" />
2211   </xs:restriction>
2212 </xs:simpleType>
2213 <xs:complexType name="RoadOrCarriagewayOrLaneManagement">
2214   <xs:complexContent>
2215     <xs:extension base="D2LogicalModel:NetworkManagement">
2216       <xs:sequence>
2217         <xs:element name="roadOrCarriagewayOrLaneManagementType" type=
2218           <xs:element name="minimumCarOccupancy" type=
2219             <xs:element name="roadOrCarriagewayOrLaneManagementExtension" type=
2220               </xs:sequence>
2221             </xs:extension>
2222           </xs:complexContent>
2223         </xs:complexType>
2224         <xs:simpleType name="RoadOrCarriagewayOrLaneManagementTypeEnum">
2225           <xs:restriction base="xs:string">
2226             <xs:enumeration value="carPoolLaneInOperation" />
2227             <xs:enumeration value="carriagewayClosures" />
2228             <xs:enumeration value="clearALaneForEmergencyVehicles" />
2229             <xs:enumeration value="clearALaneForSnowploughsAndGrittingVehicles" />
2230             <xs:enumeration value="closedPermanentlyForTheWinter" />
2231             <xs:enumeration value="contraflow" />
2232             <xs:enumeration value="doNotUseSpecifiedLanesOrCarriageways" />
2233             <xs:enumeration value="hardShoulderRunningInOperation" />
2234             <xs:enumeration value="heightRestrictionInOperation" />
2235             <xs:enumeration value="intermittentShortTermClosures" />
2236             <xs:enumeration value="keepToTheLeft" />
2237             <xs:enumeration value="keepToTheRight" />
2238             <xs:enumeration value="laneClosures" />
2239             <xs:enumeration value="lanesDeviated" />
2240             <xs:enumeration value="narrowLanes" />
2241             <xs:enumeration value="newRoadworksLayout" />
2242             <xs:enumeration value="overnightClosures" />
2243             <xs:enumeration value="roadCleared" />
2244             <xs:enumeration value="roadClosed" />
2245             <xs:enumeration value="rollingRoadBlock" />
2246             <xs:enumeration value="rushHourLaneInOperation" />
2247             <xs:enumeration value="singleAlternateLineTraffic" />
2248             <xs:enumeration value="tidalFlowLaneInOperation" />
2249             <xs:enumeration value="turnAroundInOperation" />
2250             <xs:enumeration value="useOfSpecifiedLanesOrCarriagewaysAllowed" />
2251             <xs:enumeration value="useSpecifiedLanesOrCarriageways" />
2252             <xs:enumeration value="vehicleStorageInOperation" />

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2253     <xs:enumeration value="weightRestrictionInOperation" />
2254     <xs:enumeration value="other" />
2255   </xs:restriction>
2256 </xs:simpleType>
2257 <xs:complexType name="RoadsideAssistance">
2258   <xs:complexContent>
2259     <xs:extension base="D2LogicalModel:OperatorAction">
2260       <xs:sequence>
2261         <xs:element name="roadsideAssistanceType" type=
2262 ↪ "D2LogicalModel:RoadsideAssistanceTypeEnum" minOccurs="1" maxOccurs="1" />
2263         <xs:element name="roadsideAssistanceExtension" type="D2LogicalModel:_
2264 ↪ ExtensionType" minOccurs="0" />
2265       </xs:sequence>
2266     </xs:extension>
2267   </xs:complexContent>
2268 </xs:complexType>
2269 <xs:simpleType name="RoadsideAssistanceTypeEnum">
2270   <xs:restriction base="xs:string">
2271     <xs:enumeration value="airAmbulance" />
2272     <xs:enumeration value="busPassengerAssistance" />
2273     <xs:enumeration value="emergencyServices" />
2274     <xs:enumeration value="firstAid" />
2275     <xs:enumeration value="foodDelivery" />
2276     <xs:enumeration value="helicopterRescue" />
2277     <xs:enumeration value="vehicleRepair" />
2278     <xs:enumeration value="vehicleRecovery" />
2279     <xs:enumeration value="other" />
2280   </xs:restriction>
2281 </xs:simpleType>
2282 <xs:complexType name="RoadsideServiceDisruption">
2283   <xs:complexContent>
2284     <xs:extension base="D2LogicalModel:NonRoadEventInformation">
2285       <xs:sequence>
2286         <xs:element name="roadsideServiceDisruptionType" type=
2287 ↪ "D2LogicalModel:RoadsideServiceDisruptionTypeEnum" minOccurs="1" maxOccurs=
2288 ↪ "unbounded" />
2289         <xs:element name="roadsideServiceDisruptionExtension" type="D2LogicalModel:_
2290 ↪ ExtensionType" minOccurs="0" />
2291       </xs:sequence>
2292     </xs:extension>
2293   </xs:complexContent>
2294 </xs:complexType>
2295 <xs:simpleType name="RoadsideServiceDisruptionTypeEnum">
2296   <xs:restriction base="xs:string">
2297     <xs:enumeration value="barClosed" />
2298     <xs:enumeration value="dieselShortage" />
2299     <xs:enumeration value="fuelShortage" />
2300     <xs:enumeration value="lpgShortage" />
2301     <xs:enumeration value="methaneShortage" />
2302     <xs:enumeration value="noDieselForHeavyVehicles" />
2303     <xs:enumeration value="noDieselForLightVehicles" />
2304     <xs:enumeration value="noPublicTelephones" />
2305     <xs:enumeration value="noToiletFacilities" />
2306     <xs:enumeration value="noVehicleRepairFacilities" />
2307     <xs:enumeration value="petrolShortage" />
2308     <xs:enumeration value="restAreaBusy" />

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2304     <xs:enumeration value="restAreaClosed" />
2305     <xs:enumeration value="restAreaOvercrowdedDriveToAnotherRestArea" />
2306     <xs:enumeration value="serviceAreaBusy" />
2307     <xs:enumeration value="serviceAreaClosed" />
2308     <xs:enumeration value="serviceAreaFuelStationClosed" />
2309     <xs:enumeration value="serviceAreaOvercrowdedDriveToAnotherServiceArea" />
2310     <xs:enumeration value="serviceAreaRestaurantClosed" />
2311     <xs:enumeration value="someCommercialServicesClosed" />
2312     <xs:enumeration value="waterShortage" />
2313   </xs:restriction>
2314 </xs:simpleType>
2315 <xs:complexType name="RoadSurfaceConditionMeasurements">
2316   <xs:sequence>
2317     <xs:element name="roadSurfaceTemperature" type="D2LogicalModel:TemperatureValue
2318 ↪ " minOccurs="0" />
2319     <xs:element name="protectionTemperature" type="D2LogicalModel:TemperatureValue"
2320 ↪ minOccurs="0" />
2321     <xs:element name="deIcingApplicationRate" type=
2322 ↪ "D2LogicalModel:ApplicationRateValue" minOccurs="0" />
2323     <xs:element name="deIcingConcentration" type=
2324 ↪ "D2LogicalModel:KilogramsConcentrationValue" minOccurs="0" />
2325     <xs:element name="depthOfSnow" type=
2326 ↪ "D2LogicalModel:FloatingPointMetreDistanceValue" minOccurs="0" />
2327     <xs:element name="waterFilmThickness" type=
2328 ↪ "D2LogicalModel:FloatingPointMetreDistanceValue" minOccurs="0" />
2329     <xs:element name="roadSurfaceConditionMeasurementsExtension" type=
2330 ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
2331   </xs:sequence>
2332 </xs:complexType>
2333 <xs:complexType name="Roadworks" abstract="true">
2334   <xs:complexContent>
2335     <xs:extension base="D2LogicalModel:OperatorAction">
2336       <xs:sequence>
2337         <xs:element name="roadworksDuration" type=
2338 ↪ "D2LogicalModel:RoadworksDurationEnum" minOccurs="0" maxOccurs="1" />
2339         <xs:element name="roadworksScale" type="D2LogicalModel:RoadworksScaleEnum"
2340 ↪ minOccurs="0" maxOccurs="1" />
2341         <xs:element name="underTraffic" type="D2LogicalModel:Boolean" minOccurs="0"
2342 ↪ maxOccurs="1" />
2343         <xs:element name="urgentRoadworks" type="D2LogicalModel:Boolean" minOccurs=
2344 ↪ "0" maxOccurs="1" />
2345         <xs:element name="mobility" type="D2LogicalModel:Mobility" minOccurs="0" />
2346         <xs:element name="subjects" type="D2LogicalModel:Subjects" minOccurs="0" />
2347         <xs:element name="maintenanceVehicles" type=
2348 ↪ "D2LogicalModel:MaintenanceVehicles" minOccurs="0" />
2349         <xs:element name="roadworksExtension" type="D2LogicalModel:_ExtensionType"
2350 ↪ minOccurs="0" />
2351       </xs:sequence>
2352     </xs:extension>
2353   </xs:complexContent>
2354 </xs:complexType>
2355 <xs:simpleType name="RoadworksDurationEnum">
2356   <xs:restriction base="xs:string">
2357     <xs:enumeration value="longTerm" />
2358     <xs:enumeration value="mediumTerm" />
2359     <xs:enumeration value="shortTerm" />

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2347     </xs:restriction>
2348 </xs:simpleType>
2349 <xs:simpleType name="RoadworksScaleEnum">
2350     <xs:restriction base="xs:string">
2351         <xs:enumeration value="major" />
2352         <xs:enumeration value="medium" />
2353         <xs:enumeration value="minor" />
2354     </xs:restriction>
2355 </xs:simpleType>
2356 <xs:simpleType name="Seconds">
2357     <xs:restriction base="D2LogicalModel:Float" />
2358 </xs:simpleType>
2359 <xs:complexType name="Situation">
2360     <xs:sequence>
2361         <xs:element name="situationVersionTime" type="D2LogicalModel:DateTime"
2362         ↪ minOccurs="0" maxOccurs="1" />
2363         <xs:element name="headerInformation" type="D2LogicalModel:HeaderInformation" />
2364         <xs:element name="situationRecord" type="D2LogicalModel:SituationRecord"
2365         ↪ maxOccurs="unbounded" />
2366         <xs:element name="situationExtension" type="D2LogicalModel:_ExtensionType"
2367         ↪ minOccurs="0" />
2368     </xs:sequence>
2369     <xs:attribute name="id" type="xs:string" use="required" />
2370     <xs:attribute name="version" type="xs:string" use="required" />
2371 </xs:complexType>
2372 <xs:complexType name="SituationPublication">
2373     <xs:complexContent>
2374         <xs:extension base="D2LogicalModel:PayloadPublication">
2375             <xs:sequence>
2376                 <xs:element name="situation" type="D2LogicalModel:Situation" minOccurs="0"
2377                 ↪ maxOccurs="unbounded" />
2378                 <xs:element name="situationPublicationExtension" type="D2LogicalModel:_
2379                 ↪ ExtensionType" minOccurs="0" />
2380             </xs:sequence>
2381         </xs:extension>
2382     </xs:complexContent>
2383 </xs:complexType>
2384 <xs:complexType name="SituationRecord" abstract="true">
2385     <xs:sequence>
2386         <xs:element name="situationRecordCreationTime" type="D2LogicalModel:DateTime"
2387         ↪ minOccurs="1" maxOccurs="1" />
2388         <xs:element name="situationRecordVersionTime" type="D2LogicalModel:DateTime"
2389         ↪ minOccurs="1" maxOccurs="1" />
2390         <xs:element name="probabilityOfOccurrence" type=
2391         ↪ "D2LogicalModel:ProbabilityOfOccurrenceEnum" minOccurs="1" maxOccurs="1" />
2392         <xs:element name="source" type="D2LogicalModel:Source" minOccurs="0" />
2393         <xs:element name="validity" type="D2LogicalModel:Validity" />
2394         <xs:element name="impact" type="D2LogicalModel:Impact" minOccurs="0" />
2395         <xs:element name="cause" type="D2LogicalModel:Cause" minOccurs="0" />
2396         <xs:element name="generalPublicComment" type="D2LogicalModel:Comment" minOccurs=
2397         ↪ "0" maxOccurs="unbounded" />
2398         <xs:element name="groupOfLocations" type="D2LogicalModel:GroupOfLocations" />
2399         <xs:element name="situationRecordExtension" type="D2LogicalModel:_ExtensionType
2400         ↪ " minOccurs="0" />
2401     </xs:sequence>
2402     <xs:attribute name="id" type="xs:string" use="required" />

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2393     <xs:attribute name="version" type="xs:string" use="required" />
2394   </xs:complexType>
2395   <xs:complexType name="SJTSKPointCoordinates">
2396     <xs:sequence>
2397       <xs:element name="sjtskX" type="D2LogicalModel:Float" minOccurs="1" maxOccurs="1"
2398 ↪ " />
2399       <xs:element name="sjtskY" type="D2LogicalModel:Float" minOccurs="1" maxOccurs="1"
2400 ↪ " />
2401       <xs:element name="sjtskPointCoordinatesExtension" type="D2LogicalModel:_
2402 ↪ ExtensionType" minOccurs="0" />
2403     </xs:sequence>
2404   </xs:complexType>
2405   <xs:complexType name="Source">
2406     <xs:sequence>
2407       <xs:element name="sourceCountry" type="D2LogicalModel:CountryEnum" minOccurs="0
2408 ↪ " maxOccurs="1" />
2409       <xs:element name="sourceIdentification" type="D2LogicalModel:String" minOccurs=
2410 ↪ "0" maxOccurs="1" />
2411       <xs:element name="sourceName" type="D2LogicalModel:MultilingualString"
2412 ↪ minOccurs="0" maxOccurs="1" />
2413       <xs:element name="sourceType" type="D2LogicalModel:SourceTypeEnum" minOccurs="0
2414 ↪ " maxOccurs="1" />
2415       <xs:element name="reliable" type="D2LogicalModel:Boolean" minOccurs="0"
2416 ↪ maxOccurs="1" />
2417       <xs:element name="sourceExtension" type="D2LogicalModel:_ExtensionType"
2418 ↪ minOccurs="0" />
2419     </xs:sequence>
2420   </xs:complexType>
2421   <xs:simpleType name="SourceTypeEnum">
2422     <xs:restriction base="xs:string">
2423       <xs:enumeration value="automobileClubPatrol" />
2424       <xs:enumeration value="cameraObservation" />
2425       <xs:enumeration value="freightVehicleOperator" />
2426       <xs:enumeration value="inductionLoopMonitoringStation" />
2427       <xs:enumeration value="infraredMonitoringStation" />
2428       <xs:enumeration value="microwaveMonitoringStation" />
2429       <xs:enumeration value="mobileTelephoneCaller" />
2430       <xs:enumeration value="nonPoliceEmergencyServicePatrol" />
2431       <xs:enumeration value="otherInformation" />
2432       <xs:enumeration value="otherOfficialVehicle" />
2433       <xs:enumeration value="policePatrol" />
2434       <xs:enumeration value="privateBreakdownService" />
2435       <xs:enumeration value="publicAndPrivateUtilities" />
2436       <xs:enumeration value="registeredMotoristObserver" />
2437       <xs:enumeration value="roadAuthorities" />
2438       <xs:enumeration value="roadOperatorPatrol" />
2439       <xs:enumeration value="roadsideTelephoneCaller" />
2440       <xs:enumeration value="spotterAircraft" />
2441       <xs:enumeration value="trafficMonitoringStation" />
2442       <xs:enumeration value="transitOperator" />
2443       <xs:enumeration value="vehicleProbeMeasurement" />
2444       <xs:enumeration value="videoProcessingMonitoringStation" />
2445     </xs:restriction>
2446   </xs:simpleType>
2447   <xs:complexType name="SpeedManagement">
2448     <xs:complexContent>

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2440     <xs:extension base="D2LogicalModel:NetworkManagement">
2441       <xs:sequence>
2442         <xs:element name="speedManagementType" type=
↪ "D2LogicalModel:SpeedManagementTypeEnum" minOccurs="0" maxOccurs="1" />
2443         <xs:element name="temporarySpeedLimit" type=
↪ "D2LogicalModel:KilometresPerHour" minOccurs="0" maxOccurs="1" />
2444         <xs:element name="speedManagementExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
2445       </xs:sequence>
2446     </xs:extension>
2447   </xs:complexContent>
2448 </xs:complexType>
2449 <xs:simpleType name="SpeedManagementTypeEnum">
2450   <xs:restriction base="xs:string">
2451     <xs:enumeration value="activeSpeedControlInOperation" />
2452     <xs:enumeration value="doNotSlowdownUnnecessarily" />
2453     <xs:enumeration value="observeSpeedLimit" />
2454     <xs:enumeration value="policeSpeedChecksInOperation" />
2455     <xs:enumeration value="reduceYourSpeed" />
2456     <xs:enumeration value="speedRestrictionInOperation" />
2457     <xs:enumeration value="other" />
2458   </xs:restriction>
2459 </xs:simpleType>
2460 <xs:complexType name="SpeedValue">
2461   <xs:complexContent>
2462     <xs:extension base="D2LogicalModel:DataValue">
2463       <xs:sequence>
2464         <xs:element name="speed" type="D2LogicalModel:KilometresPerHour" minOccurs=
↪ "1" maxOccurs="1" />
2465         <xs:element name="speedValueExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
2466       </xs:sequence>
2467     </xs:extension>
2468   </xs:complexContent>
2469 </xs:complexType>
2470 <xs:simpleType name="String">
2471   <xs:restriction base="xs:string">
2472     <xs:maxLength value="1024" />
2473   </xs:restriction>
2474 </xs:simpleType>
2475 <xs:complexType name="Subjects">
2476   <xs:sequence>
2477     <xs:element name="subjectTypeOfWorks" type=
↪ "D2LogicalModel:SubjectTypeOfWorksEnum" minOccurs="1" maxOccurs="1" />
2478     <xs:element name="numberOfSubjects" type="D2LogicalModel:NonNegativeInteger"
↪ minOccurs="0" maxOccurs="1" />
2479     <xs:element name="subjectsExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
2480   </xs:sequence>
2481 </xs:complexType>
2482 <xs:simpleType name="SubjectTypeOfWorksEnum">
2483   <xs:restriction base="xs:string">
2484     <xs:enumeration value="bridge" />
2485     <xs:enumeration value="buriedCables" />
2486     <xs:enumeration value="buriedServices" />
2487     <xs:enumeration value="crashBarrier" />

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2488     <xs:enumeration value="gallery" />
2489     <xs:enumeration value="gantry" />
2490     <xs:enumeration value="gasMainWork" />
2491     <xs:enumeration value="interchange" />
2492     <xs:enumeration value="junction" />
2493     <xs:enumeration value="levelCrossing" />
2494     <xs:enumeration value="lightingSystem" />
2495     <xs:enumeration value="measurementEquipment" />
2496     <xs:enumeration value="noiseProtection" />
2497     <xs:enumeration value="road" />
2498     <xs:enumeration value="roadsideDrains" />
2499     <xs:enumeration value="roadsideEmbankment" />
2500     <xs:enumeration value="roadsideEquipment" />
2501     <xs:enumeration value="roadSigns" />
2502     <xs:enumeration value="roundabout" />
2503     <xs:enumeration value="tollGate" />
2504     <xs:enumeration value="tunnel" />
2505     <xs:enumeration value="waterMain" />
2506     <xs:enumeration value="other" />
2507   </xs:restriction>
2508 </xs:simpleType>
2509 <xs:complexType name="SupplementaryPositionalDescription">
2510   <xs:sequence>
2511     <xs:element name="locationDescriptor" type=
2512 ↪ "D2LogicalModel:LocationDescriptorEnum" minOccurs="0" maxOccurs="unbounded" />
2513     <xs:element name="sequentialRampNumber" type="D2LogicalModel:NonNegativeInteger
2514 ↪ " minOccurs="0" maxOccurs="1" />
2515     <xs:element name="affectedCarriagewayAndLanes" type=
2516 ↪ "D2LogicalModel:AffectedCarriagewayAndLanes" minOccurs="0" maxOccurs="unbounded" />
2517     <xs:element name="supplementaryPositionalDescriptionExtension" type=
2518 ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
2519   </xs:sequence>
2520   <xs:attribute name="locationPrecision" type=
2521 ↪ "D2LogicalModel:MetresAsNonNegativeInteger" use="optional" />
2522 </xs:complexType>
2523 <xs:complexType name="Temperature">
2524   <xs:sequence>
2525     <xs:element name="airTemperature" type="D2LogicalModel:TemperatureValue" ↪
2526 ↪ minOccurs="0" />
2527     <xs:element name="dewPointTemperature" type="D2LogicalModel:TemperatureValue" ↪
2528 ↪ minOccurs="0" />
2529     <xs:element name="maximumTemperature" type="D2LogicalModel:TemperatureValue" ↪
2530 ↪ minOccurs="0" />
2531     <xs:element name="minimumTemperature" type="D2LogicalModel:TemperatureValue" ↪
2532 ↪ minOccurs="0" />
2533     <xs:element name="temperatureExtension" type="D2LogicalModel:_ExtensionType" ↪
2534 ↪ minOccurs="0" />
2535   </xs:sequence>
2536 </xs:complexType>
2537 <xs:simpleType name="TemperatureCelsius">
2538   <xs:restriction base="D2LogicalModel:Float" />
2539 </xs:simpleType>
2540 <xs:complexType name="TemperatureValue">
2541   <xs:complexContent>
2542     <xs:extension base="D2LogicalModel:DataValue">
2543       <xs:sequence>

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2534     <xs:element name="temperature" type="D2LogicalModel:TemperatureCelsius"
↪ minOccurs="1" maxOccurs="1" />
2535     <xs:element name="temperatureValueExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
2536   </xs:sequence>
2537 </xs:extension>
2538 </xs:complexContent>
2539 </xs:complexType>
2540 <xs:simpleType name="Tonnes">
2541   <xs:restriction base="D2LogicalModel:Float" />
2542 </xs:simpleType>
2543 <xs:complexType name="TpegDescriptor" abstract="true">
2544   <xs:sequence>
2545     <xs:element name="descriptor" type="D2LogicalModel:MultilingualString"
↪ minOccurs="1" maxOccurs="1" />
2546     <xs:element name="tpegDescriptorExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
2547   </xs:sequence>
2548 </xs:complexType>
2549 <xs:complexType name="TpegIlcPointDescriptor">
2550   <xs:complexContent>
2551     <xs:extension base="D2LogicalModel:TpegPointDescriptor">
2552       <xs:sequence>
2553         <xs:element name="tpegIlcPointDescriptorType" type=
↪ "D2LogicalModel:TpegLoc03IlcPointDescriptorSubtypeEnum" minOccurs="1" maxOccurs="1"
↪ />
2554         <xs:element name="tpegIlcPointDescriptorExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
2555       </xs:sequence>
2556     </xs:extension>
2557   </xs:complexContent>
2558 </xs:complexType>
2559 <xs:complexType name="TpegJunctionPointDescriptor">
2560   <xs:complexContent>
2561     <xs:extension base="D2LogicalModel:TpegPointDescriptor">
2562       <xs:sequence>
2563         <xs:element name="tpegJunctionPointDescriptorType" type=
↪ "D2LogicalModel:TpegLoc03JunctionPointDescriptorSubtypeEnum" minOccurs="1"
↪ maxOccurs="1" />
2564         <xs:element name="tpegJunctionPointDescriptorExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
2565       </xs:sequence>
2566     </xs:extension>
2567   </xs:complexContent>
2568 </xs:complexType>
2569 <xs:simpleType name="TpegLoc03IlcPointDescriptorSubtypeEnum">
2570   <xs:restriction base="xs:string">
2571     <xs:enumeration value="tpegIlcName1" />
2572     <xs:enumeration value="tpegIlcName2" />
2573     <xs:enumeration value="tpegIlcName3" />
2574   </xs:restriction>
2575 </xs:simpleType>
2576 <xs:simpleType name="TpegLoc03JunctionPointDescriptorSubtypeEnum">
2577   <xs:restriction base="xs:string">
2578     <xs:enumeration value="junctionName" />
2579   </xs:restriction>

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2580 </xs:simpleType>
2581 <xs:simpleType name="TpegLoc030therPointDescriptorSubtypeEnum">
2582   <xs:restriction base="xs:string">
2583     <xs:enumeration value="administrativeAreaName" />
2584     <xs:enumeration value="administrativeReferenceName" />
2585     <xs:enumeration value="airportName" />
2586     <xs:enumeration value="areaName" />
2587     <xs:enumeration value="buildingName" />
2588     <xs:enumeration value="busStopIdentifier" />
2589     <xs:enumeration value="busStopName" />
2590     <xs:enumeration value="canalName" />
2591     <xs:enumeration value="countyName" />
2592     <xs:enumeration value="ferryPortName" />
2593     <xs:enumeration value="intersectionName" />
2594     <xs:enumeration value="lakeName" />
2595     <xs:enumeration value="linkName" />
2596     <xs:enumeration value="localLinkName" />
2597     <xs:enumeration value="metroStationName" />
2598     <xs:enumeration value="nationName" />
2599     <xs:enumeration value="nonLinkedPointName" />
2600     <xs:enumeration value="parkingFacilityName" />
2601     <xs:enumeration value="pointName" />
2602     <xs:enumeration value="pointOfInterestName" />
2603     <xs:enumeration value="railwayStation" />
2604     <xs:enumeration value="regionName" />
2605     <xs:enumeration value="riverName" />
2606     <xs:enumeration value="seaName" />
2607     <xs:enumeration value="serviceAreaName" />
2608     <xs:enumeration value="tidalRiverName" />
2609     <xs:enumeration value="townName" />
2610     <xs:enumeration value="other" />
2611   </xs:restriction>
2612 </xs:simpleType>
2613 <xs:complexType name="TpegOtherPointDescriptor">
2614   <xs:complexContent>
2615     <xs:extension base="D2LogicalModel:TpegPointDescriptor">
2616       <xs:sequence>
2617         <xs:element name="tpegOtherPointDescriptorType" type=
2618 ↪ "D2LogicalModel:TpegLoc030therPointDescriptorSubtypeEnum" minOccurs="1" maxOccurs="1"
2619 ↪ />
2620         <xs:element name="tpegOtherPointDescriptorExtension" type="D2LogicalModel:_
2621 ↪ ExtensionType" minOccurs="0" />
2622       </xs:sequence>
2623     </xs:extension>
2624   </xs:complexContent>
2625 </xs:complexType>
2626 <xs:complexType name="TpegPointDescriptor" abstract="true">
2627   <xs:complexContent>
2628     <xs:extension base="D2LogicalModel:TpegDescriptor">
2629       <xs:sequence>
2630         <xs:element name="tpegPointDescriptorExtension" type="D2LogicalModel:_
2631 ↪ ExtensionType" minOccurs="0" />
2632       </xs:sequence>
2633     </xs:extension>
2634   </xs:complexContent>
2635 </xs:complexType>

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2632 <xs:simpleType name="TrafficConstrictionTypeEnum">
2633   <xs:restriction base="xs:string">
2634     <xs:enumeration value="carriagewayBlocked" />
2635     <xs:enumeration value="carriagewayPartiallyObstructed" />
2636     <xs:enumeration value="lanesBlocked" />
2637     <xs:enumeration value="lanesPartiallyObstructed" />
2638     <xs:enumeration value="roadBlocked" />
2639     <xs:enumeration value="roadPartiallyObstructed" />
2640   </xs:restriction>
2641 </xs:simpleType>
2642 <xs:complexType name="TrafficElement" abstract="true">
2643   <xs:complexContent>
2644     <xs:extension base="D2LogicalModel:SituationRecord">
2645       <xs:sequence>
2646         <xs:element name="trafficElementExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
2647       </xs:sequence>
2648     </xs:extension>
2649   </xs:complexContent>
2650 </xs:complexType>
2651 <xs:simpleType name="TrafficStatusEnum">
2652   <xs:restriction base="xs:string">
2653     <xs:enumeration value="impossible" />
2654     <xs:enumeration value="congested" />
2655     <xs:enumeration value="heavy" />
2656     <xs:enumeration value="freeFlow" />
2657     <xs:enumeration value="unknown" />
2658   </xs:restriction>
2659 </xs:simpleType>
2660 <xs:complexType name="TrafficStatusValue">
2661   <xs:complexContent>
2662     <xs:extension base="D2LogicalModel:DataValue">
2663       <xs:sequence>
2664         <xs:element name="trafficStatusValue" type="D2LogicalModel:TrafficStatusEnum
↪ " minOccurs="1" maxOccurs="1" />
2665         <xs:element name="trafficStatusValueExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
2666       </xs:sequence>
2667     </xs:extension>
2668   </xs:complexContent>
2669 </xs:complexType>
2670 <xs:simpleType name="TrafficTrendTypeEnum">
2671   <xs:restriction base="xs:string">
2672     <xs:enumeration value="trafficBuildingUp" />
2673     <xs:enumeration value="trafficEasing" />
2674     <xs:enumeration value="trafficStable" />
2675     <xs:enumeration value="unknown" />
2676   </xs:restriction>
2677 </xs:simpleType>
2678 <xs:simpleType name="TrafficTypeEnum">
2679   <xs:restriction base="xs:string">
2680     <xs:enumeration value="accessOnlyTraffic" />
2681     <xs:enumeration value="destinedForAirport" />
2682     <xs:enumeration value="destinedForAirportArrivals" />
2683     <xs:enumeration value="destinedForAirportDepartures" />
2684     <xs:enumeration value="destinedForFerryService" />

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2685     <xs:enumeration value="destinedForRailService" />
2686     <xs:enumeration value="holidayTraffic" />
2687     <xs:enumeration value="localTraffic" />
2688     <xs:enumeration value="longDistanceTraffic" />
2689     <xs:enumeration value="regionalTraffic" />
2690     <xs:enumeration value="residentsOnlyTraffic" />
2691     <xs:enumeration value="throughTraffic" />
2692     <xs:enumeration value="visitorTraffic" />
2693 </xs:restriction>
2694 </xs:simpleType>
2695 <xs:complexType name="TransitInformation">
2696   <xs:complexContent>
2697     <xs:extension base="D2LogicalModel:NonRoadEventInformation">
2698       <xs:sequence>
2699         <xs:element name="transitServiceInformation" type=
2700 ↪ "D2LogicalModel:TransitServiceInformationEnum" minOccurs="1" maxOccurs="1" />
2701         <xs:element name="transitServiceType" type=
2702 ↪ "D2LogicalModel:TransitServiceTypeEnum" minOccurs="1" maxOccurs="1" />
2703         <xs:element name="transitInformationExtension" type="D2LogicalModel:_
2704 ↪ ExtensionType" minOccurs="0" />
2705       </xs:sequence>
2706     </xs:extension>
2707   </xs:complexContent>
2708 </xs:complexType>
2709 <xs:simpleType name="TransitServiceInformationEnum">
2710   <xs:restriction base="xs:string">
2711     <xs:enumeration value="cancellations" />
2712     <xs:enumeration value="delayDueToBadWeather" />
2713     <xs:enumeration value="delayDueToRepairs" />
2714     <xs:enumeration value="delayedUntilFurtherNotice" />
2715     <xs:enumeration value="delaysDueToFlotsam" />
2716     <xs:enumeration value="departureOnSchedule" />
2717     <xs:enumeration value="ferryReplacedByIceRoad" />
2718     <xs:enumeration value="freeShuttleServiceOperating" />
2719     <xs:enumeration value="informationServiceNotAvailable" />
2720     <xs:enumeration value="irregularServiceDelays" />
2721     <xs:enumeration value="loadCapacityChanged" />
2722     <xs:enumeration value="restrictionsForLongerVehicles" />
2723     <xs:enumeration value="serviceDelays" />
2724     <xs:enumeration value="serviceDelaysOfUncertainDuration" />
2725     <xs:enumeration value="serviceFullyBooked" />
2726     <xs:enumeration value="serviceNotOperating" />
2727     <xs:enumeration value="serviceNotOperatingSubstituteServiceAvailable" />
2728     <xs:enumeration value="serviceSuspended" />
2729     <xs:enumeration value="serviceWithdrawn" />
2730     <xs:enumeration value="shuttleServiceOperating" />
2731     <xs:enumeration value="temporaryChangesToTimetables" />
2732     <xs:enumeration value="other" />
2733   </xs:restriction>
2734 </xs:simpleType>
2735 <xs:simpleType name="TransitServiceTypeEnum">
2736   <xs:restriction base="xs:string">
2737     <xs:enumeration value="air" />
2738     <xs:enumeration value="bus" />
2739     <xs:enumeration value="ferry" />
2740     <xs:enumeration value="hydrofoil" />

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2738     <xs:enumeration value="rail" />
2739     <xs:enumeration value="tram" />
2740     <xs:enumeration value="undergroundMetro" />
2741   </xs:restriction>
2742 </xs:simpleType>
2743 <xs:simpleType name="UrgencyEnum">
2744   <xs:restriction base="xs:string">
2745     <xs:enumeration value="extremelyUrgent" />
2746     <xs:enumeration value="urgent" />
2747     <xs:enumeration value="normalUrgency" />
2748   </xs:restriction>
2749 </xs:simpleType>
2750 <xs:complexType name="Validity">
2751   <xs:sequence>
2752     <xs:element name="validityStatus" type="D2LogicalModel:ValidityStatusEnum"
2753     ↪ minOccurs="1" maxOccurs="1" />
2754     <xs:element name="validityTimeSpecification" type="D2LogicalModel:OverallPeriod
2755     ↪ " />
2756     <xs:element name="validityExtension" type="D2LogicalModel:_ExtensionType"
2757     ↪ minOccurs="0" />
2758   </xs:sequence>
2759 </xs:complexType>
2760 <xs:simpleType name="ValidityStatusEnum">
2761   <xs:restriction base="xs:string">
2762     <xs:enumeration value="active" />
2763     <xs:enumeration value="suspended" />
2764     <xs:enumeration value="definedByValidityTimeSpec" />
2765   </xs:restriction>
2766 </xs:simpleType>
2767 <xs:complexType name="Vehicle">
2768   <xs:sequence>
2769     <xs:element name="vehicleColour" type="D2LogicalModel:MultilingualString"
2770     ↪ minOccurs="0" maxOccurs="1" />
2771     <xs:element name="vehicleCountryOfOrigin" type=
2772     ↪ "D2LogicalModel:MultilingualString" minOccurs="0" maxOccurs="1" />
2773     <xs:element name="vehicleIdentifier" type="D2LogicalModel:String" minOccurs="0"
2774     ↪ maxOccurs="1" />
2775     <xs:element name="vehicleManufacturer" type="D2LogicalModel:String" minOccurs="0"
2776     ↪ maxOccurs="1" />
2777     <xs:element name="vehicleModel" type="D2LogicalModel:String" minOccurs="0"
2778     ↪ maxOccurs="1" />
2779     <xs:element name="vehicleRegistrationPlateIdentifier" type=
2780     ↪ "D2LogicalModel:String" minOccurs="0" maxOccurs="1" />
2781     <xs:element name="vehicleStatus" type="D2LogicalModel:VehicleStatusEnum"
2782     ↪ minOccurs="0" maxOccurs="1" />
2783     <xs:element name="vehicleCharacteristics" type=
2784     ↪ "D2LogicalModel:VehicleCharacteristics" minOccurs="0" />
2785     <xs:element name="axleSpacingOnVehicle" type="D2LogicalModel:AxleSpacing"
2786     ↪ minOccurs="0" maxOccurs="unbounded" />
2787     <xs:element name="specificAxleWeight" type="D2LogicalModel:AxleWeight"
2788     ↪ minOccurs="0" maxOccurs="unbounded" />
2789     <xs:element name="hazardousGoodsAssociatedWithVehicle" type=
2790     ↪ "D2LogicalModel:HazardousMaterials" minOccurs="0" />
2791     <xs:element name="vehicleExtension" type="D2LogicalModel:_ExtensionType"
2792     ↪ minOccurs="0" />
2793   </xs:sequence>

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2779 </xs:complexType>
2780 <xs:complexType name="VehicleCharacteristics">
2781   <xs:sequence>
2782     <xs:element name="loadType" type="D2LogicalModel:LoadTypeEnum" minOccurs="0"
↪ maxOccurs="1" />
2783     <xs:element name="vehicleType" type="D2LogicalModel:VehicleTypeEnum" minOccurs=
↪ "0" maxOccurs="unbounded" />
2784     <xs:element name="vehicleUsage" type="D2LogicalModel:VehicleUsageEnum"
↪ minOccurs="0" maxOccurs="1" />
2785     <xs:element name="grossWeightCharacteristic" type=
↪ "D2LogicalModel:GrossWeightCharacteristic" minOccurs="0" maxOccurs="2" />
2786     <xs:element name="heightCharacteristic" type=
↪ "D2LogicalModel:HeightCharacteristic" minOccurs="0" maxOccurs="2" />
2787     <xs:element name="lengthCharacteristic" type=
↪ "D2LogicalModel:LengthCharacteristic" minOccurs="0" maxOccurs="2" />
2788     <xs:element name="widthCharacteristic" type="D2LogicalModel:WidthCharacteristic
↪ " minOccurs="0" maxOccurs="2" />
2789     <xs:element name="heaviestAxleWeightCharacteristic" type=
↪ "D2LogicalModel:HeaviestAxleWeightCharacteristic" minOccurs="0" maxOccurs="2" />
2790     <xs:element name="numberOfAxlesCharacteristic" type=
↪ "D2LogicalModel:NumberOfAxlesCharacteristic" minOccurs="0" maxOccurs="2" />
2791     <xs:element name="vehicleCharacteristicsExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
2792   </xs:sequence>
2793 </xs:complexType>
2794 <xs:complexType name="VehicleCountValue">
2795   <xs:complexContent>
2796     <xs:extension base="D2LogicalModel:DataValue">
2797       <xs:sequence>
2798         <xs:element name="vehicleCount" type="D2LogicalModel:NonNegativeInteger"
↪ minOccurs="1" maxOccurs="1" />
2799         <xs:element name="vehicleCountValueExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
2800       </xs:sequence>
2801     </xs:extension>
2802   </xs:complexContent>
2803 </xs:complexType>
2804 <xs:complexType name="VehicleFlowValue">
2805   <xs:complexContent>
2806     <xs:extension base="D2LogicalModel:DataValue">
2807       <xs:sequence>
2808         <xs:element name="vehicleFlowRate" type="D2LogicalModel:VehiclesPerHour"
↪ minOccurs="1" maxOccurs="1" />
2809         <xs:element name="vehicleFlowValueExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
2810       </xs:sequence>
2811     </xs:extension>
2812   </xs:complexContent>
2813 </xs:complexType>
2814 <xs:complexType name="VehicleObstruction">
2815   <xs:complexContent>
2816     <xs:extension base="D2LogicalModel:Obstruction">
2817       <xs:sequence>
2818         <xs:element name="vehicleObstructionType" type=
↪ "D2LogicalModel:VehicleObstructionTypeEnum" minOccurs="1" maxOccurs="1" />
2819         <xs:element name="obstructingVehicle" type="D2LogicalModel:Vehicle"

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2820 <minOccurs="0" maxOccurs="unbounded" />
2821   <xs:element name="vehicleObstructionExtension" type="D2LogicalModel:_
2822   <ExtensionType" minOccurs="0" />
2823   </xs:sequence>
2824 </xs:extension>
2825 </xs:complexContent>
2826 </xs:complexType>
2827 <xs:simpleType name="VehicleObstructionTypeEnum">
2828   <xs:restriction base="xs:string">
2829     <xs:enumeration value="abandonedVehicle" />
2830     <xs:enumeration value="abnormalLoad" />
2831     <xs:enumeration value="brokenDownBus" />
2832     <xs:enumeration value="brokenDownHeavyLorry" />
2833     <xs:enumeration value="brokenDownVehicle" />
2834     <xs:enumeration value="convoy" />
2835     <xs:enumeration value="damagedVehicle" />
2836     <xs:enumeration value="dangerousSlowMovingVehicle" />
2837     <xs:enumeration value="emergencyVehicle" />
2838     <xs:enumeration value="highSpeedEmergencyVehicle" />
2839     <xs:enumeration value="longLoad" />
2840     <xs:enumeration value="militaryConvoy" />
2841     <xs:enumeration value="overheightVehicle" />
2842     <xs:enumeration value="prohibitedVehicleOnTheRoadway" />
2843     <xs:enumeration value="saltingOrGrittingVehicleInUse" />
2844     <xs:enumeration value="slowMovingMaintenanceVehicle" />
2845     <xs:enumeration value="slowVehicle" />
2846     <xs:enumeration value="snowplough" />
2847     <xs:enumeration value="trackLayingVehicle" />
2848     <xs:enumeration value="unlitVehicleOnTheRoad" />
2849     <xs:enumeration value="vehicleOnFire" />
2850     <xs:enumeration value="vehicleCarryingHazardousMaterials" />
2851     <xs:enumeration value="vehicleInDifficulty" />
2852     <xs:enumeration value="vehicleOnWrongCarriageway" />
2853     <xs:enumeration value="vehicleStuck" />
2854     <xs:enumeration value="vehicleStuckUnderBridge" />
2855     <xs:enumeration value="vehicleWithOverheightLoad" />
2856     <xs:enumeration value="vehicleWithOverwideLoad" />
2857     <xs:enumeration value="other" />
2858   </xs:restriction>
2859 </xs:simpleType>
2860 <xs:simpleType name="VehiclesPerHour">
2861   <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
2862 </xs:simpleType>
2863 <xs:simpleType name="VehicleStatusEnum">
2864   <xs:restriction base="xs:string">
2865     <xs:enumeration value="abandoned" />
2866     <xs:enumeration value="brokenDown" />
2867     <xs:enumeration value="burntOut" />
2868     <xs:enumeration value="damaged" />
2869     <xs:enumeration value="damagedAndImmobilized" />
2870     <xs:enumeration value="onFire" />
2871   </xs:restriction>
2872 </xs:simpleType>
2873 <xs:simpleType name="VehicleTypeEnum">
2874   <xs:restriction base="xs:string">
2875     <xs:enumeration value="agriculturalVehicle" />

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2874     <xs:enumeration value="anyVehicle" />
2875     <xs:enumeration value="articulatedVehicle" />
2876     <xs:enumeration value="bicycle" />
2877     <xs:enumeration value="bus" />
2878     <xs:enumeration value="car" />
2879     <xs:enumeration value="caravan" />
2880     <xs:enumeration value="carOrLightVehicle" />
2881     <xs:enumeration value="carWithCaravan" />
2882     <xs:enumeration value="carWithTrailer" />
2883     <xs:enumeration value="constructionOrMaintenanceVehicle" />
2884     <xs:enumeration value="fourWheelDrive" />
2885     <xs:enumeration value="highSidedVehicle" />
2886     <xs:enumeration value="lorry" />
2887     <xs:enumeration value="moped" />
2888     <xs:enumeration value="motorcycle" />
2889     <xs:enumeration value="motorcycleWithSideCar" />
2890     <xs:enumeration value="motorscooter" />
2891     <xs:enumeration value="tanker" />
2892     <xs:enumeration value="threeWheeledVehicle" />
2893     <xs:enumeration value="trailer" />
2894     <xs:enumeration value="tram" />
2895     <xs:enumeration value="twoWheeledVehicle" />
2896     <xs:enumeration value="van" />
2897     <xs:enumeration value="vehicleWithCatalyticConverter" />
2898     <xs:enumeration value="vehicleWithoutCatalyticConverter" />
2899     <xs:enumeration value="vehicleWithCaravan" />
2900     <xs:enumeration value="vehicleWithTrailer" />
2901     <xs:enumeration value="withEvenNumberedRegistrationPlates" />
2902     <xs:enumeration value="withOddNumberedRegistrationPlates" />
2903     <xs:enumeration value="other" />
2904 </xs:restriction>
2905 </xs:simpleType>
2906 <xs:simpleType name="VehicleUsageEnum">
2907     <xs:restriction base="xs:string">
2908         <xs:enumeration value="agricultural" />
2909         <xs:enumeration value="commercial" />
2910         <xs:enumeration value="emergencyServices" />
2911         <xs:enumeration value="military" />
2912         <xs:enumeration value="nonCommercial" />
2913         <xs:enumeration value="patrol" />
2914         <xs:enumeration value="recoveryServices" />
2915         <xs:enumeration value="roadMaintenanceOrConstruction" />
2916         <xs:enumeration value="roadOperator" />
2917         <xs:enumeration value="taxi" />
2918     </xs:restriction>
2919 </xs:simpleType>
2920 <xs:complexType name="Visibility">
2921     <xs:sequence>
2922         <xs:element name="minimumVisibilityDistance" type=
2923         ↪ "D2LogicalModel:IntegerMetreDistanceValue" />
2924         <xs:element name="visibilityExtension" type="D2LogicalModel:_ExtensionType"
2925         ↪ minOccurs="0" />
2926     </xs:sequence>
2927 </xs:complexType>
2928 <xs:complexType name="WeatherRelatedRoadConditions">
2929     <xs:complexContent>

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2928     <xs:extension base="D2LogicalModel:RoadConditions">
2929         <xs:sequence>
2930             <xs:element name="weatherRelatedRoadConditionType" type=
↪ "D2LogicalModel:WeatherRelatedRoadConditionTypeEnum" minOccurs="1" maxOccurs=
↪ "unbounded" />
2931             <xs:element name="roadSurfaceConditionMeasurements" type=
↪ "D2LogicalModel:RoadSurfaceConditionMeasurements" minOccurs="0" />
2932             <xs:element name="weatherRelatedRoadConditionsExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
2933         </xs:sequence>
2934     </xs:extension>
2935 </xs:complexContent>
2936 </xs:complexType>
2937 <xs:simpleType name="WeatherRelatedRoadConditionTypeEnum">
2938     <xs:restriction base="xs:string">
2939         <xs:enumeration value="blackIce" />
2940         <xs:enumeration value="deepSnow" />
2941         <xs:enumeration value="dry" />
2942         <xs:enumeration value="freezingOfWetRoads" />
2943         <xs:enumeration value="freezingPavements" />
2944         <xs:enumeration value="freezingRain" />
2945         <xs:enumeration value="freshSnow" />
2946         <xs:enumeration value="ice" />
2947         <xs:enumeration value="iceBuildUp" />
2948         <xs:enumeration value="iceWithWheelBarTracks" />
2949         <xs:enumeration value="icyPatches" />
2950         <xs:enumeration value="looseSnow" />
2951         <xs:enumeration value="normalWinterConditionsForPedestrians" />
2952         <xs:enumeration value="packedSnow" />
2953         <xs:enumeration value="roadSurfaceMelting" />
2954         <xs:enumeration value="slipperyRoad" />
2955         <xs:enumeration value="slushOnRoad" />
2956         <xs:enumeration value="slushStrings" />
2957         <xs:enumeration value="snowDrifts" />
2958         <xs:enumeration value="snowOnPavement" />
2959         <xs:enumeration value="snowOnTheRoad" />
2960         <xs:enumeration value="surfaceWater" />
2961         <xs:enumeration value="wet" />
2962         <xs:enumeration value="wetAndIcyRoad" />
2963         <xs:enumeration value="wetIcyPavement" />
2964         <xs:enumeration value="other" />
2965     </xs:restriction>
2966 </xs:simpleType>
2967 <xs:complexType name="WidthCharacteristic">
2968     <xs:sequence>
2969         <xs:element name="comparisonOperator" type=
↪ "D2LogicalModel:ComparisonOperatorEnum" minOccurs="1" maxOccurs="1" />
2970         <xs:element name="vehicleWidth" type="D2LogicalModel:MetresAsFloat" minOccurs="1"
↪ " maxOccurs="1" />
2971         <xs:element name="widthCharacteristicExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
2972     </xs:sequence>
2973 </xs:complexType>
2974 <xs:complexType name="Wind">
2975     <xs:sequence>
2976         <xs:element name="windMeasurementHeight" type=

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2977 <↪ "D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="0" maxOccurs="1" />
      <xs:element name="windSpeed" type="D2LogicalModel:SpeedValue" minOccurs="0" />
2978 <xs:element name="maximumWindSpeed" type="D2LogicalModel:SpeedValue" minOccurs=
      <↪ "0" />
2979 <xs:element name="windDirectionBearing" type=
      <↪ "D2LogicalModel:DirectionBearingValue" minOccurs="0" />
2980 <xs:element name="windDirectionCompass" type=
      <↪ "D2LogicalModel:DirectionCompassValue" minOccurs="0" />
2981 <xs:element name="windExtension" type="D2LogicalModel:_ExtensionType" minOccurs=
      <↪ "0" />
2982 </xs:sequence>
2983 </xs:complexType>
2984 <xs:complexType name="WinterDrivingManagement">
2985 <xs:complexContent>
2986 <xs:extension base="D2LogicalModel:NetworkManagement">
2987 <xs:sequence>
2988 <xs:element name="winterEquipmentManagementType" type=
      <↪ "D2LogicalModel:WinterEquipmentManagementTypeEnum" minOccurs="1" maxOccurs="1" />
2989 <xs:element name="winterDrivingManagementExtension" type="D2LogicalModel:_
      <↪ ExtensionType" minOccurs="0" />
2990 </xs:sequence>
2991 </xs:extension>
2992 </xs:complexContent>
2993 </xs:complexType>
2994 <xs:simpleType name="WinterEquipmentManagementTypeEnum">
2995 <xs:restriction base="xs:string">
2996 <xs:enumeration value="doNoUseStudTyres" />
2997 <xs:enumeration value="useSnowChains" />
2998 <xs:enumeration value="useSnowChainsOrTyres" />
2999 <xs:enumeration value="useSnowTyres" />
3000 <xs:enumeration value="winterEquipmentOnBoardRequired" />
3001 <xs:enumeration value="other" />
3002 </xs:restriction>
3003 </xs:simpleType>
3004 </xs:schema>

```

CHANGELOG

version 1.1.1

- minor textual revisions, update of references

version 1.1.0

- removed pointByCoordinates and pointByMultiCoordinates as a sole location (without linear reference) since it is not supported by NDIC at the moment. Respective samples has been removed.
- removed day-time validity selector to concept and schema
- confirmed feedtype addition to schema and samples
- confirmed OpenLR location addition to schema

version 1.1.0rc

- added pointByCoordinates to concepts, schema and samples
- added day-time validity selector to concept and schema
- added feedtype to schema and samples
- added OpenLR location to schema

version 1.0.1

- revision of the documentation, proofreading, changes to examples

version 1.0.0

- Initial format and documentation

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